

Advanced Combustion

—Chapter 4. Pollutant Formation and Control in Combustion

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Outline

1. Introduction

2. NOx Formation

3. PAH and Soot Formation

Combustion Pollutants

❑ Pollutant Formation During Combustion



COMBUSTION

Carbon
Hydrogen
Nitrogen
Sulfur
Oxygen

Primary Pollutants

CO - Carbon Monoxide
CO_2 - Carbon Dioxide
SO_2 - Sulfur
NO_x - Nitrogen Dioxide
N_2O - Nitric Oxide

VOCs - Volatile Organic Compounds
or
HCs - Hydrocarbons

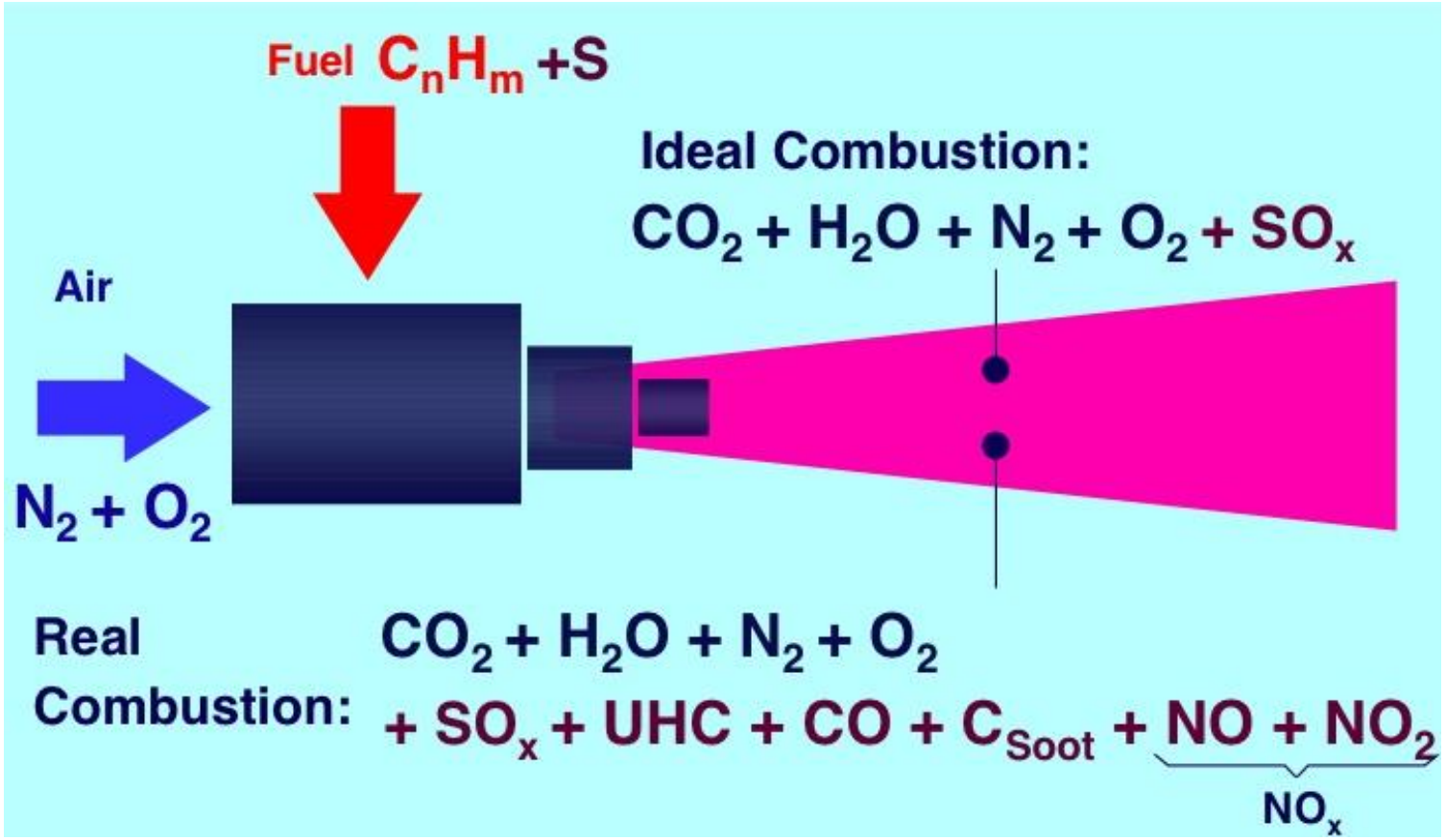
Particulate Matter

PM_{10} - Course Particles - less than 10 microns
$PM_{2.5}$ - Fine Particles - less than 2 microns

NH_3 - Ammonia

Combustion Pollutants

❑ Pollutant Formation During Combustion



❑ Water is the only environmentally friendly combustion product

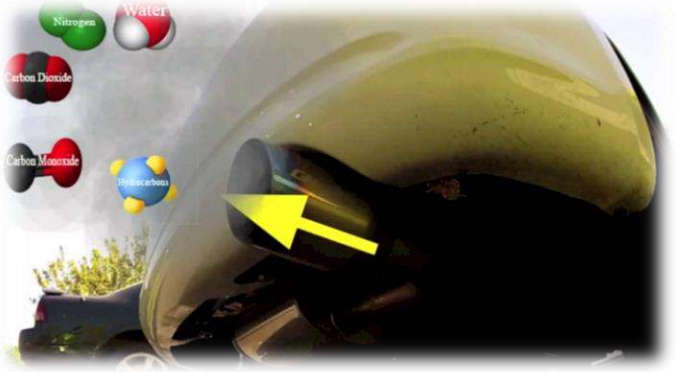
Combustion Pollutants

Pollutant Formation During Combustion

- NO_X – Reactions with nitrogen in air and/or fuel
- CO – Incomplete or rich combustion
- UHC – Incomplete combustion
- SO_X – Sulfur in fuels
- Particulates (soot, smoke) – Incomplete combustion
- CO_2 – Major product of hydrocarbon combustion



Source







Combustion Pollutants

Pollutant Formation During Combustion

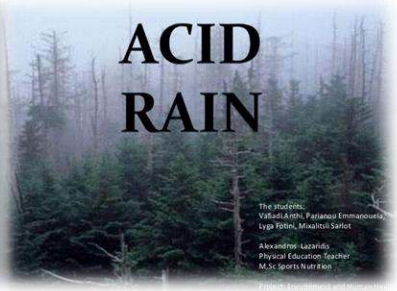
- NO, NO_2 – Causing photochemical smog and acid rain in the troposphere
- N_2O – Greenhouse gas
- CO – Poisoning
- SO_X – Acid rain
- Particulates (soot, smoke) – Haze
- CO_2 – Greenhouse gas

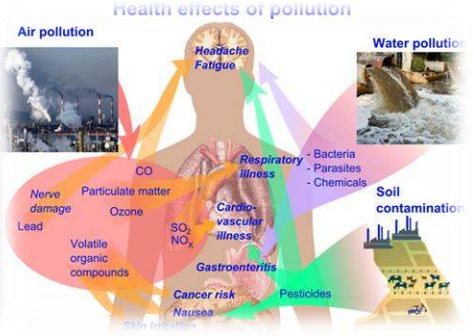
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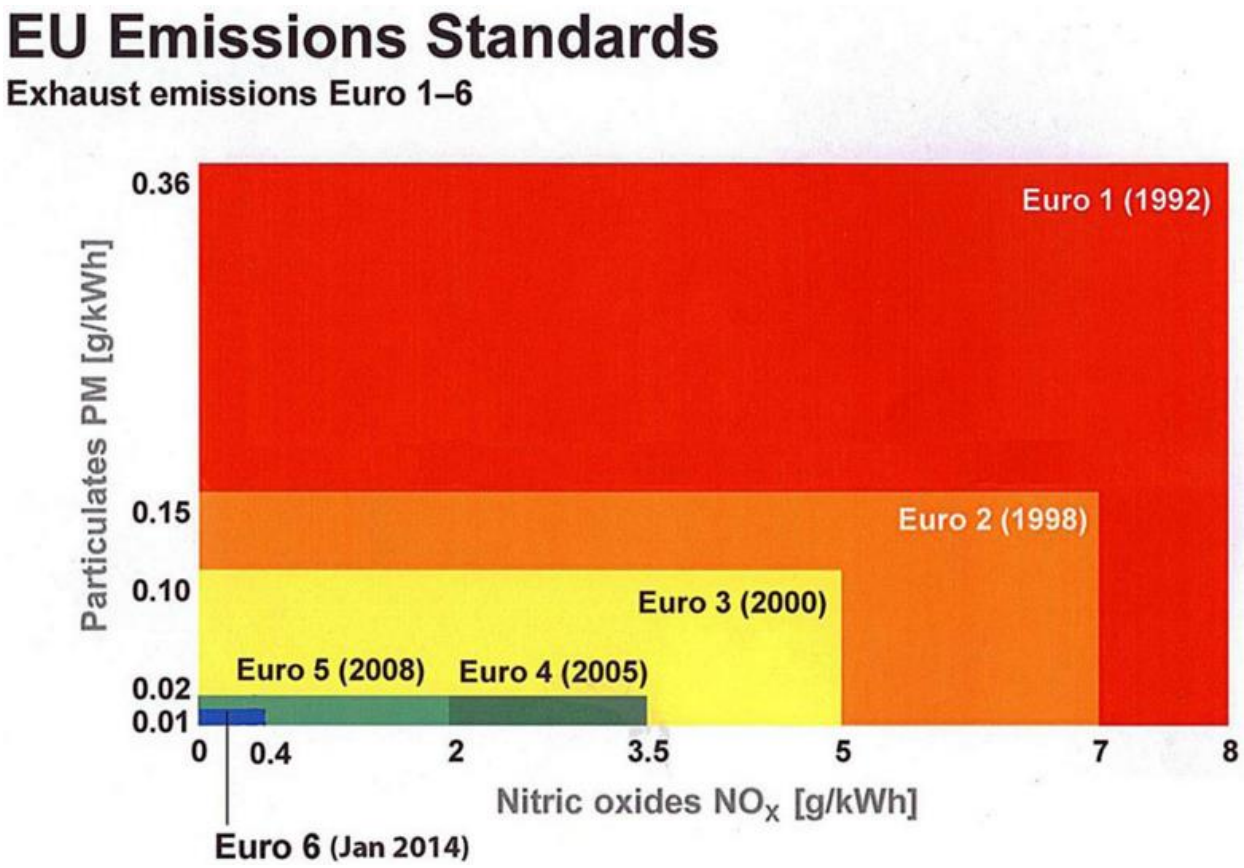






Combustion Pollutants

❑ Pollutant Formation During Combustion

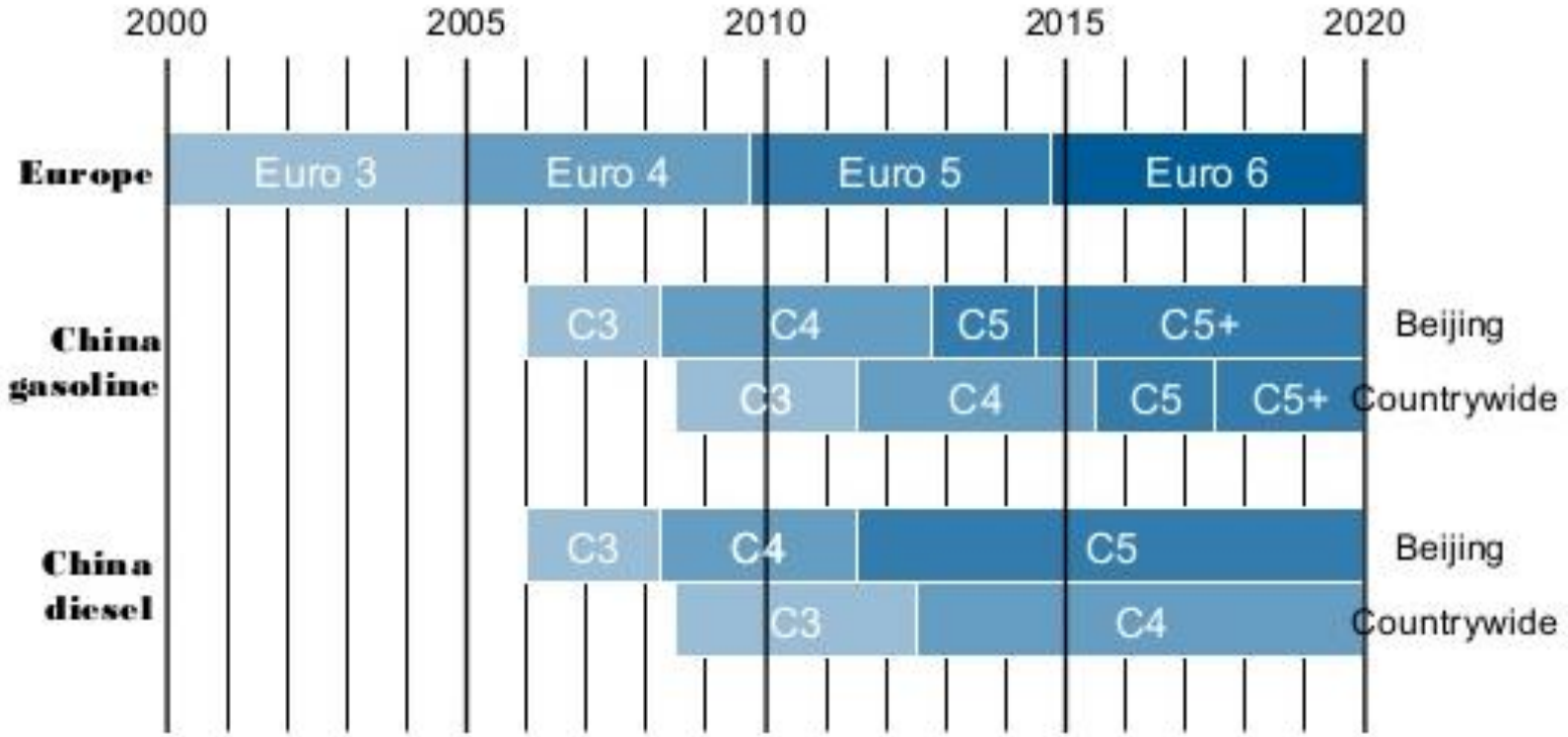


❖ **Among all the pollutants, NOx and soot are the two most important pollutants that are need strictly controlled.**

Combustion Pollutants

❑ Pollutant Formation During Combustion

☐ China emission standards



❖ More and more strict emission regulations have been made worldwide.

Combustion Pollutants

Solving Methods

- Catalytic converter
- New combustion devices
- New combustion technology
- ……

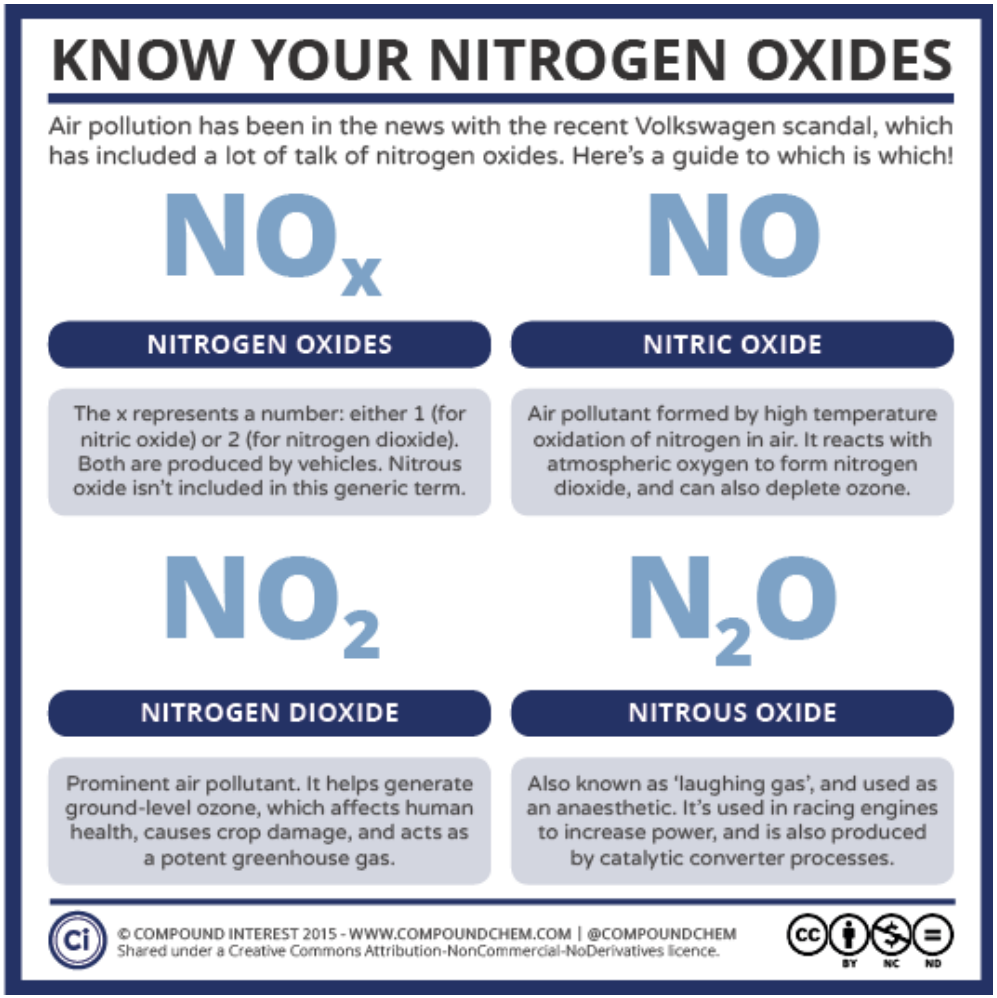
- Knowing the combustion kinetics is the first and the most important step
- Need detailed chemistry while not reduced or global chemistry
- Gas-solid interaction kinetics

1. Introduction

2. NOx Formation

3. PAH and Soot Formation

NOx



- **NO and NO_2 are important NOx pollutants.**

NOx Formation Mechanism

- Nitrogen oxides are mainly formed from atmospheric N_2 through mechanisms:
  - Thermal NO or Zeldovich-NO
  - Prompt NO or Fenimore-NO
  - NO generated via N_2O
  - NO generated via NNH
  - NO generated from fuel-bounded nitrogen
- Thermal, prompt and fuel N are the most important three sources
- Detailed chemical modeling plays a major role in the implementation of such mechanisms

Thermal/Zeldovich NOx

- Suggested by Y. B. Zeldovich, 1946, consists of three major reactions:

  - $O\bullet + N_2 \leftrightarrow NO + N\bullet$ R1
  - $N\bullet + O_2 \leftrightarrow NO + O\bullet$ R2
  - $N\bullet + OH\bullet \leftrightarrow NO + H\bullet$ R3
  - Net reaction: $N_2 + O_2 \leftrightarrow 2NO$

- N atom reacts extremely fast, therefore the quasi-steady-state approximation (QSSA) is applicable:

$$\frac{d[\mathrm{NO}]}{dt} = 2k_1 [\mathrm{N_2}] [\mathrm{O}]$$

- The first reaction is the rate controlling step
- High [O] combustion condition is favored

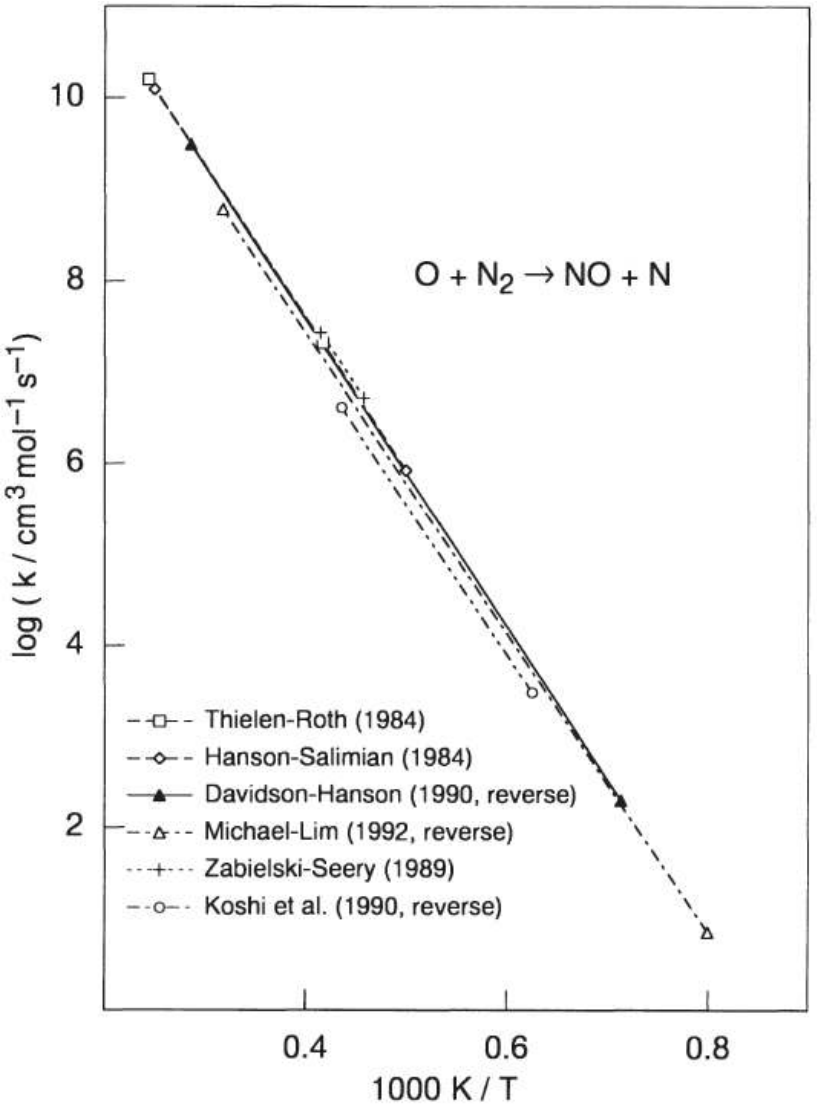
Thermal/Zeldovich NOx

- $O\bullet + N_2 \leftrightarrow NO + N\bullet$ R1 $E_1 = 314$ kJ/mol
- $N\bullet + O_2 \leftrightarrow NO + O\bullet$ R2
- $N\bullet + OH\bullet \leftrightarrow NO + H\bullet$ R3
- Net reaction: $N_2 + O_2 \leftrightarrow 2NO$

☐ Thermal NO production requires:

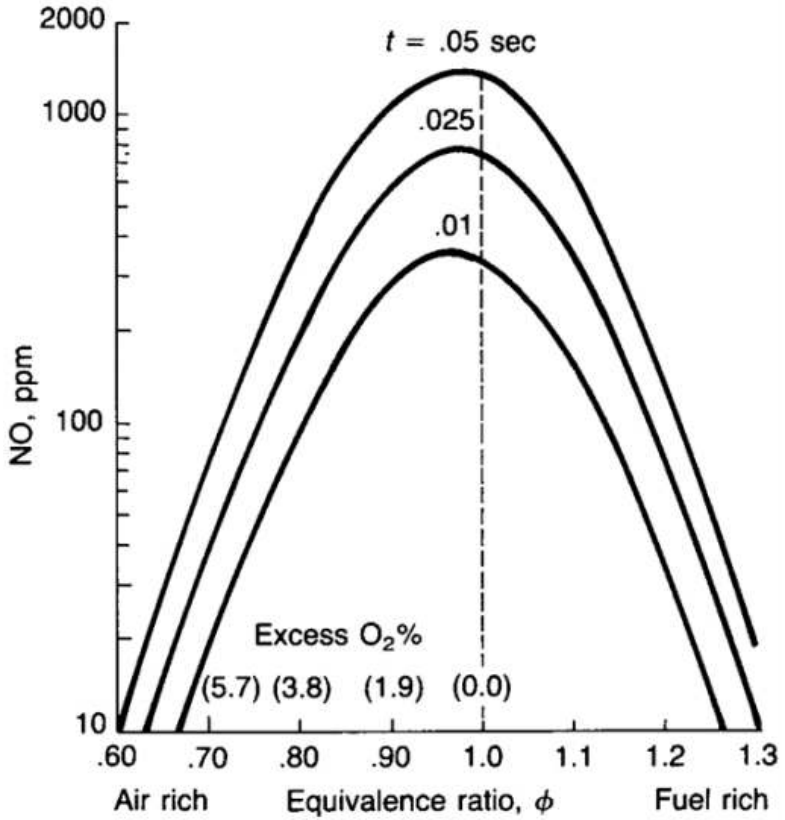
- High temperature T > 1900 K, because the activation energy of reaction R1 is high, it requires the breaking of the tight N-N triple bond
- Long residence time (slow reaction, post-flame gases)
- High [O], the concentration of oxygen atoms in flames increases rapidly with increasing temperature
- These conditions are present in the post-flame region

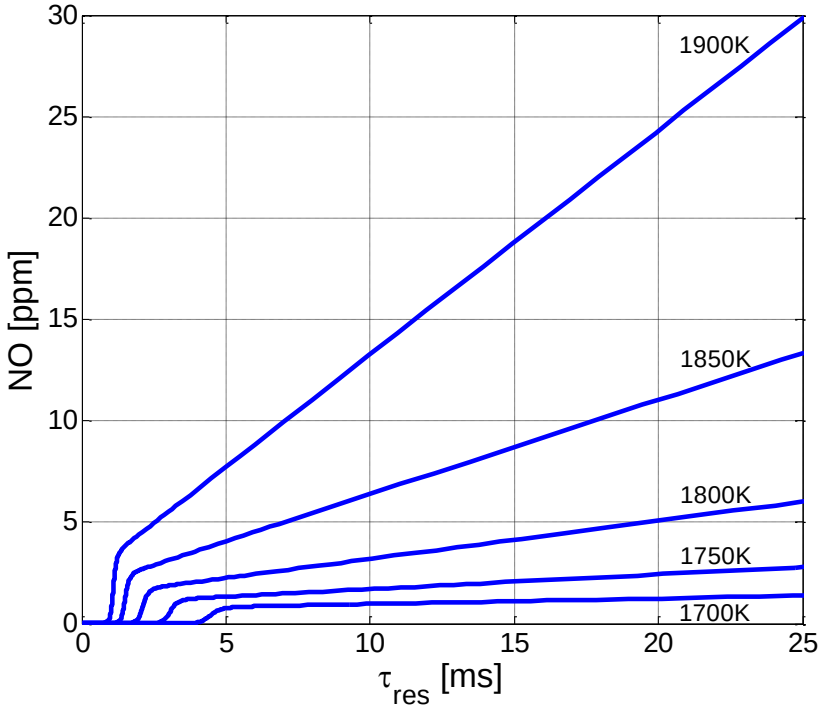
Thermal/Zeldovich NOx



- Since the rate coefficient for reaction R1 is well established, computation of the thermal NO production rate is straightforward if the oxygen atom concentration is known
- Thermal NO is sensitive to both T and [O]
- One must describe the kinetics of flame front chemistry properly in order to predict Zeldovich NO production accurately, even in the post-flame region
- Assuming that the [O] corresponds to the thermodynamic equilibrium may get wrong (higher) NO value

Thermal/Zeldovich NOx





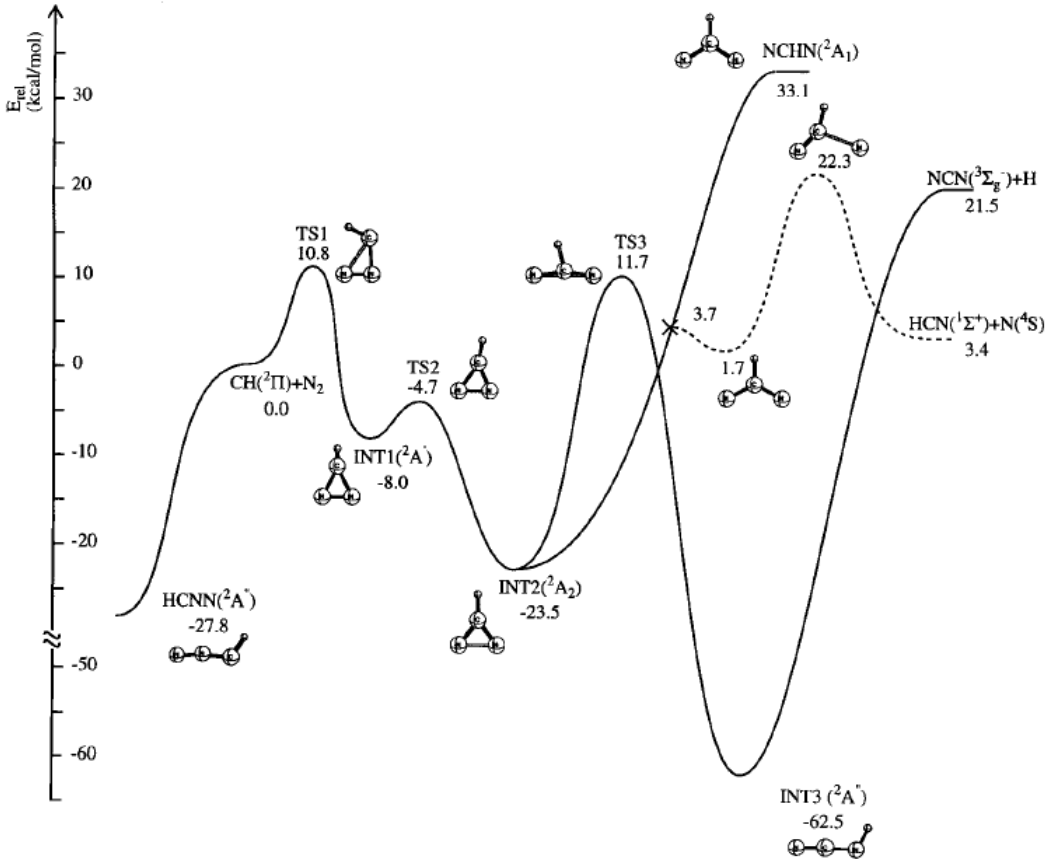
- **Two strategies may be used to reduce thermal NOx: Reduce peak temperatures of the flame zone; Reduce gas residence time in the flame zone**

Prompt/Fenimore NOx

- Discovered by C.P. Fenimore, 1976
- Observations:
  - NO formed earlier in HC combustion than possible from thermal mechanism
- Rate limiting step: $N_2 \rightarrow$ NO catalyzed by radical CH and CH_2
  - $CH\bullet + N_2 \leftrightarrow HCN + N\bullet$
  - $CH_2\bullet + N2 \leftrightarrow HCN + NH\bullet$
- Conversion to NO:
  - $HCN + O \leftrightarrow NCO + H$
  - $NCO + H \leftrightarrow NH + CO$
  - $NH + R \leftrightarrow N + RH$ $HCN + R \leftrightarrow CN + RH$ $CN + O \leftrightarrow N + CO$
  - $N + OH \leftrightarrow NO + H$

Prompt/Fenimore NOx

- New reaction proposed by Moskaleva and Lin, 2000

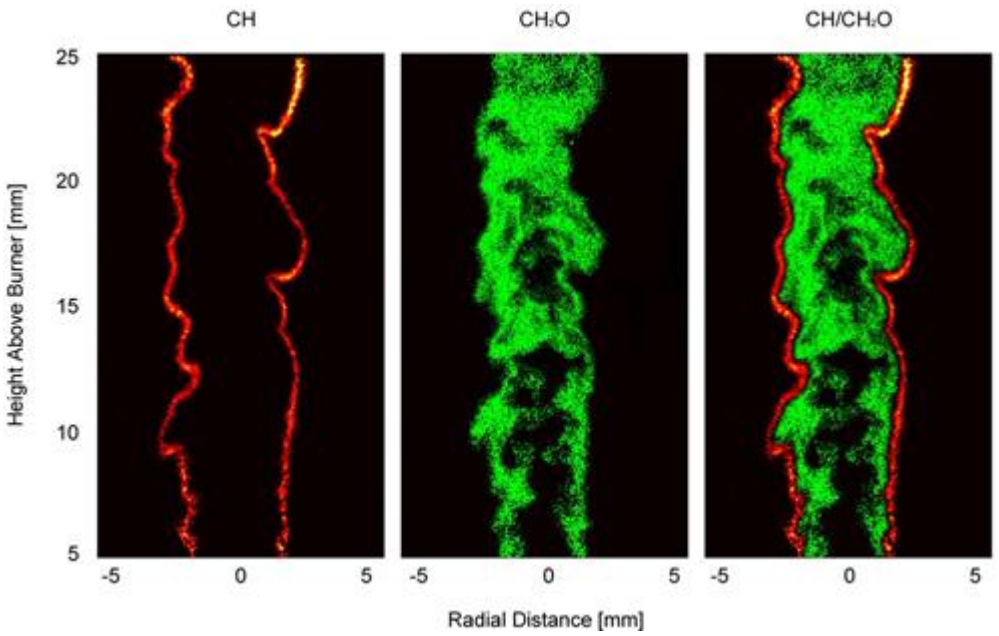


- Most important reaction:
  - CH• + N_2 ↔ NCN + H•
- Conversion to NO:
  - NCN + O ↔ NCO + N
  - NCO + H ↔ NH + CO
  - NH + R ↔ N + RH
  - N + OH ↔ NO + H

Prompt/Fenimore NOx

- Origin of CH

  - $CH_3 + OH \leftrightarrow CH_2 + H_2O$
  - $CH_3 + H \leftrightarrow CH_2 + H_2$
  - $CH_2 + OH \leftrightarrow CH + H_2O$
  - $CH_2 + H \leftrightarrow CH + H_2$



- Features of prompt NOx:

  - Important in fuel rich flames, due to the high [CH]
  - Important only at not very high temperatures (T < 2000K), because at high temperature thermal NO production is faster

NOx Production Via N_2O

- Important reactions:

N_2O production from N_2

- $N_2 + O + M \leftrightarrow N_2O + M$
- $N_2 + OH\ (+M) \leftrightarrow N_2O + H\ (+M)$
- $N_2 + HO_2\ (+M) \leftrightarrow N_2O + OH\ (+M)$
- $N_2 + O_2\ (+M) \leftrightarrow N_2O + O\ (+M)$

NO production from N_2O

- $N_2O + O \leftrightarrow 2NO$
- $N_2O + H \leftrightarrow NO + NH$

- Features:

- Important in lean combustion in gas turbines
- The lifetime of N_2O is less than 10 ms at 1500 K, then the mechanism is active only at low temperature
- Termolecular reactions (三分子反应): rate increases with pressure increases

NOx Production Via NNH

- Important reactions:

In the first reaction steps, NH and NNH are produced from N_2

- $N_2 + H \leftrightarrow NH + N$
- $N_2 + OH \leftrightarrow NH + NO$
- $N_2 + H \leftrightarrow NNH$
- $N_2 + OH \leftrightarrow NNH + O$

NO production from NH and NNH

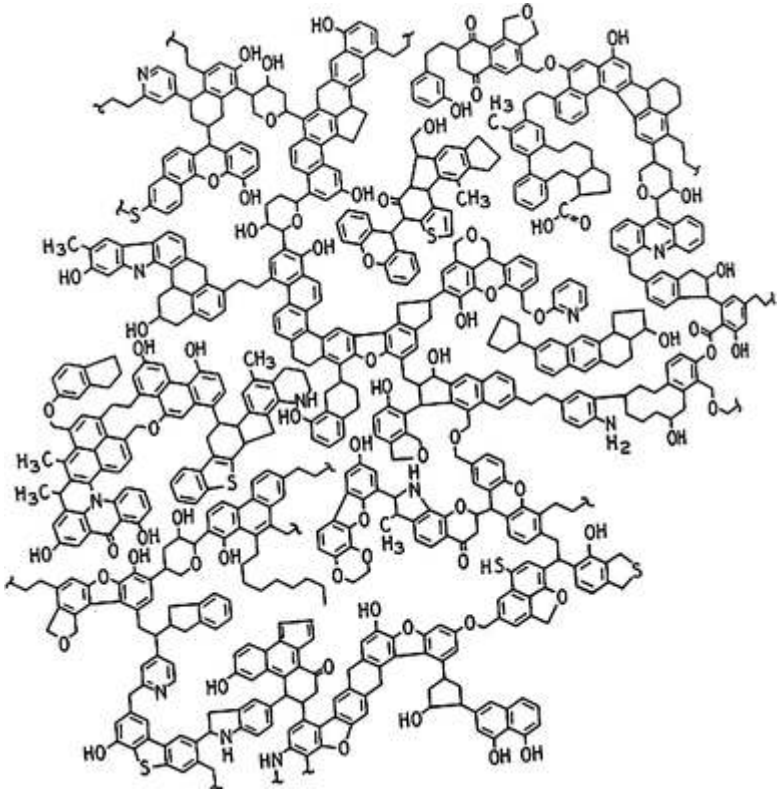
- $NNH + O \leftrightarrow NO + NH$
- $NH + R \leftrightarrow N + RH$
- $N + OH \leftrightarrow NO + H$

- Features:

- Important in lean and low-temperature flames, where thermal NO and prompt NO routes are slow
- Important at low pressure (at high pressure the "via N_2O" route is fast)

NOx Production From Fuel-N

- The nitrogen of NO may also come from the fuel. It is especially important at coal, biomass or oil sand combustion
- The formation processes of NOx are very complex
  - Model of bituminous coal



Fuel	C	H	O	N
Wood	49.65	6.23	43.20	0.92
Peat	55.44	6.28	35.56	1.72
Lignite	72.95	5.24	20.50	1.31
Bituminous	84.24	5.55	8.69	1.52
Anthracite	93.50	2.81	2.72	0.97

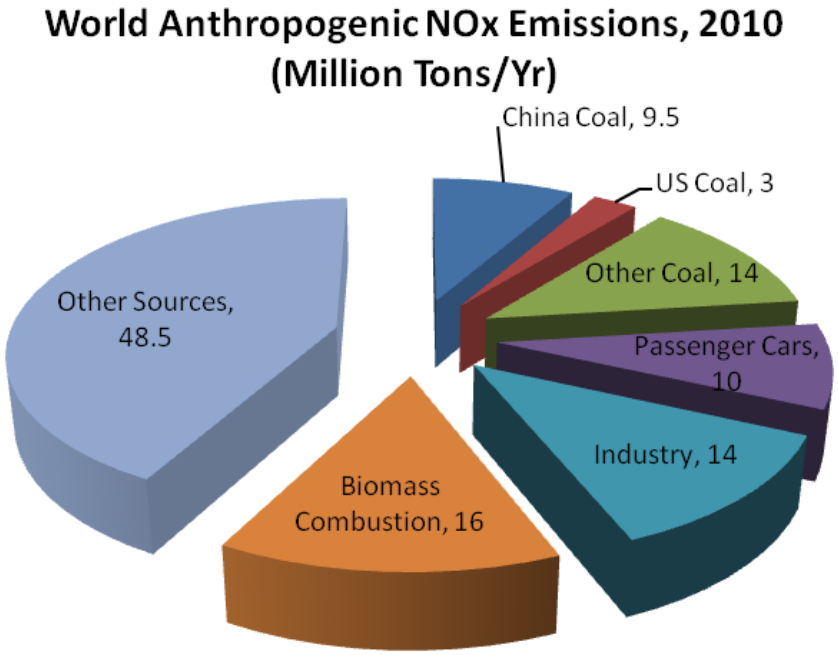
❖ Around 80% of NOx emission is from the fuel N routes

NOx Production From Fuel-N

- Important source: Fuel N, maybe different for different coal or biomass

Table 1
Typical nitrogen content in selected fuels [3,12,20,21,29]

Fuel	N content (wt%)
Straw	0.3–1.5
Other agricultural residues	0.4–3.5
Wood	0.03–1.0
Peat	0.5–2.5
Coal	0.5–2.5
Paper	0.1–0.2
RDF	0.8
Tires	0.3
Household waste	0.5–1.0
Plastic waste	0.0
Sewage sludge	2.5–6.5



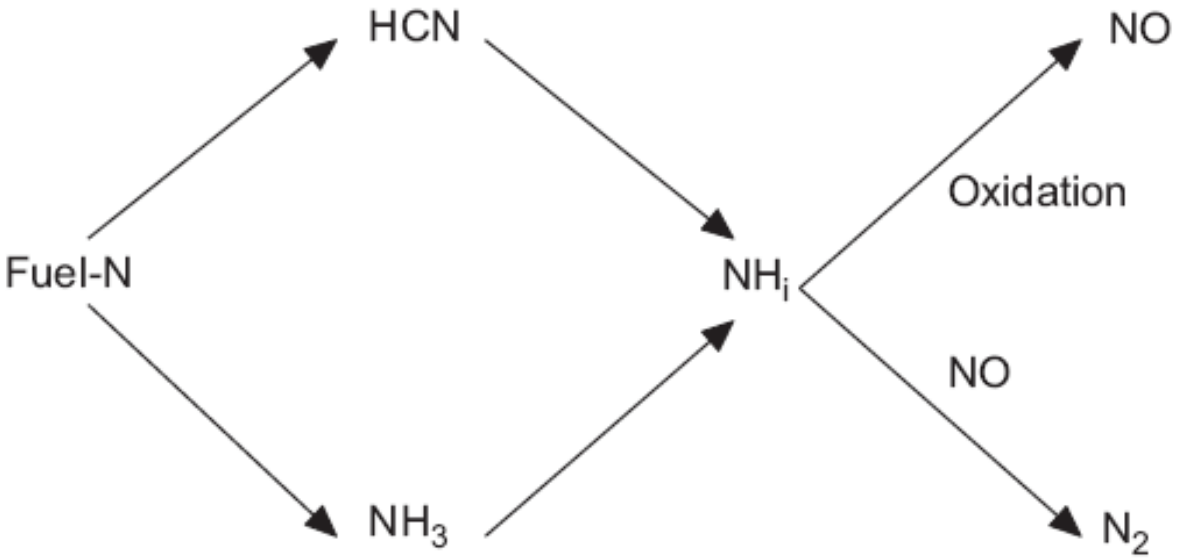


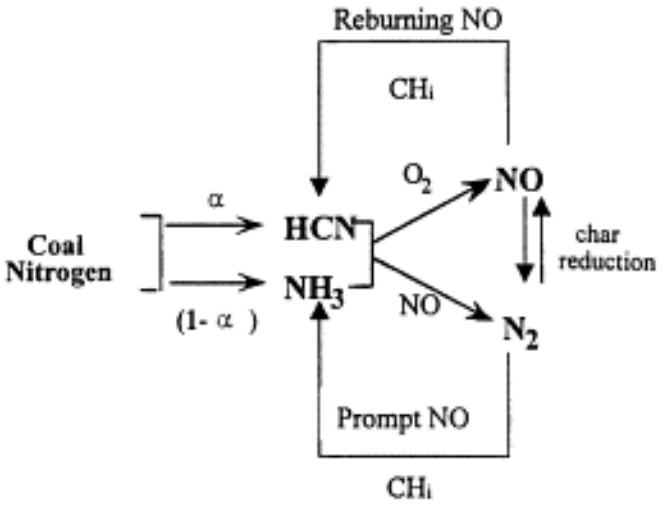
- Coal-fired power plant

- Biomass combustion



NOx Production From Fuel-N

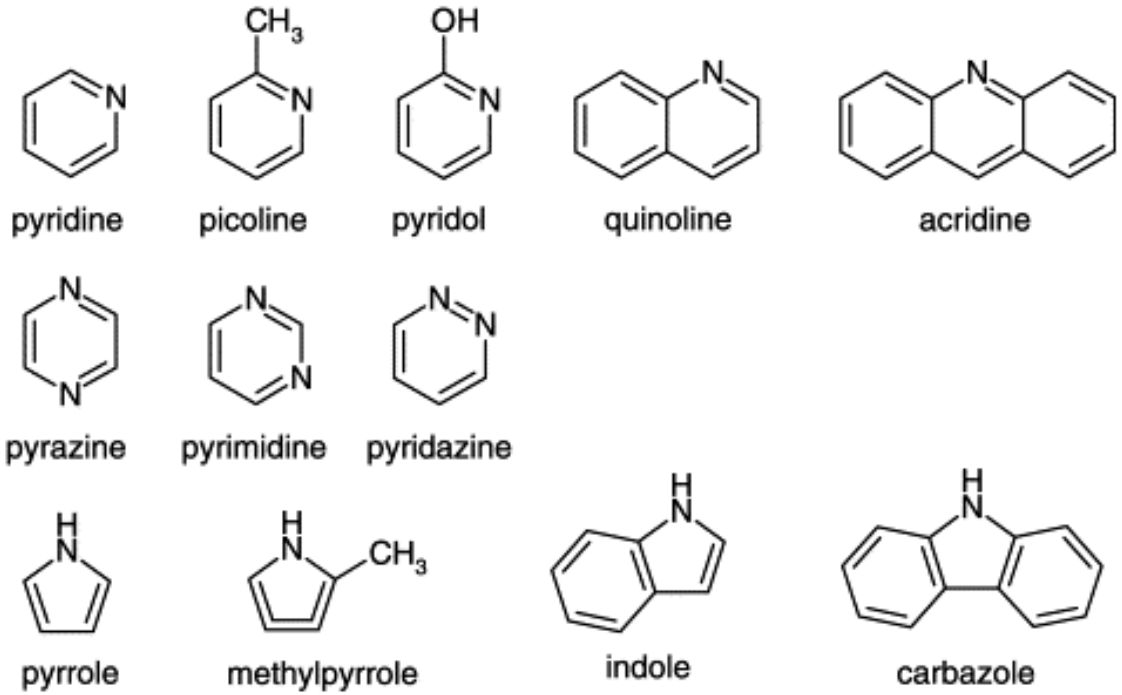


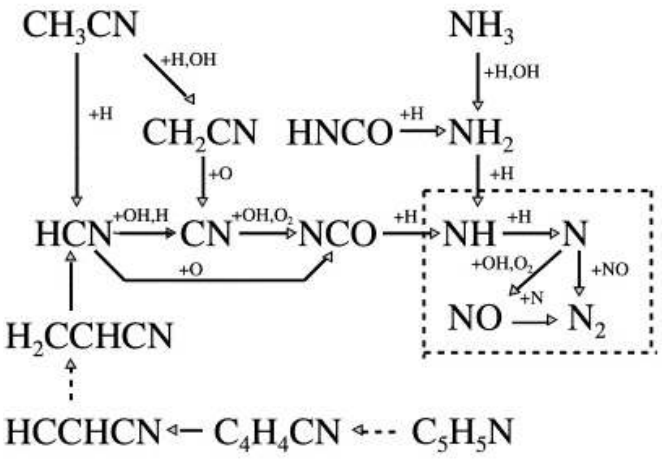


- Heterocyclic compounds → HCN → NO
- Amines (胺) → NH_3 → NH_i → NO
- Branching ratio towards NO or N2 is influenced via NH_i species, addition of ammonia or urea
- The transformation of fuel N to NOx depends on conditions of combustion (temperature and stoichiometry) and the concentration of N-compound

NOx Production From Fuel-N

- **Model compounds**

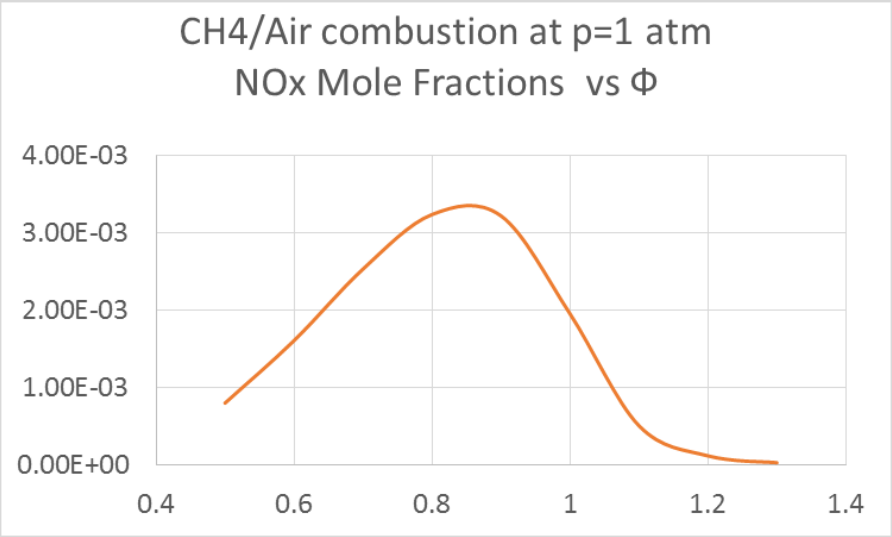




- **Usually, model compounds are chosen for combustion kinetic researches**

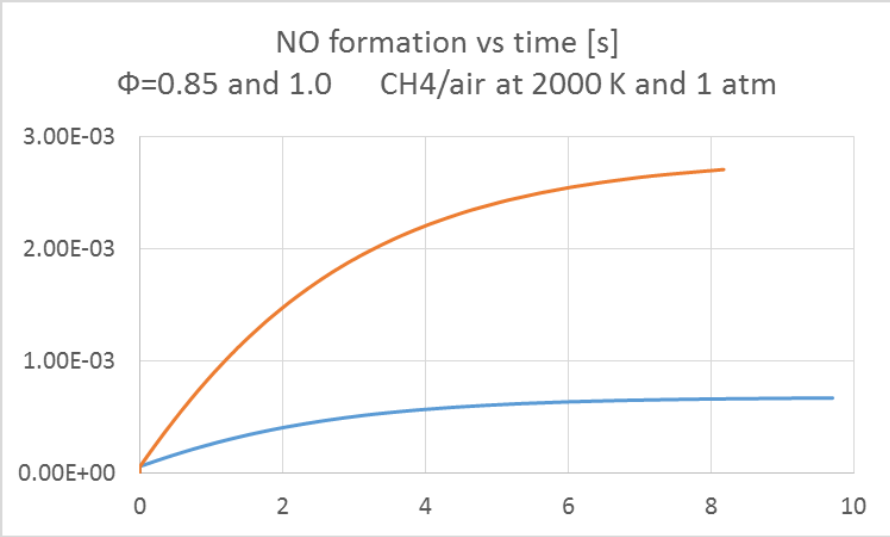
Thermodynamics vs Kinetics

Thermodynamics



NO at equilibrium

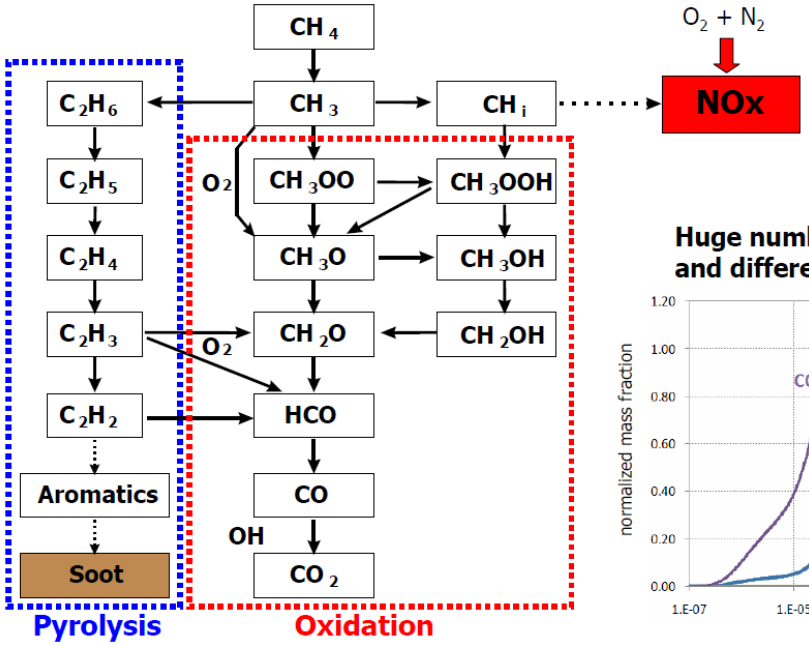
Chemical Kinetics

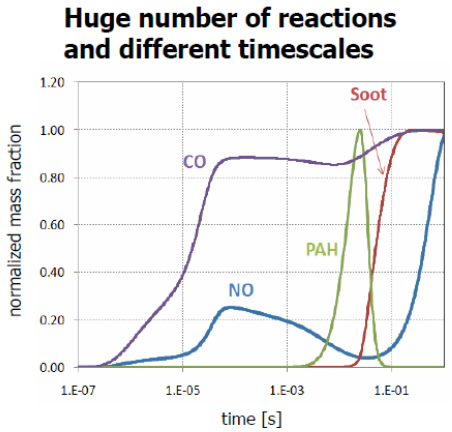


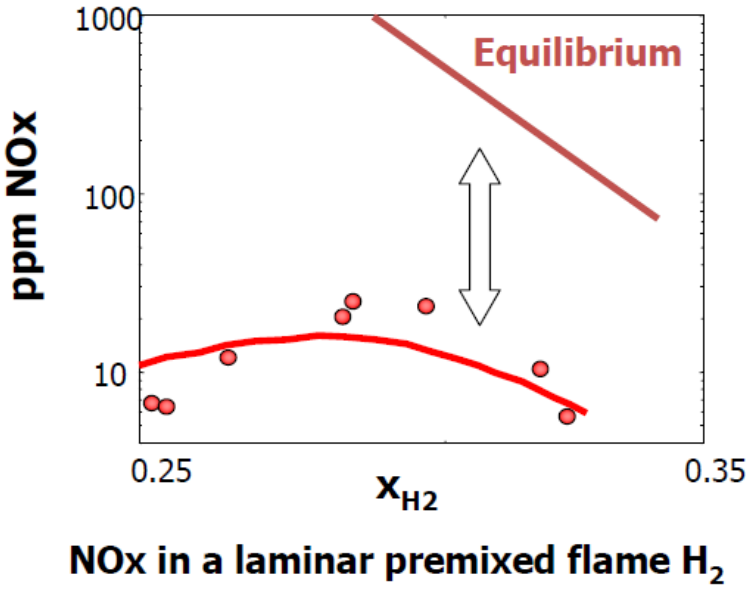
Rate of NO formation

- Equilibrium concentration is the maximum concentration
- As time increases, the concentration approaches the equilibrium

Thermodynamics vs Kinetics



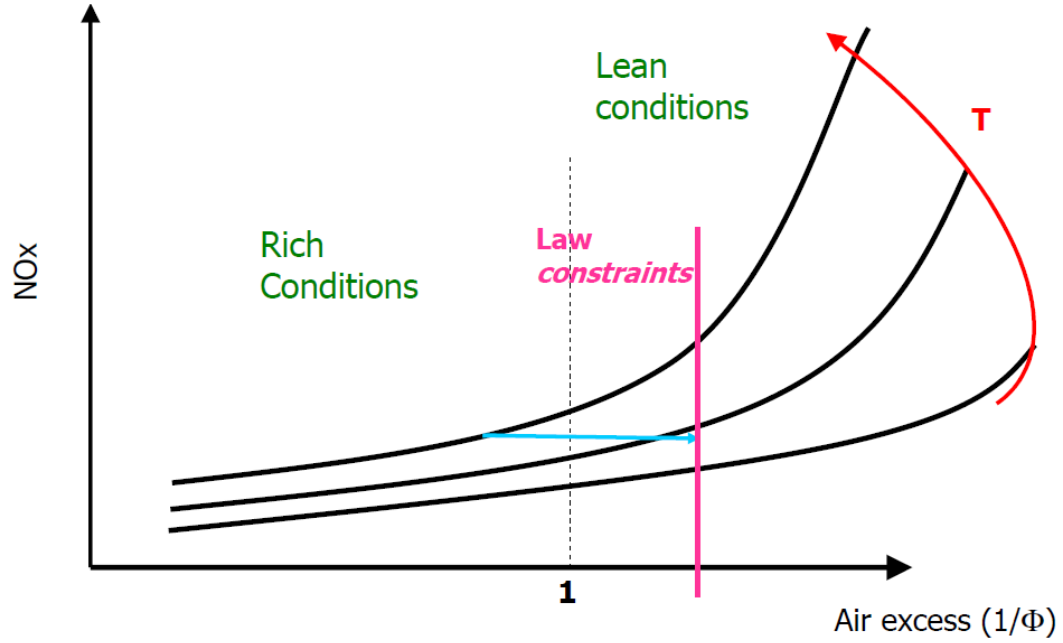




- NOx formation is chemically controlled

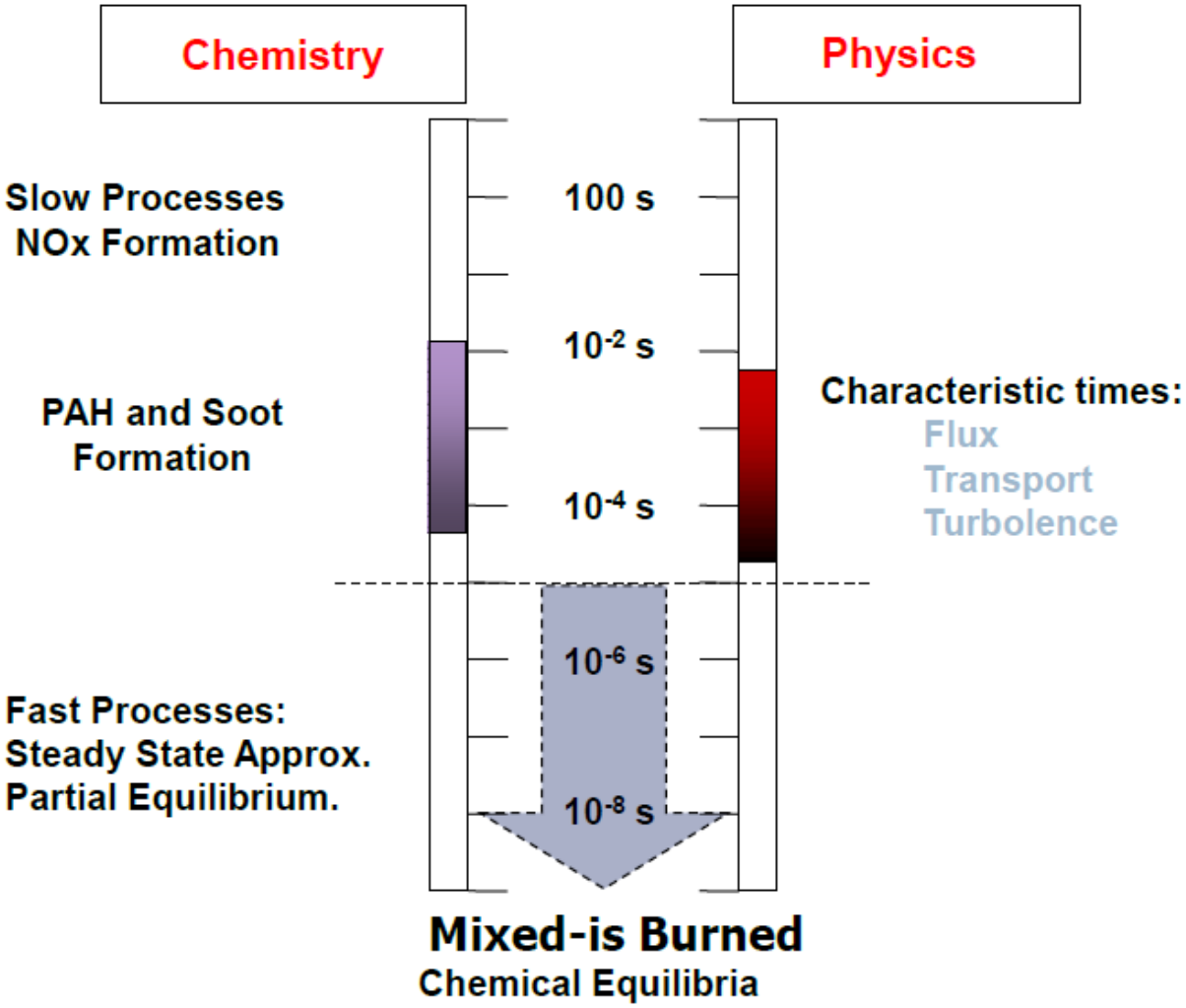
Thermodynamics vs Kinetics

- NOx formation is influenced by:
  - Flame temperature
  - Air excess
  - Amount of nitrogen chemically bound in the fuel (Fuel NOx)
  - Residence times (at high temperatures)



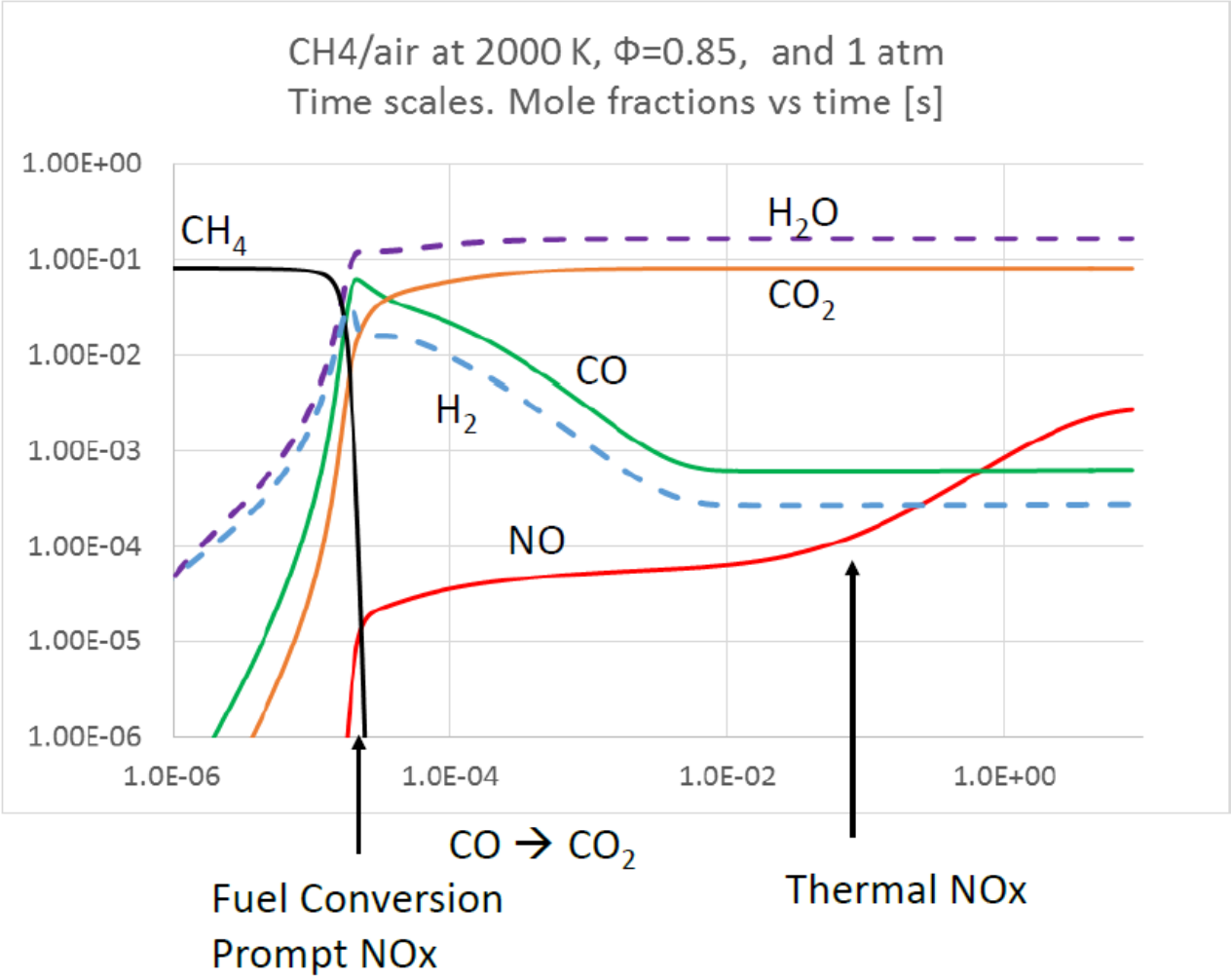
- T and ϕ effect on NOx formation

Time Scales



- The NOx formation processes is a slow chemical processes

Time Scales



- **Prompt important in reaction zone; thermal important in downstream**

Methods for Decreasing NO Emission

Abatement or Emission Control Principle or Method	Successful Technologies	Pollution Prevention Method (P2) or Add-on Technology (A)
1. Reducing peak temperature	Flue Gas Recirculation (FGR)	P2
	Natural Gas Reburning	P2
	Low NOx Burners (LNB)	P2
	Combustion Optimization	P2
	Burners Out Of Service (BOOS)	P2
	Less Excess Air (LEA)	P2
	Inject Water or Steam	P2
	Over Fire Air (OFA)	P2
	Air Staging	P2
	Reduced Air Preheat	P2
	Catalytic Combustion	P2
2.Reducing residence time at peak temperature	Inject Air	P2
	Inject Fuel	P2
	Inject Steam	P2

- Many technologies are used to control NOx emission

Methods for Decreasing NO Emission

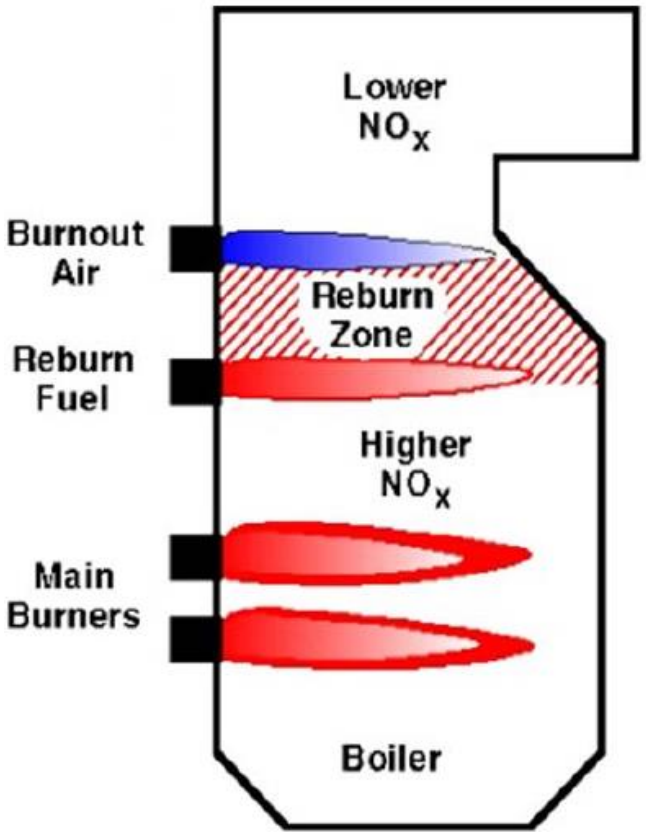
3. Chemical reduction of NOx	Fuel Reburning (FR) Low NOx Burners (LNB) Selective Catalytic Reduction (SCR) Selective Non-Catalytic Reduction (SNCR)	P2 P2 A A
4. Oxidation of NOx with subsequent absorption	Non-Thermal Plasma Reactor Inject Oxidant	A A
5. Removal of nitrogen	Oxygen Instead Of Air Ultra-Low Nitrogen Fuel	P2 P2
6. Using a sorbent	Sorbent In Combustion Chambers Sorbent In Ducts	A A
7. Combinations of these Methods	All Commercial Products	P2 and A

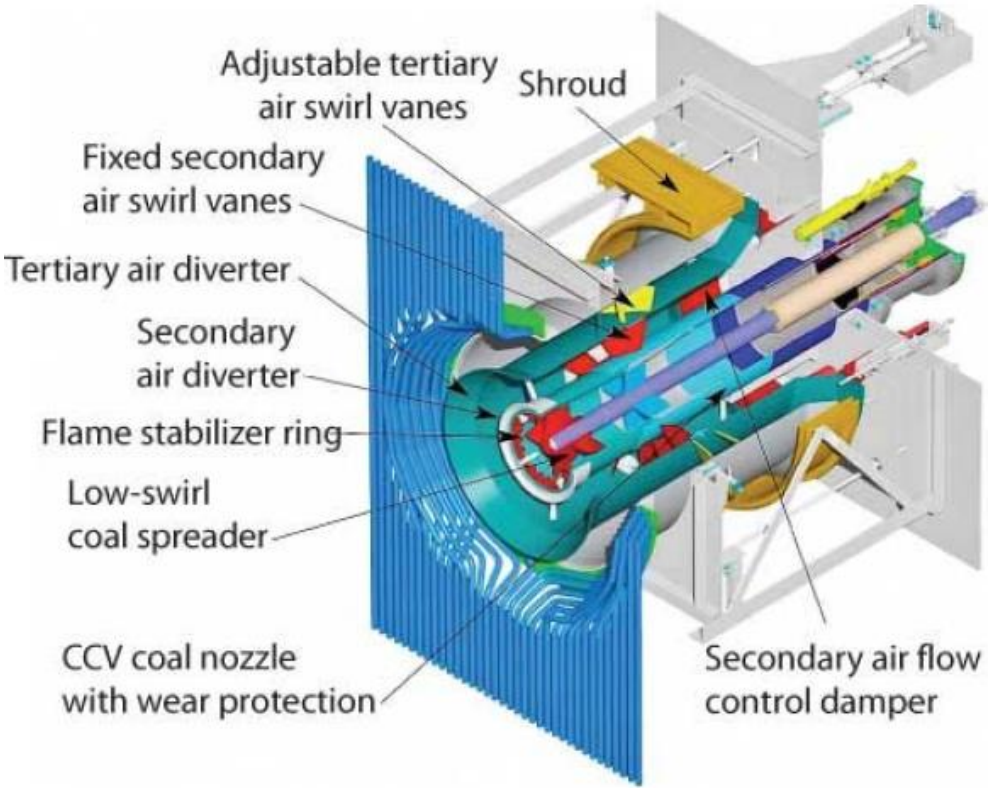
Methods for Decreasing NO Emission

- Primary methods: combustion modification-low NOx combustion
  - Staged combustion (分级燃烧): fuel staging; air staging
  - Reburning
  - Exhaust Gas Recirculation
  - .....
  - Usually extensive modification of the furnace is needed, these principles are mainly used at the design of new furnaces
- NO removal without the modification of the furnace
  - Aftertreatment of the exhaust gas
  - Selective heterogeneous catalytic reduction
  - Selective catalytic reduction: Thermal De NOx; AdBlue
  - Selective non-catalytic reduction

Methods for Decreasing NO Emission

- Staged combustion

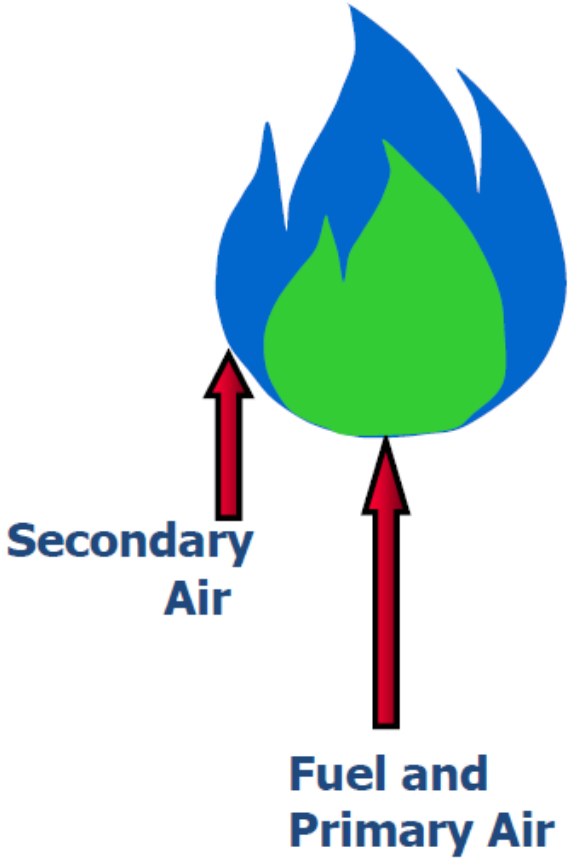




- Staged Combustion Thermal Power Plant
- Low-NOx Staged Burner of pulverized Coal

Methods for Decreasing NO Emission

- Chemistry of staged combustion
  - The fuel rich flame has a reduction capacity → NO is reduced to N_2, main reactions:
    - $CH_2 + NO = HCNO + H$
    - $CH + NO = HCN + O$
    - $C + NO = CN + O$
  - And then, $CN \rightarrow N_2$:
    - $CN \rightarrow HCN \rightarrow NCO \rightarrow NH \rightarrow N \rightarrow N_2$
    - $HNCO \rightarrow NH_2 \rightarrow NH \rightarrow N \rightarrow N_2$



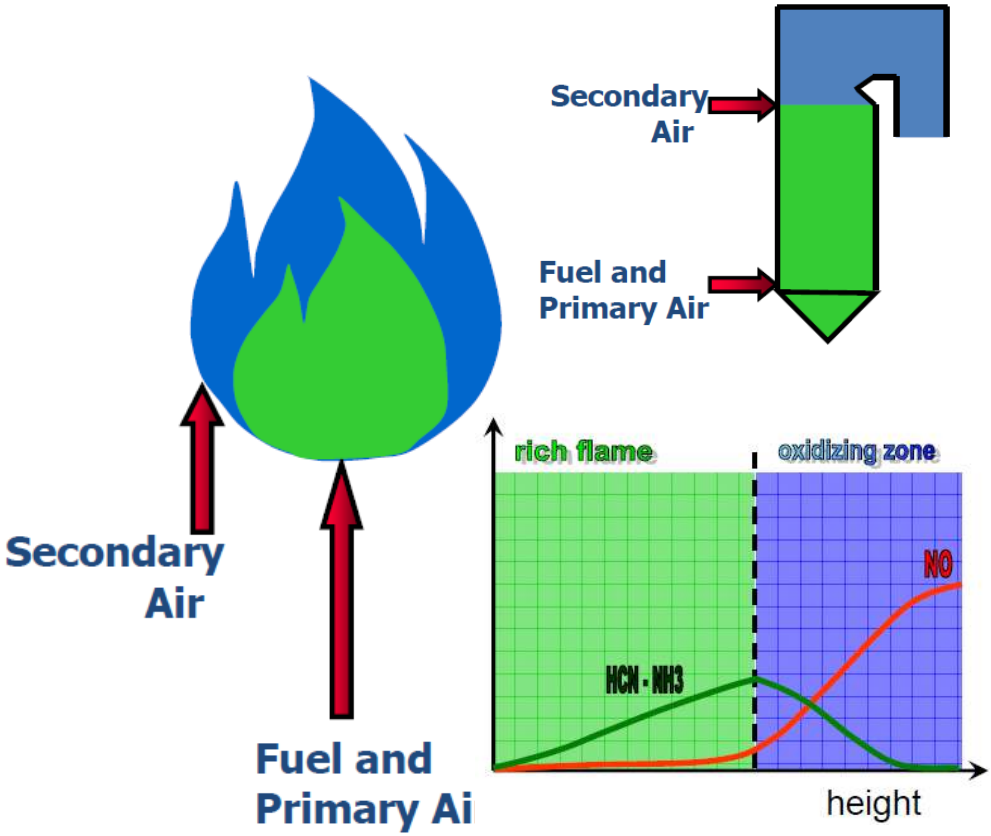
Methods for Decreasing NO Emission

- Chemistry of staged combustion

  - Problem: if ϕ >>1 → the furnace has a low efficiency

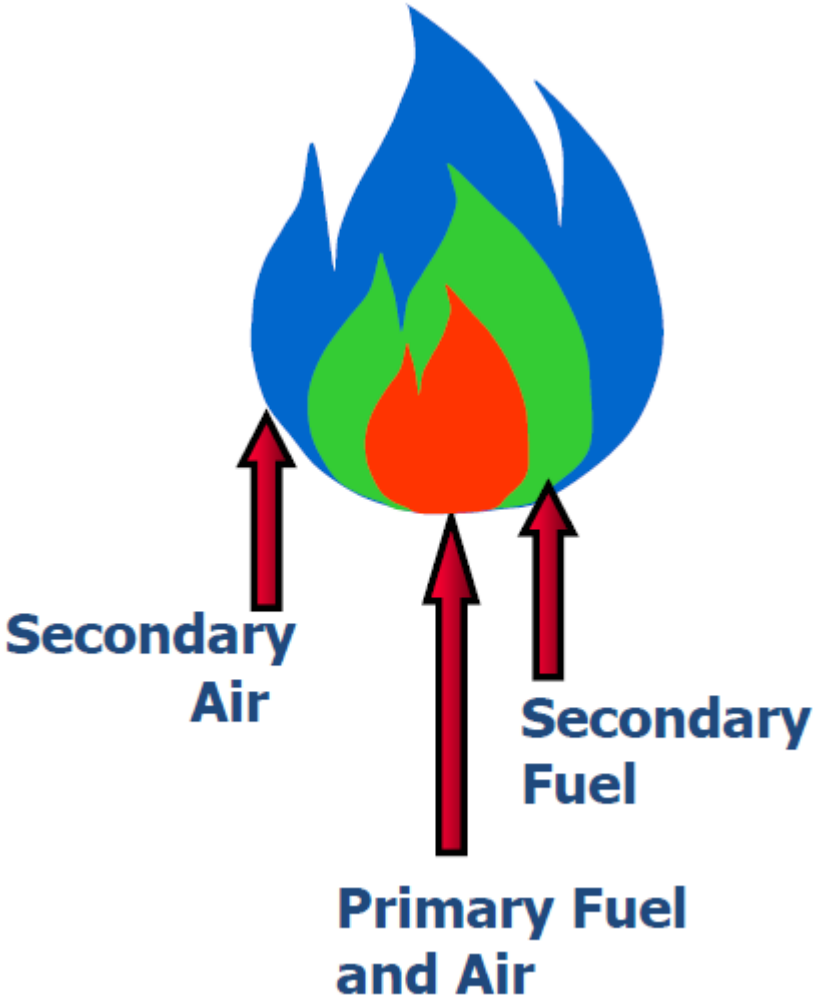
- Solution:

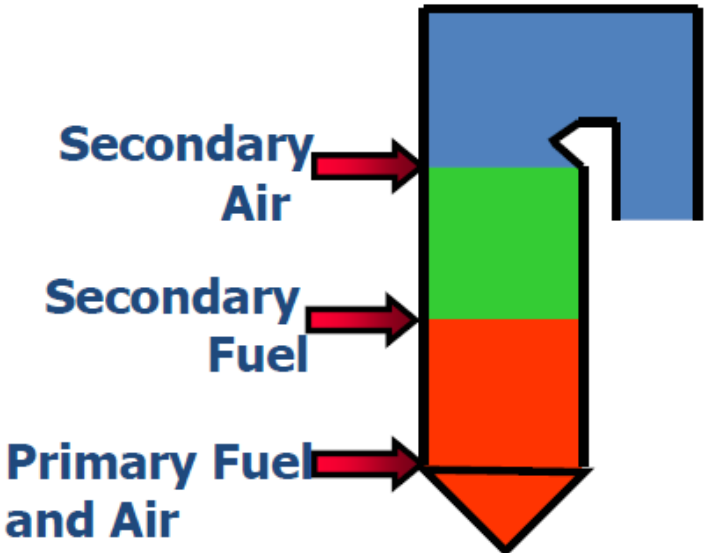
  - First phase: fuel rich stage ϕ = 1.4, NO reduction to N_2 (including the NO of fuel origin)
  - Second phase: air stage (complete fuel conversion, high efficiency, but here the temperature is low)

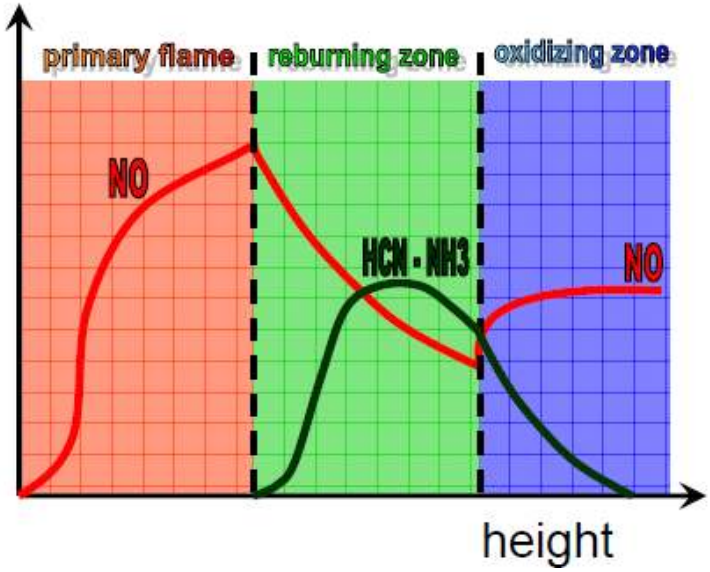


Methods for Decreasing NO Emission

- Reburning Technique

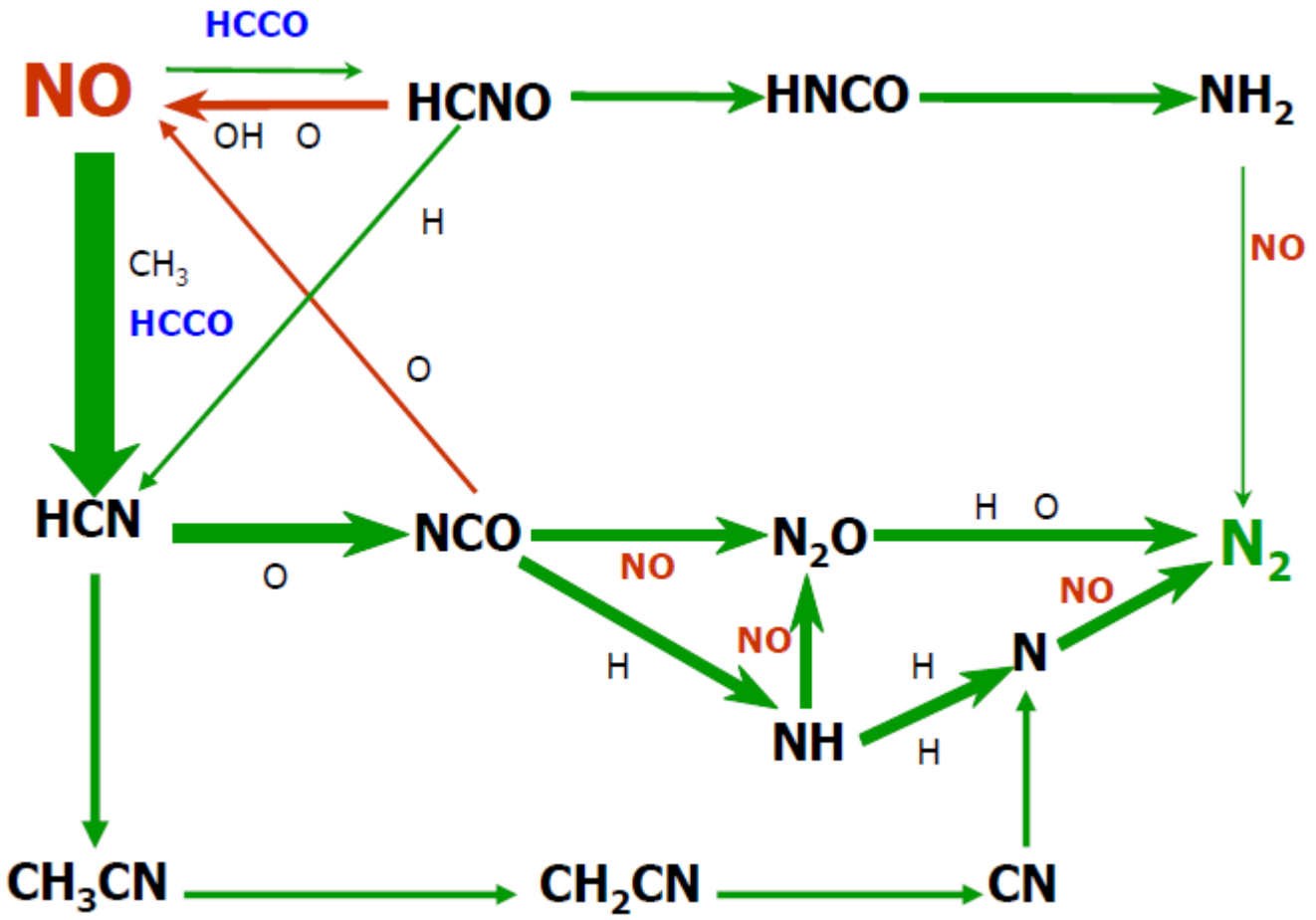






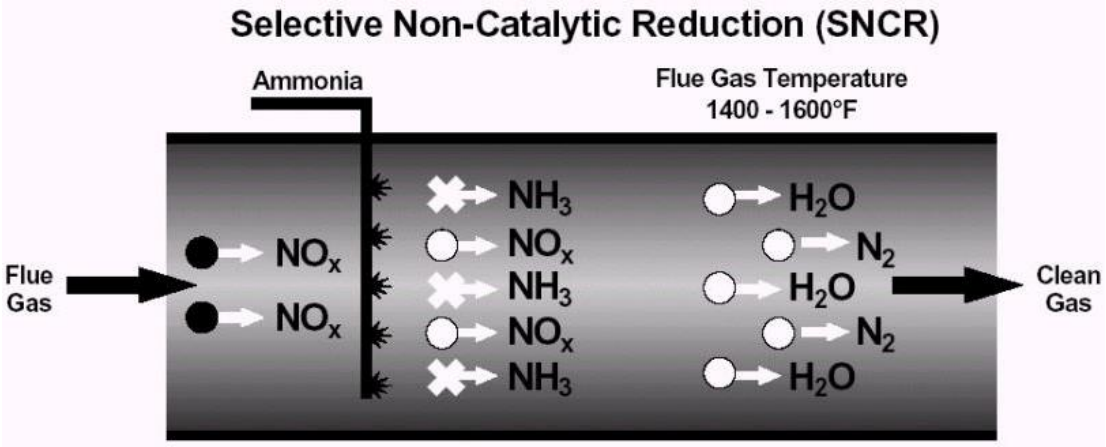
Methods for Decreasing NO Emission

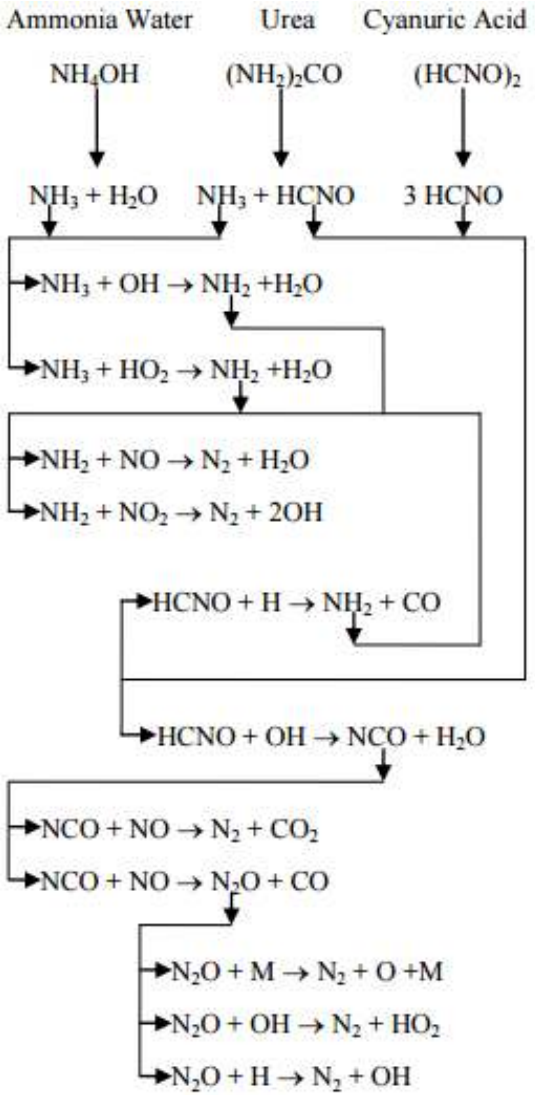
Chemistry of Reburning



Methods for Decreasing NO Emission

☐ Selective Non-Catalytic Reduction (SNCR)

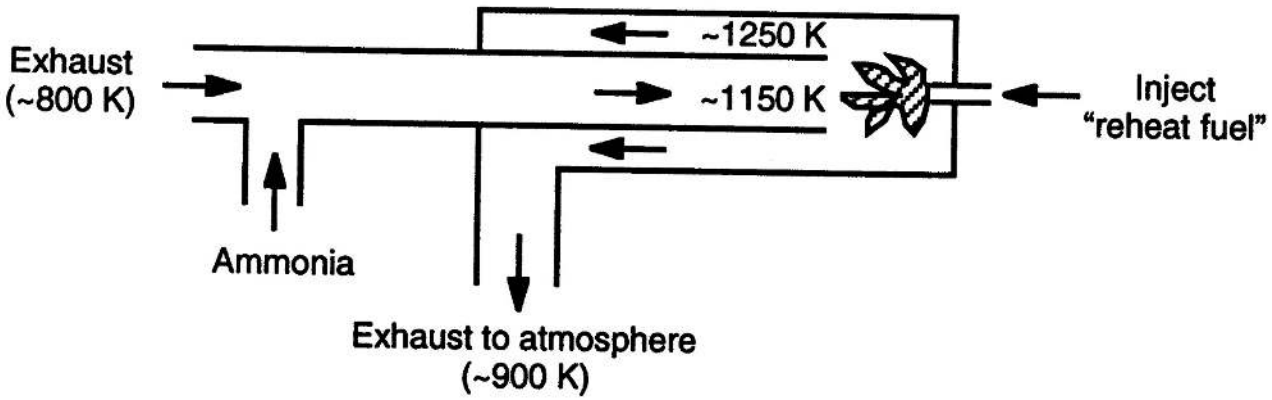


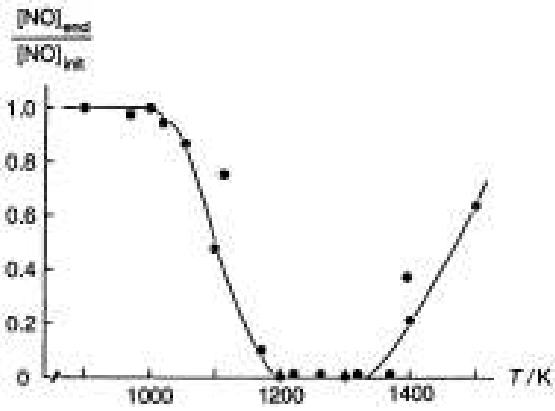


- ❖ In SNCR ammonia or urea is injected within a boiler or in ducts in a region where temperature is between 900 ℃ and 1100 ℃
- ❖ This technology is based on temperature ionizing the ammonia or urea instead of using a catalyst or non-thermal plasma

Methods for Decreasing NO Emission

- Selective Non-Catalytic Reduction (SNCR)





- The most practical exhaust gas treating method, relatively cheap
- Works in a narrow temperature window around T=1250 K (1100 K < T < 1400 K), The temperature of the exhaust gas should be in the good temperature interval
- + H_2 or + H_2O_2 : temperature window moves downwards, its width remains identical; +H_2O: temperature window moves upwards, its width remains identical

Methods for Decreasing NO Emission

Chemistry of SNCR

- $NH_3 + OH = NH_2 + H_2O$
- $NH_2 + NO = NNH + OH$
- $NNH + NO = N_2 + HNO$
- $HNO + M = H + NO + M$
- $H + O_2 = OH + O$
- $O + H_2O = 2OH$
- ……

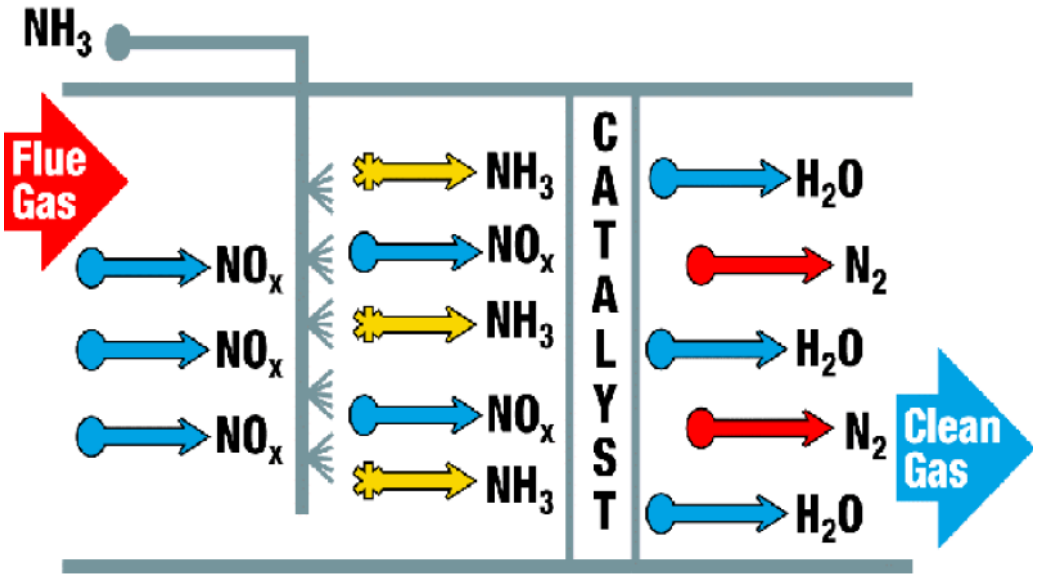
Chain termination reactions

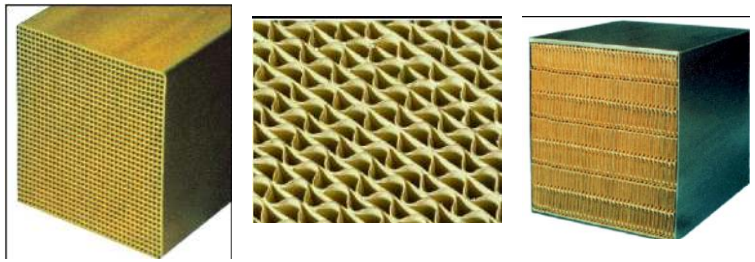
- $NH_2 + NO = N_2 + H_2O$
- $NH_2 + NHO = NH_3 + NO$
- $OH + HNO = H_2O + NO$

- Low temperature ($T < 1100$ K): the chain terminating reactions above are fast; high temperature ($T > 1400$ K): oxidation of NH_3 to NO
- without water: $NH_3+O = NH_2+OH$ instead $H_2O+O = 2OH$ (chain branching reaction), less OH is produced
- Much ammonia is present: $NH_3+H = NH_2+H_2$ instead $O_2+H = OH+O$ chain branching

Methods for Decreasing NO Emission

☐ Selective Catalytic Reduction (SCR)





$4NO + 4NH_3 + O_2 \rightarrow 4N_2 + 6H_2O$

$6NO + 8NH_3 \rightarrow 7N_2 + 12H_2O$

$6NO + 4NH_3 \rightarrow 5N_2 + 6H_2O$

$2NO + 4NH_3 + O_2 \rightarrow 7N_2 + 6H_2O$

$2NO + NO_2 + 8NH_3 \rightarrow 2N_2 + 3H_2O$

Catalyst

Nitrogen Oxides
Ammonia
Oxygen

Nitrogen
Water

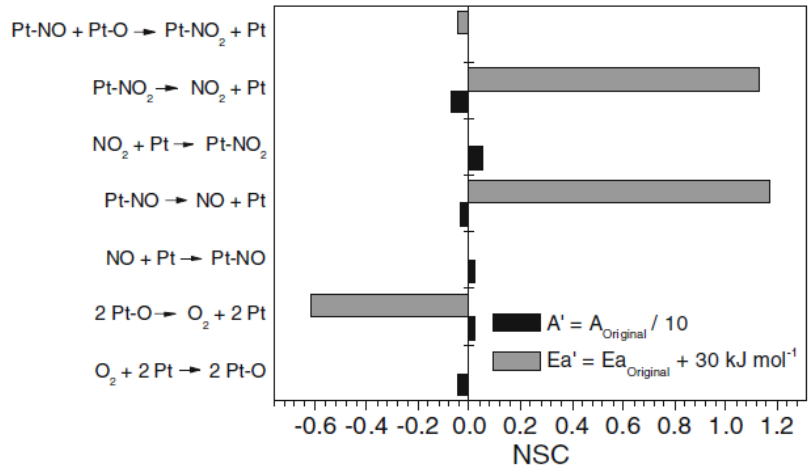
Ammonia (gaseous)
Ammonia as aqueous solution
Urea as aqueous solution

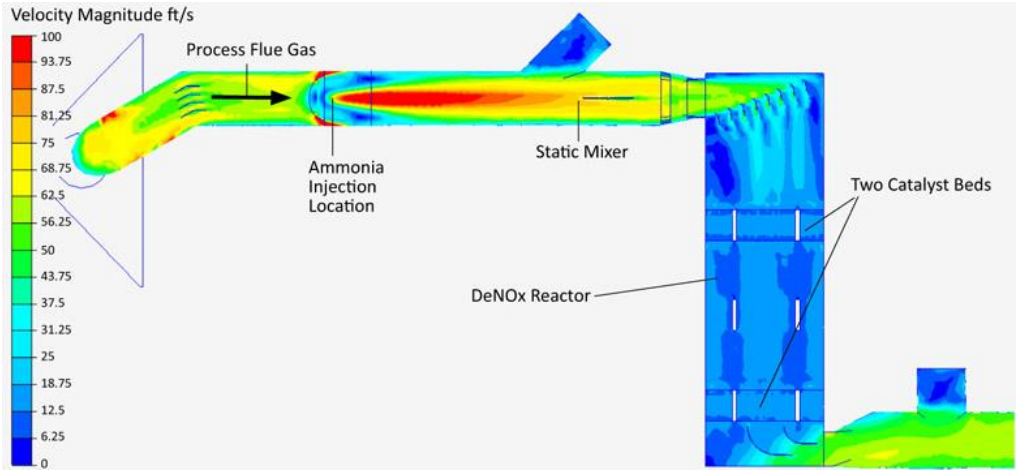
- ❖ Uses urea solution (ammonia) as a reductant to convert NOx into nitrogen and water
- ❖ In SCR process, a catalyst facilitates a chemical reaction between NOx and ammonia to produce nitrogen and water

Methods for Decreasing NO Emission

☐ Chemistry of SCR

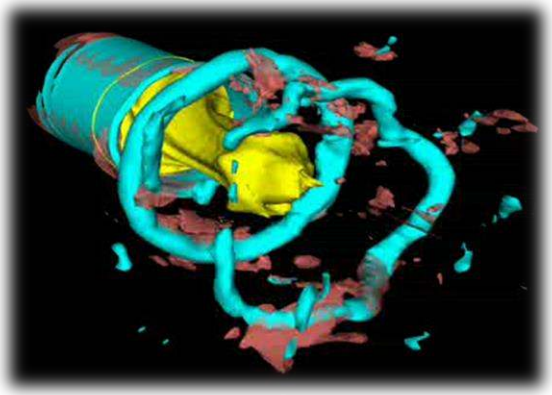
No.	Reaction	s_0 (unitless)[a] or A/s^{-1}	$E_a/(kJ\ mol^{-1})$
R1	$O_2 + 2\ Pt \rightarrow 2\ Pt{-}O$	5.42×10^{-2}	0
R2	$2\ Pt{-}O \rightarrow O_2 + 2\ Pt$	8.41×10^{12}	$207 - 134\theta_O$
R3	$NO + Pt \rightarrow Pt{-}NO$	8.80×10^{-1}	0
R4	$Pt{-}NO \rightarrow NO + Pt$	1.00×10^{16}	$138 - 30\theta_O - 33\theta_{NO}$ $(138 - 67\theta_O - 33\theta_{NO})$[b]
R5	$NO_2 + Pt \rightarrow Pt{-}NO_2$	9.70×10^{-1}	0
R6	$Pt{-}NO_2 \rightarrow NO_2 + Pt$	1.00×10^{16}	$119 - 45\theta_O$
R7	$Pt{-}NO + Pt{-}O \rightarrow Pt{-}NO_2 + Pt$	1.00×10^{11}	$120 - 34\theta_O - 26\theta_{NO}$ $(82 - 65\theta_O - 26\theta_{NO})$[b]
R8	$Pt{-}NO_2 + Pt \rightarrow Pt{-}NO + Pt{-}O$	9.65×10^{11}[c]	$56 + 19\theta_O + 7\theta_{NO}$[c] $(17 + 23\theta_O + 7\theta_{NO})$[b]

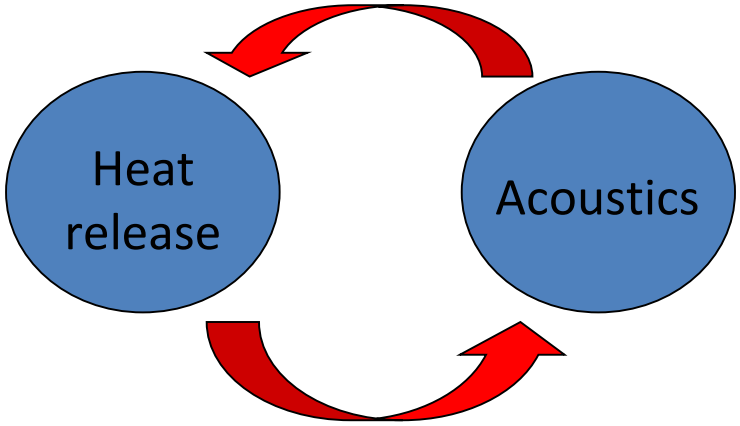




Low-NOx Design and Combustion Instabilities







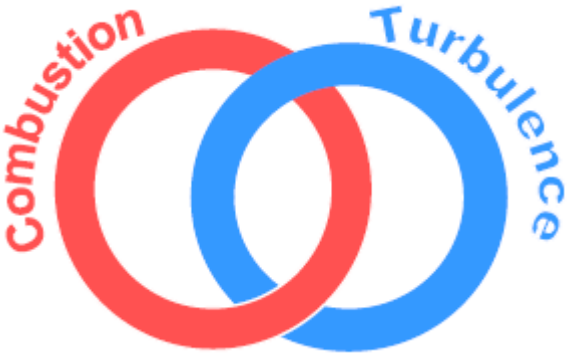


- Low NOx Designs make systems susceptible to large amplitude acoustic pulsations
- Low NOX designs make flame stabilization more problematic

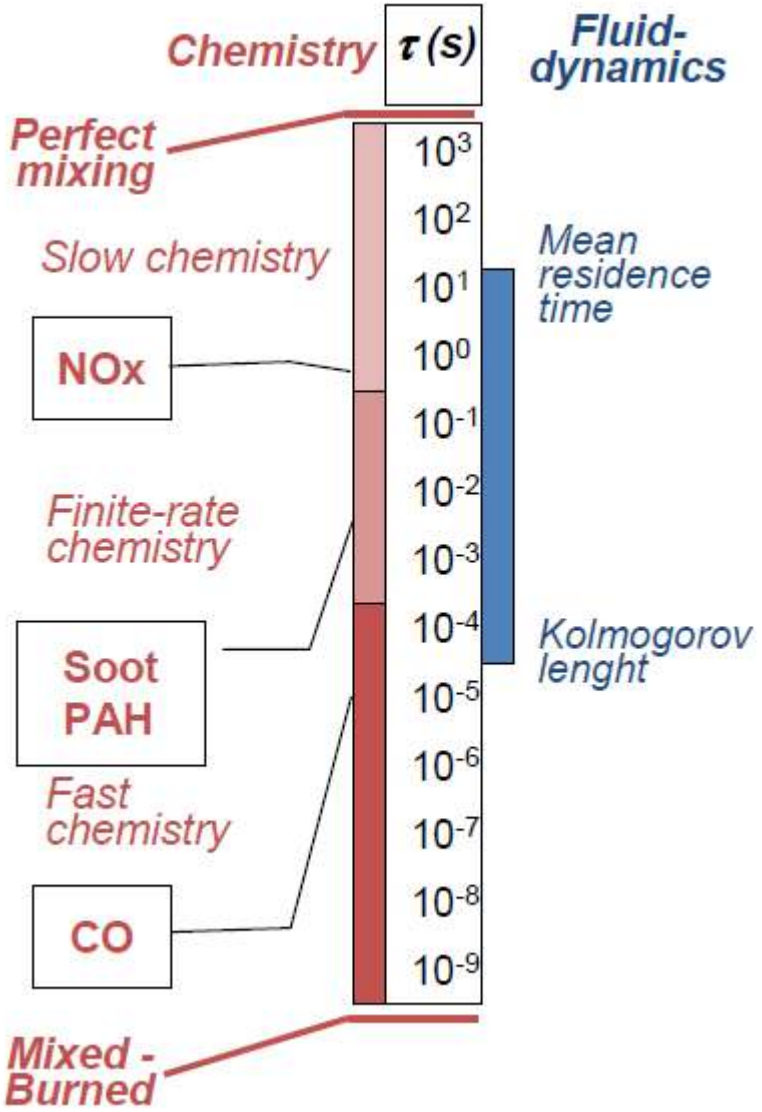
Approaches to Modeling

- Turbulence-kinetics interactions
  - Pollutant formation in turbulent flames

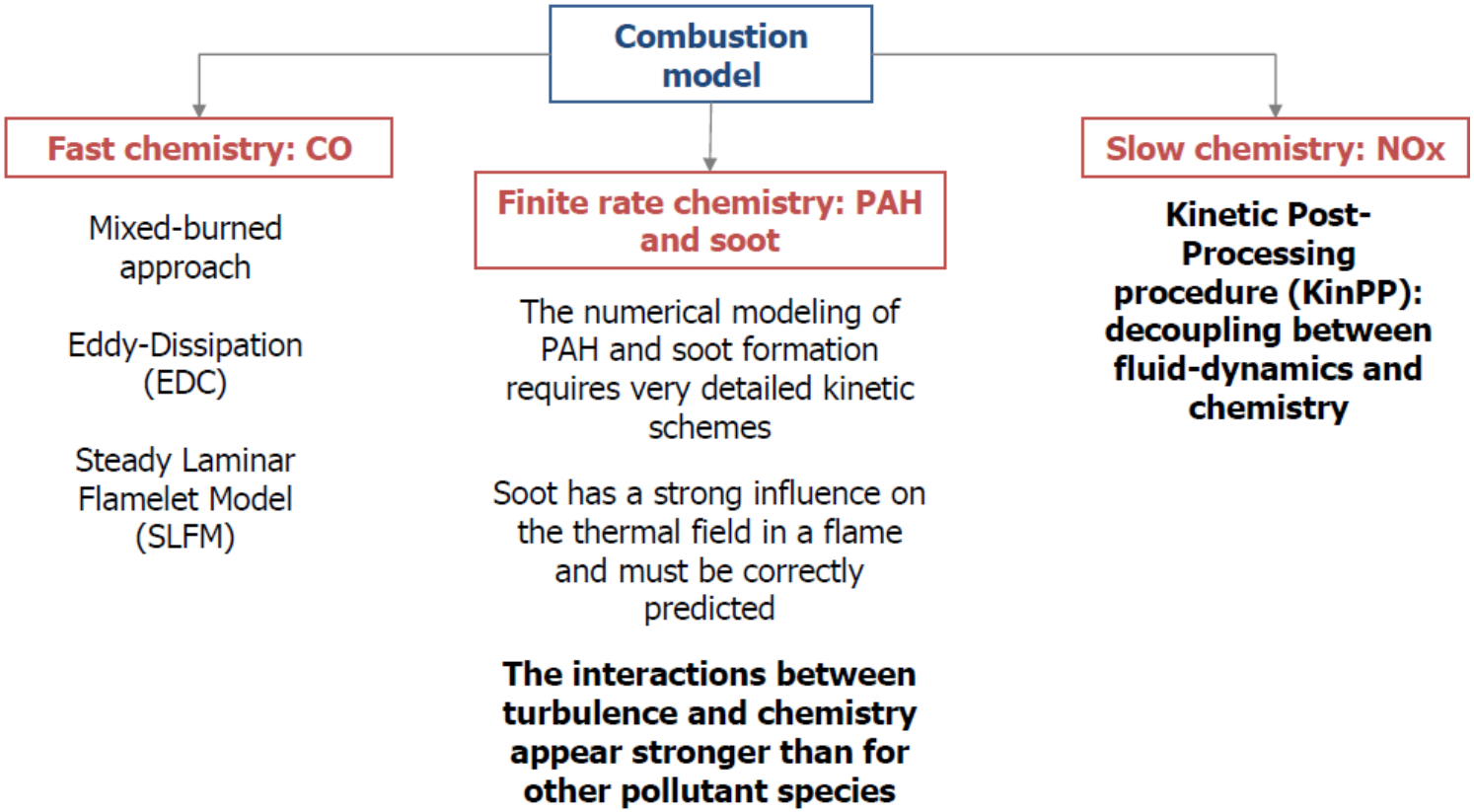




❖ For PAH and soot the decoupling between chemistry and fluid dynamics is not possible



Approaches to Modeling



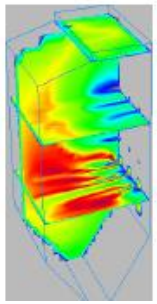
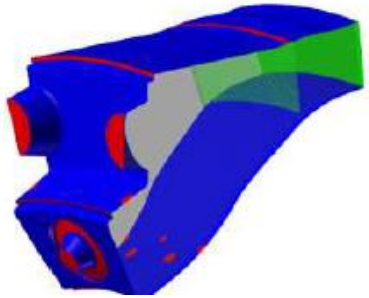
- **Accurate predictions of pollutant emissions from turbulent flames require simplified approaches, specifically conceived for each class of pollutant species, according to the characteristic times of its chemistry**

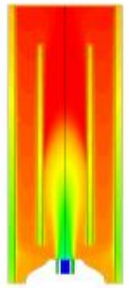
Approaches to Modeling

- Main Methods

- Different levels of detail are possible:
  - Empirical and semi-empirical models
  - Simplified physics-based models
  - Detailed models

- There are currently several techniques used in practice to predict the emissions from real combustors



Approaches to Modeling

Empirical models

Equations 1.3 and 1.4 are examples of typical empirical NO_x models.

$$\mathrm{EINO}_x = 0.0042 \left(\frac{P_3}{439}\right)^{.37} \exp\left(\frac{T_3 - 1471}{345}\right) T_4, \qquad (1.3)$$

where P_3 is the total pressure at the combustor inlet in psia, T_3 is the total temperature at the combustor inlet in ^oR, and T_4 is the total temperature at the combustor exit in ^oR [11].

[D.L. Allaire – Thesis, MIT, 2006]

- Empirical models are very simple and the least computationally intensive
- Require empirically determined constants and are useful for correlating known historical NOx emissions for a specific combustor
- are generated from fitting a certain number of constants using historical data, they cannot be expected to perform well when the combustor undergoes a design change

Approaches to Modeling

☐ Simplified Physics-based models

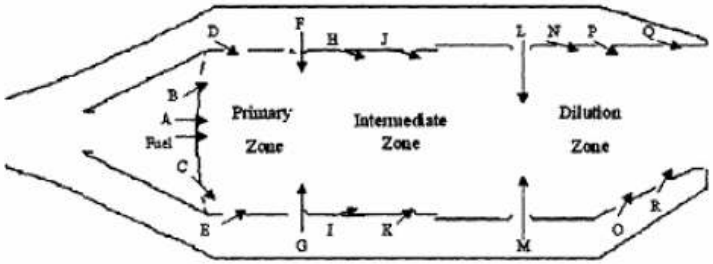


Figure 2-9: Typical Combustor Flow Splits

D.L. Allaire - MIT



A Physics-Based Emissions Model for Aircraft Gas Turbine Combustors

by

Douglas L. Allaire

Submitted to the Department of Aeronautics and Astronautics in partial fulfillment of the requirements for the degree of

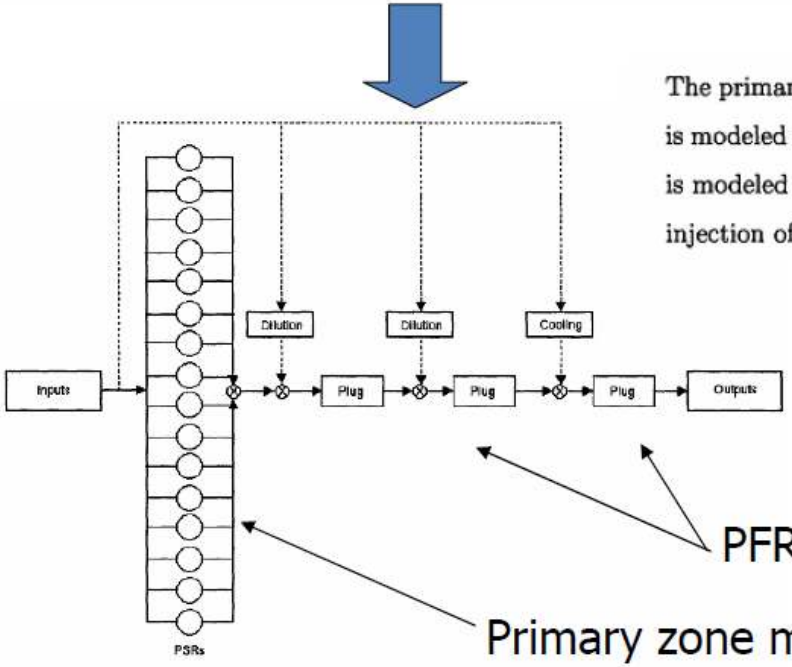
Master of Science in Aerospace Engineering

at the

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

May 2006

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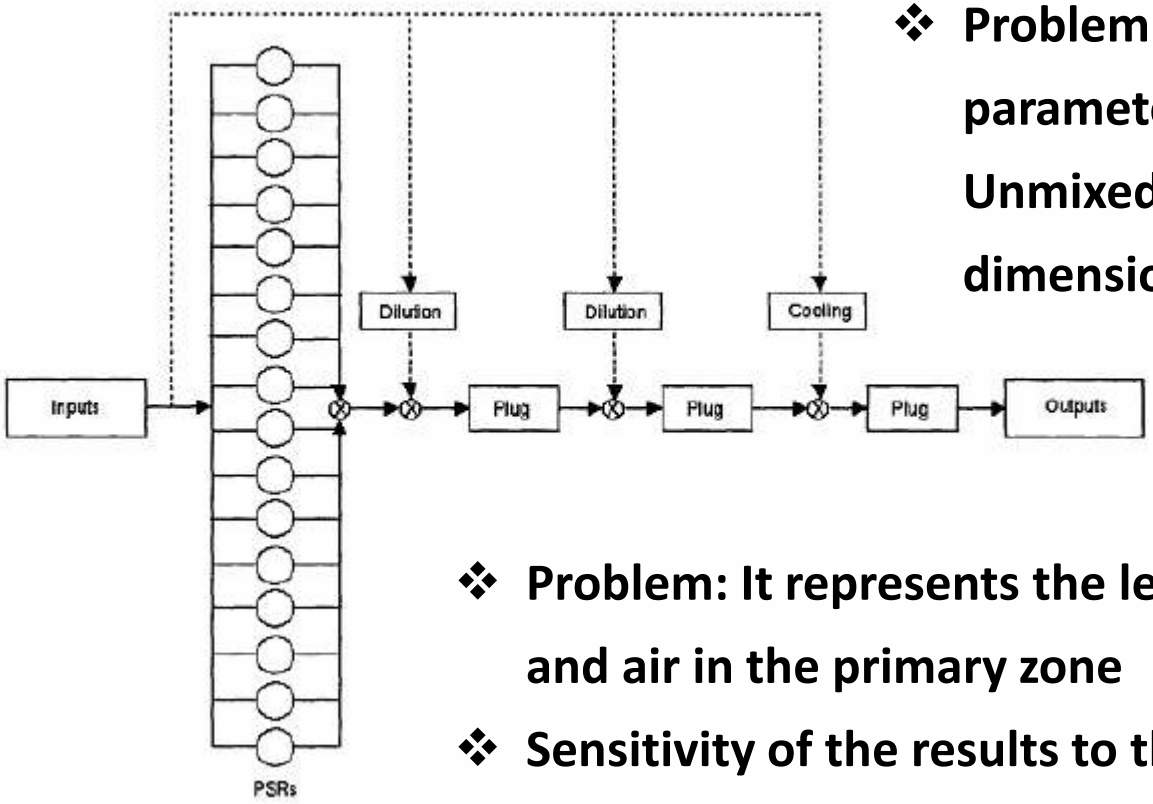
The primary zone is modeled as a network of 16 parallel PSRs, the intermediate zone is modeled as an injection of air followed by a plug flow reactor, and the dilution zone is modeled as an injection of air followed by a plug flow reactor, followed by another injection of air and another plug flow reactor.

PFR for other zones

Primary zone modeled using 16 parallel PSRs

Approaches to Modeling

- Simplified Physics-based models

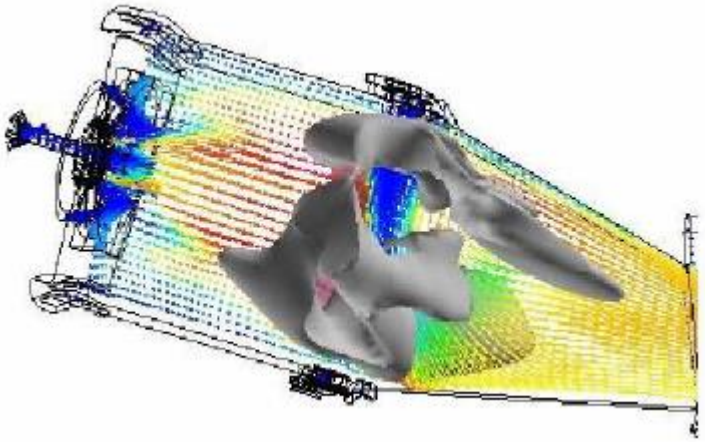


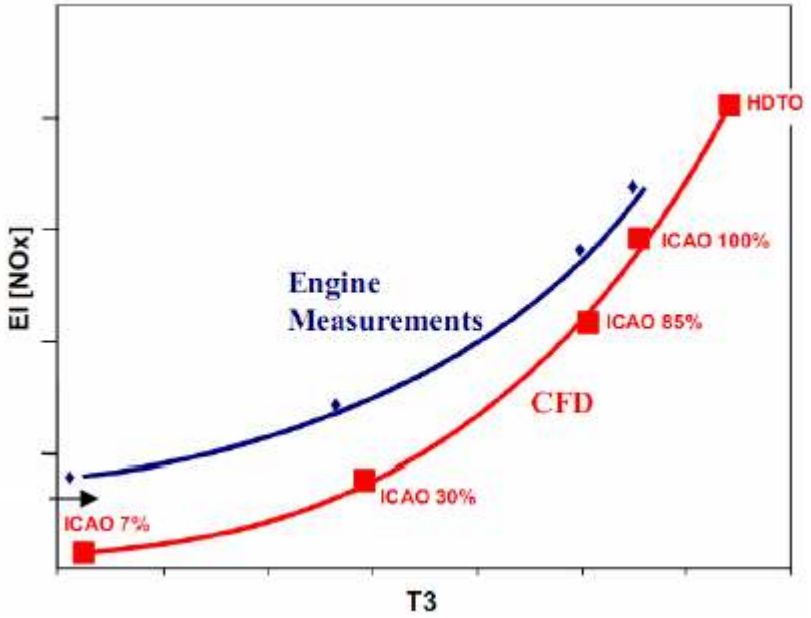
- Problem: it is necessary to define parameters: Primary zone Unmixedness, temperature and dimensions of the PFRs

- Problem: It represents the level of mixing between fuel and air in the primary zone
- Sensitivity of the results to this parameter -> tuned using experimental data obtained in some operating conditions

Approaches to Modeling

Detailed Models





- CFD model but simplified kinetics
- Only Thermal NOx formation is considered
- NOx underestimated due to the exclusion of the N_2O and prompt NOx mechanism

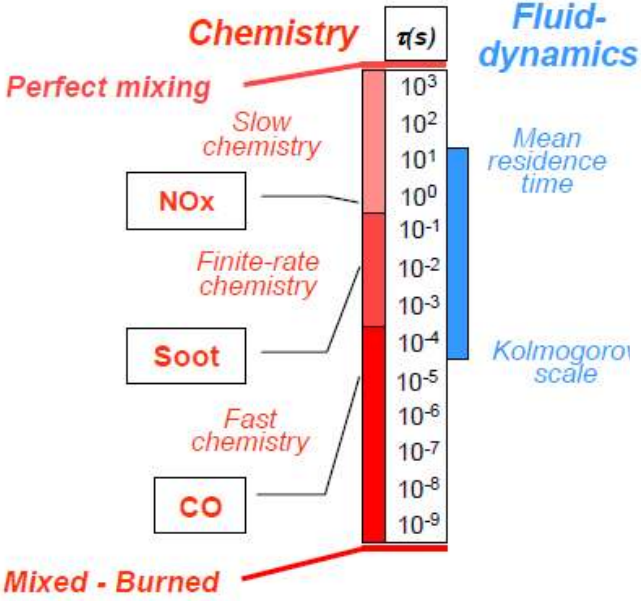
Approaches to Modeling

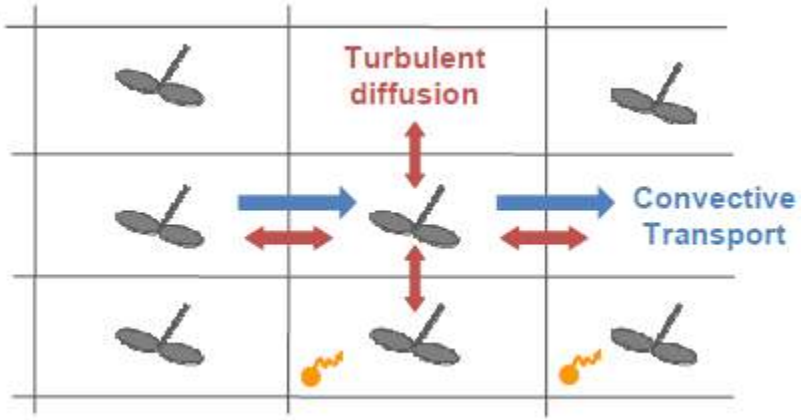
☐ Kinetic Post Processor Methods

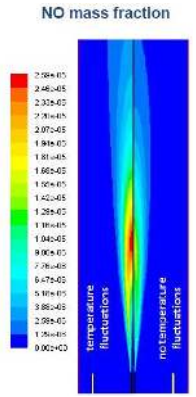
Pollutant species (such as NOx) affect only marginally the main combustion process and consequently do not influence the overall temperature and flow field

The prediction of pollutant species (**NOx**) can be **decoupled** from the CFD simulation

Kinetic Post Processor (KinPP)







Approaches to Modeling

- Kinetic Post Processor Methods
  - Advantages:
    - ❖ Assumes the temperature and flow fields as predicted by the CFD
    - ❖ Solves mass balances in the cells with detailed chemistry at fixed temperature. Each cell is a PSR reactor
    - ❖ Possibility to group several cells. Clustering equivalent cells reduces the dimensions of the problem
    - ❖ Fix the temperature inside the cells. This reduces the high non linearity of the system, mainly related to the reaction rates and to the coupling between mass and energy balances
  - The most popular methods!

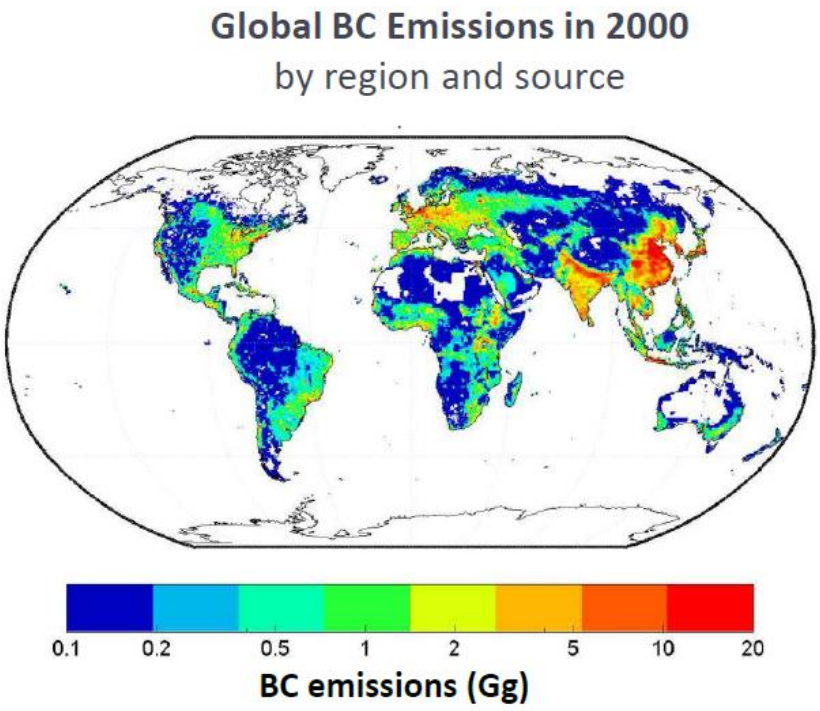
1. Introduction

2. NOx Formation

3. PAH and Soot Formation

Soot Emissions

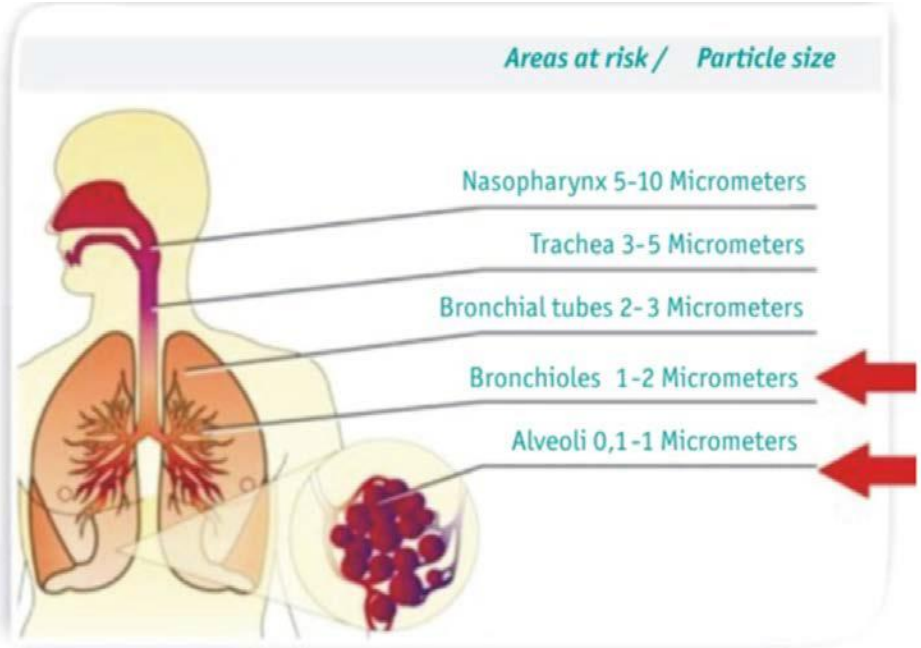
- Combustion is the main source of PAH and Soot or PM
- On-road and non-road diesel engines are leading emitters (about 70%) in Europe, North America, and Latin America; In China, the combustion of coal is the major source
- Soot forms at high temperatures and rich conditions

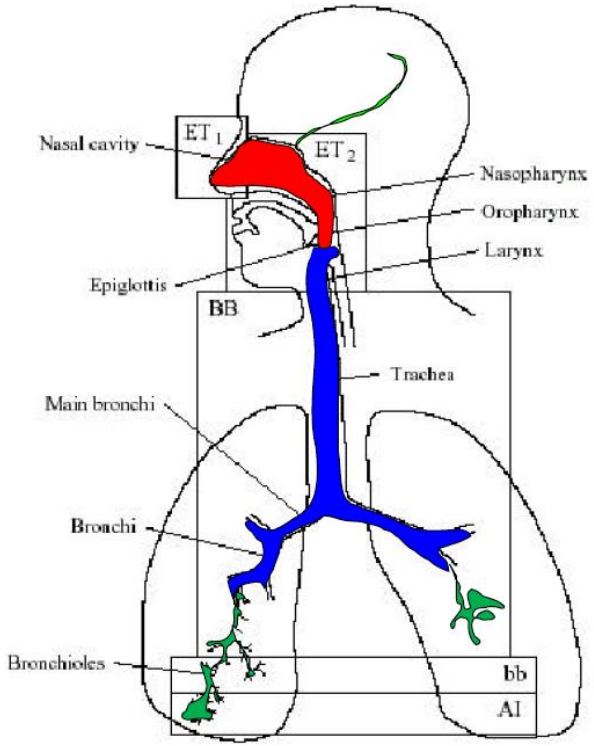






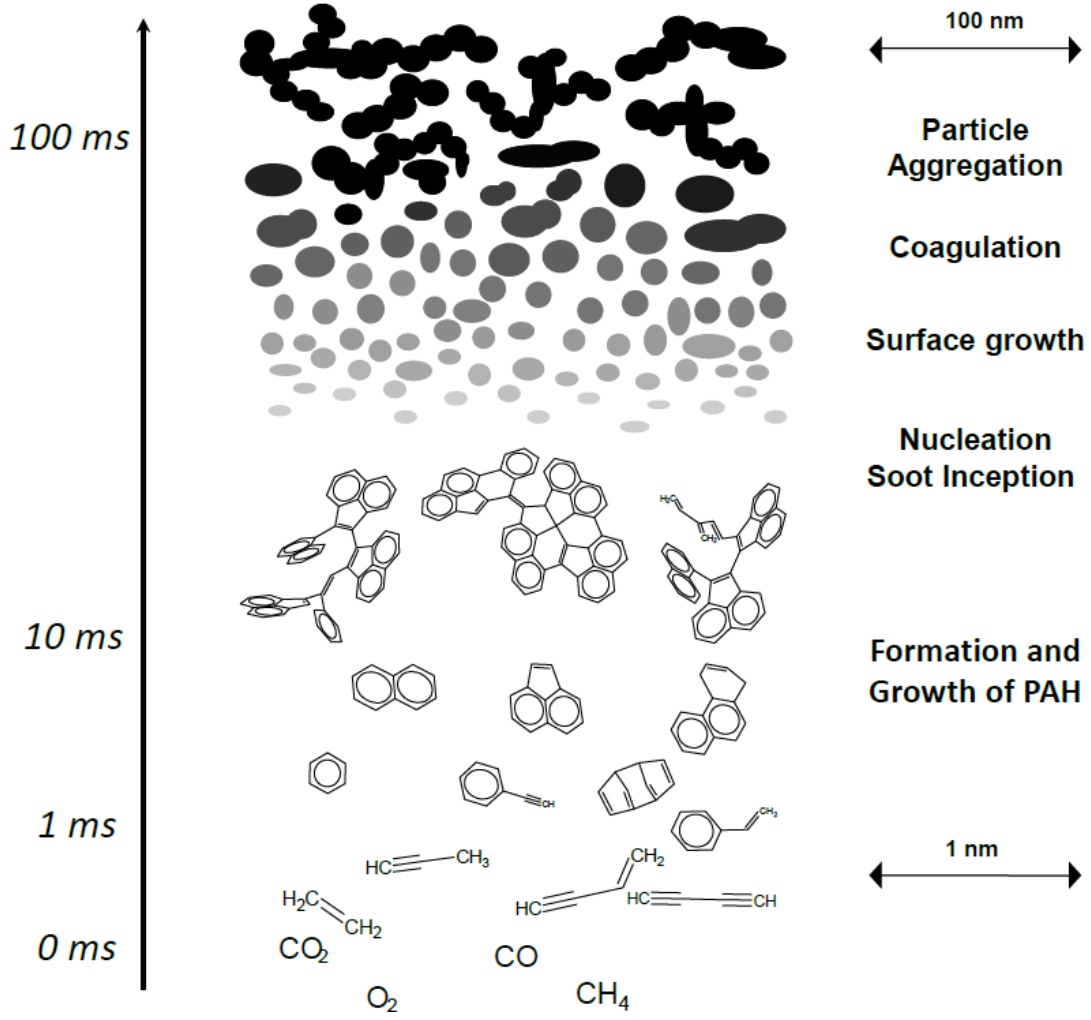
Soot Impact on Human Health

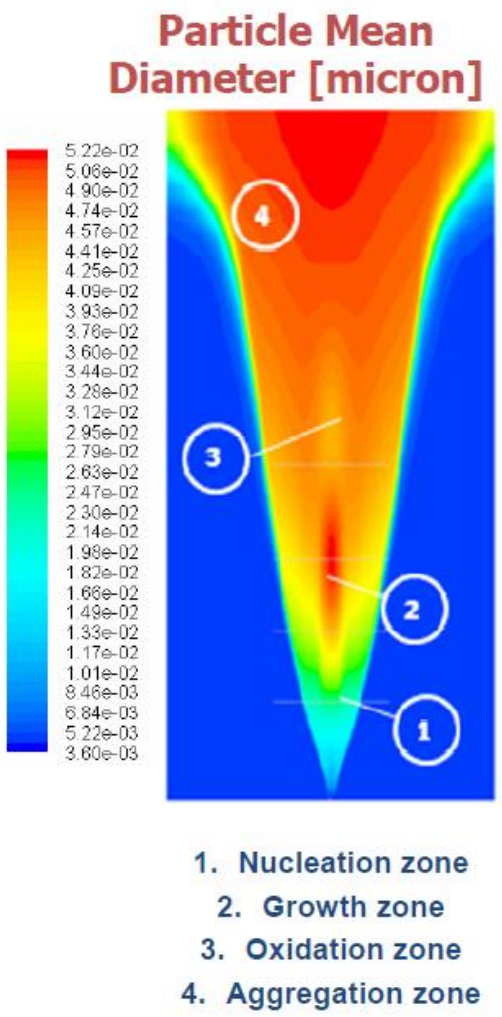




- Particle dimension is the critical issue: PM 10 (coarse particles): 10 μm; PM 1.0 (fine particles): 1 μm; PM 0.1 (ultrafine particles): 100 nm
- Ultrafine Particles are more dangerous than larger ones
- Ultrafine Particles are more numerous and with larger surface areas

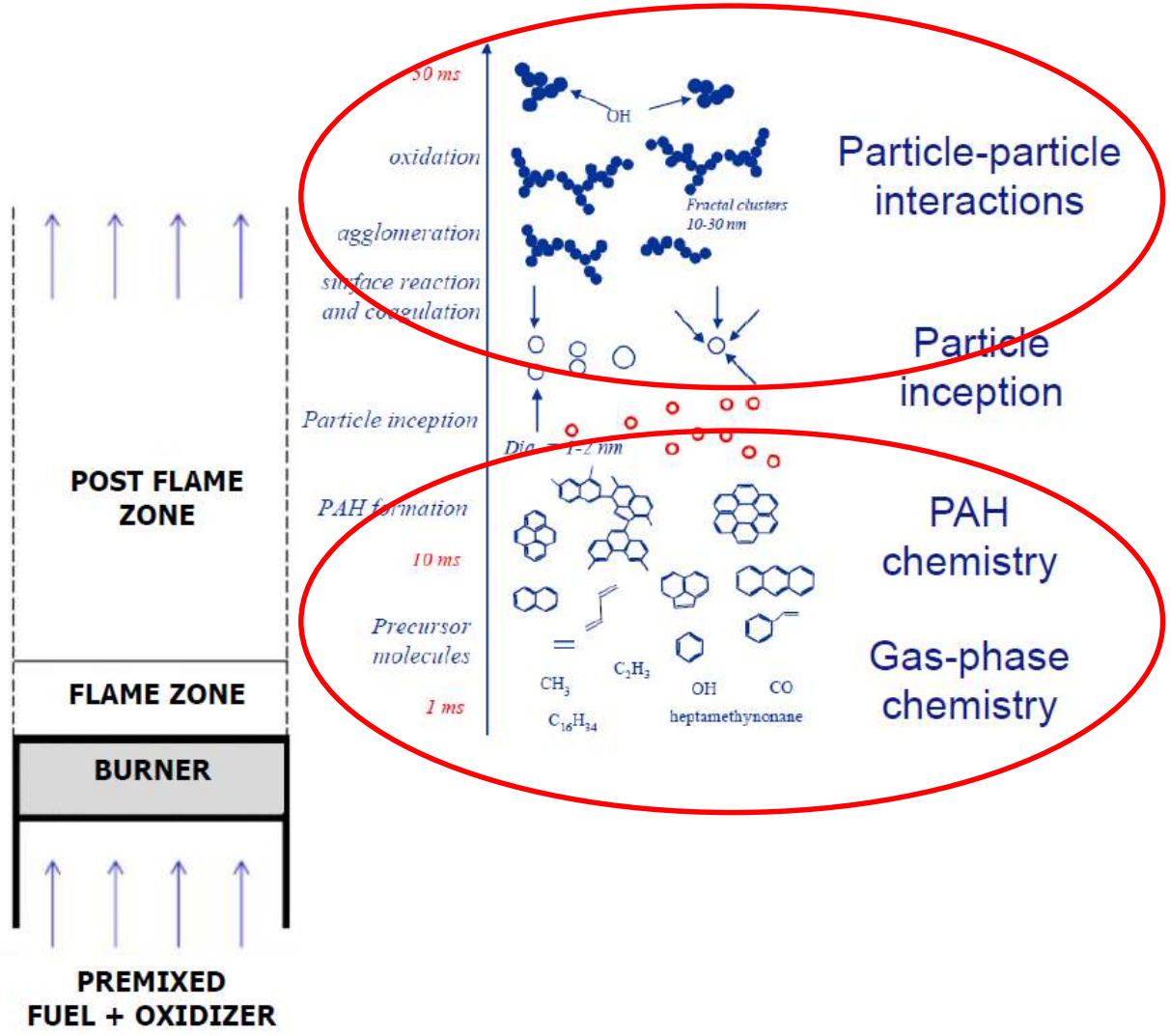
Soot Formation in Combustion

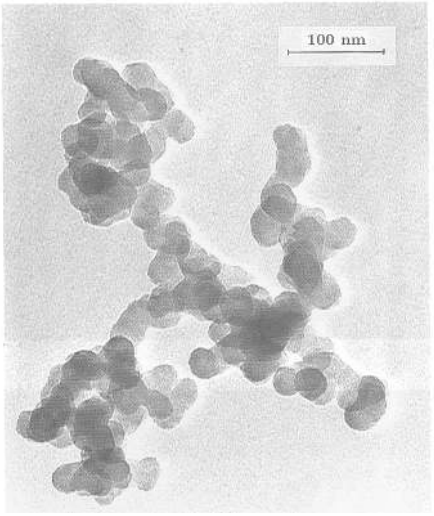




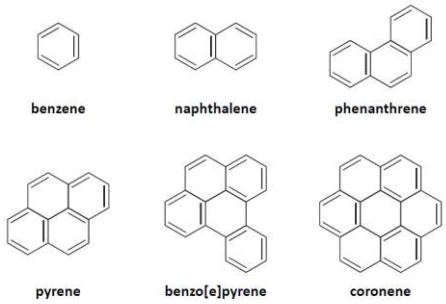
❖ **Steady growth in molecular size leads to visible soot**

Soot Formation in Combustion



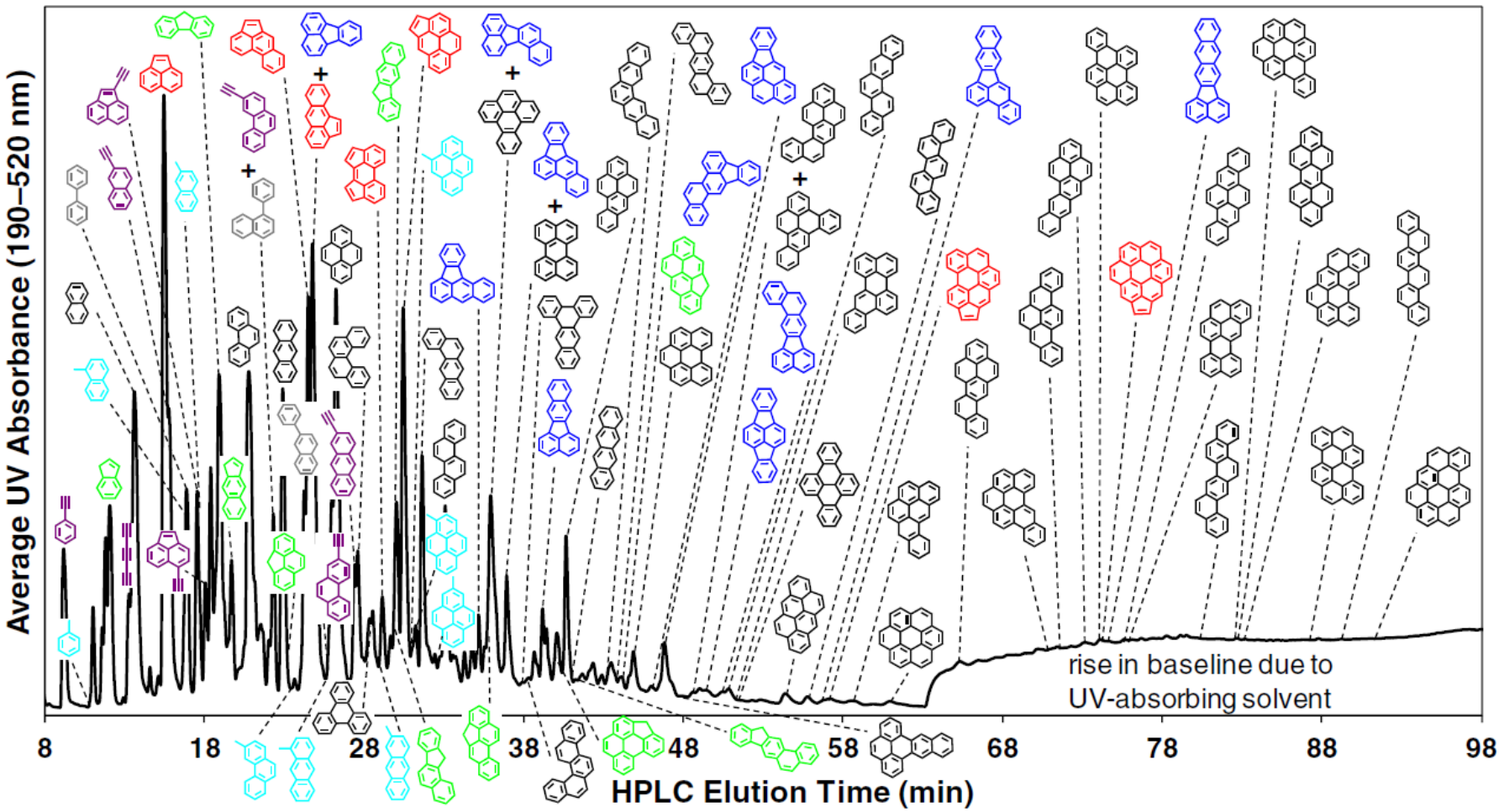


❖ **PAH**



❖ **Steady growth in molecular size leads to visible soot**

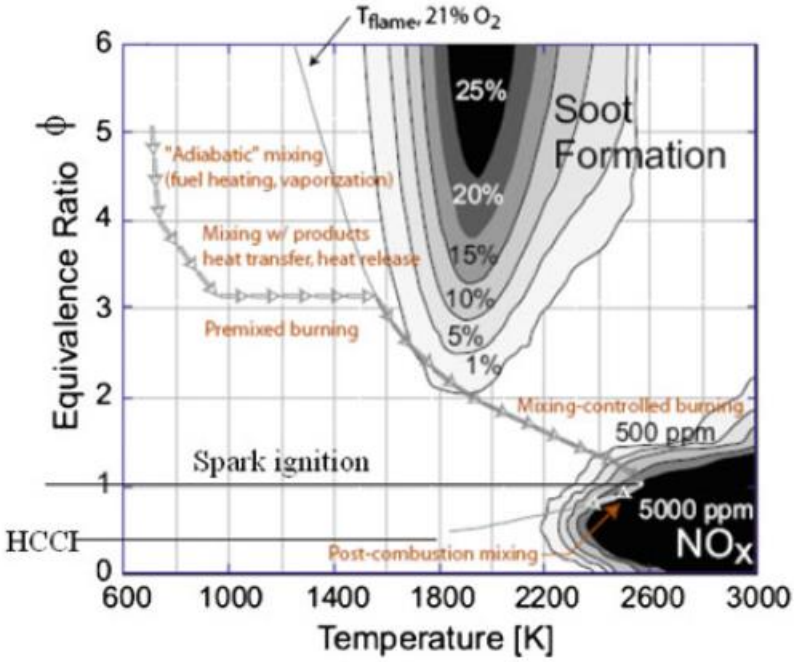
PAH



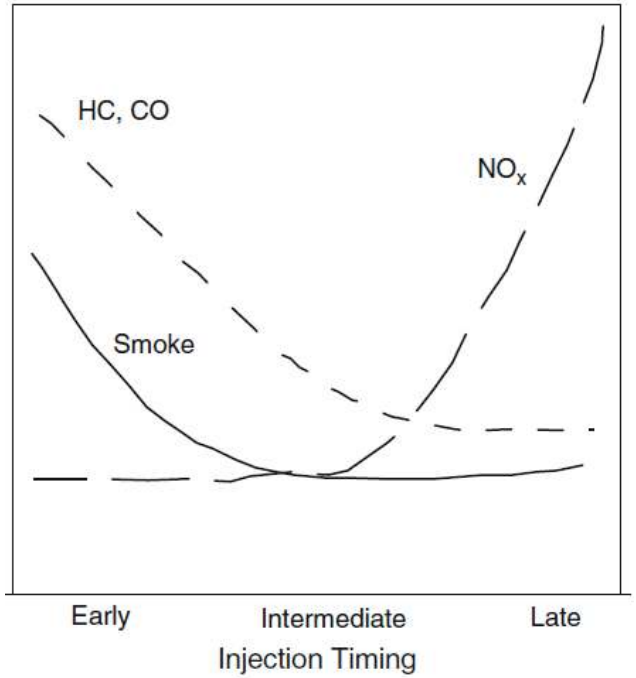
- HPLC chromatogram of the PAH products of 1,3-butadiene/propyne co-pyrolysis

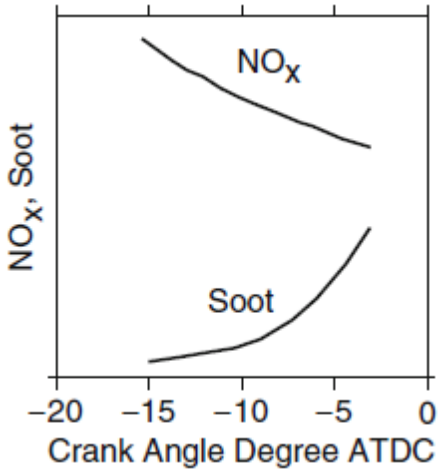
Tradeoff Relation Between Soot and NOx

- Soot and NOx relation, ϕ,T map



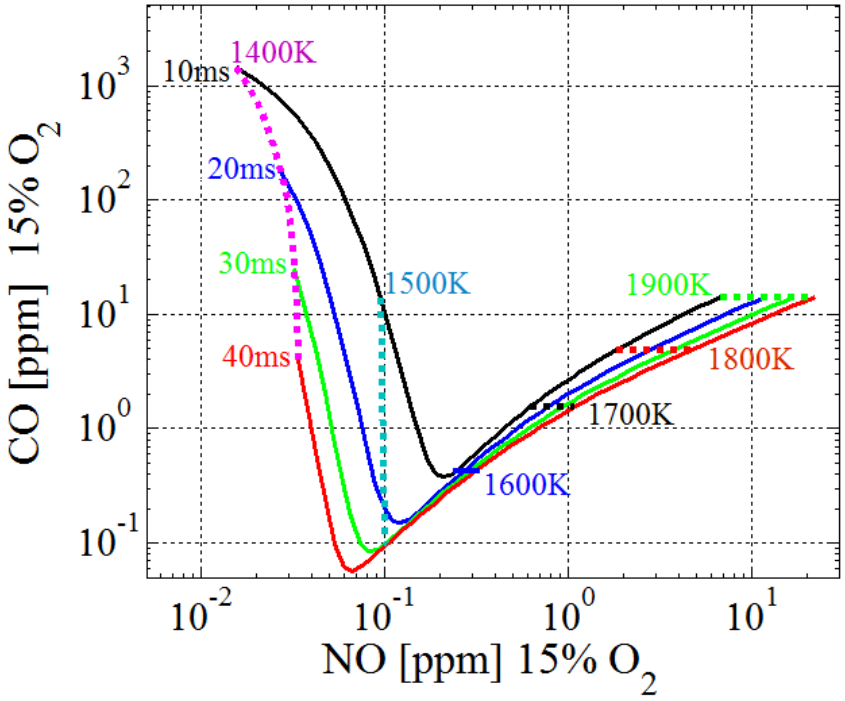
- Tradeoff between NOx and soot as function of injection timing

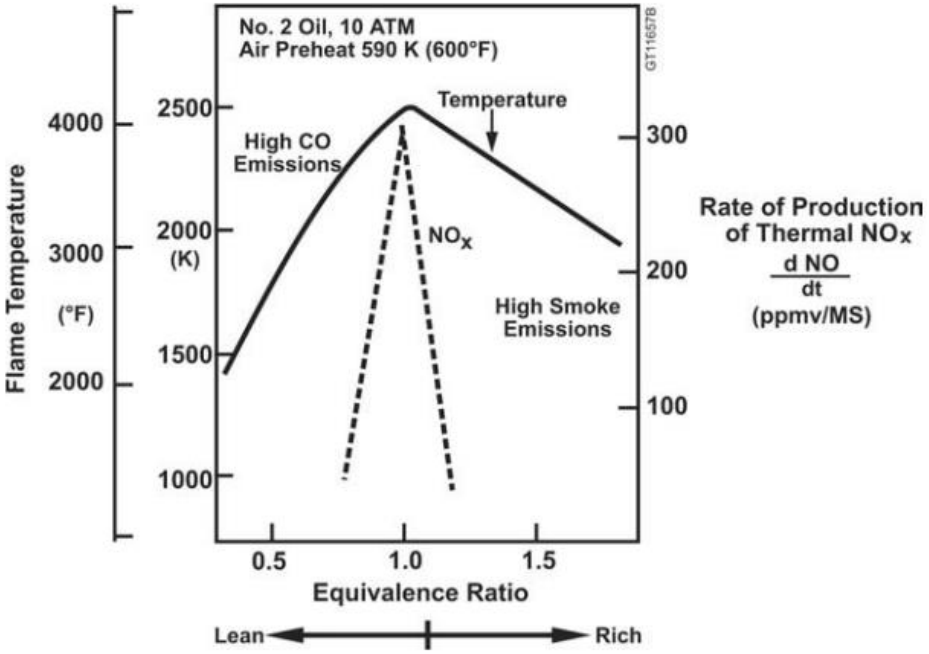




- In SI engine, NOx and soot are formed in different regions in the (ϕ, T) map; NOx and soot often exhibit a trade-off relation
- Soot oxidation is heavily dependent on radicals, such as OH, and the temperature must be high enough. In addition, soot oxidation requires sufficient residence time. NOx is formed primarily at the diffusion flame front

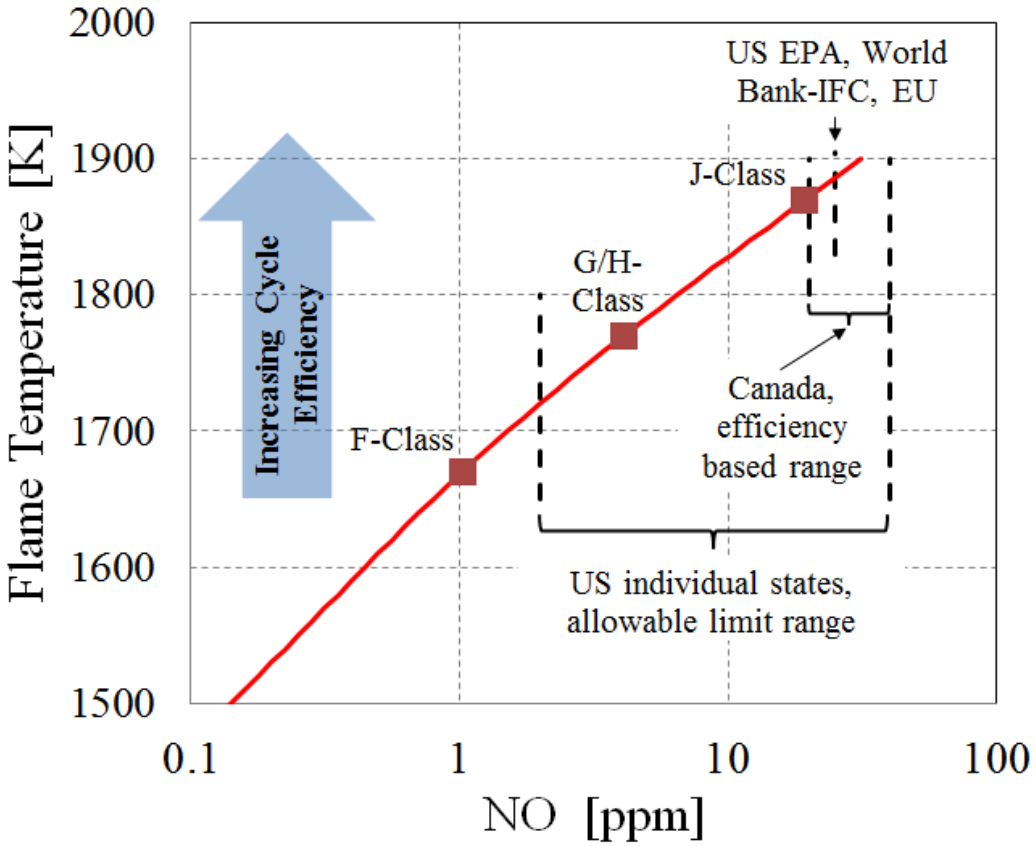
Tradeoff Relation Between Soot and NOx





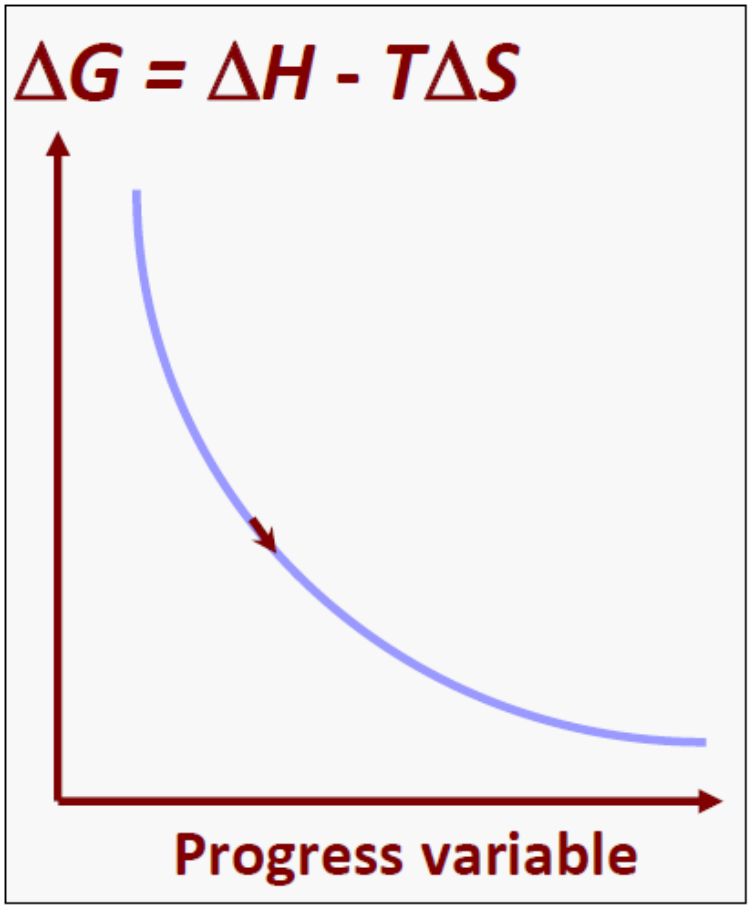
- A reduction in temperature tends to increase CO emissions due to incomplete combustion; Conversely, an increase in temperature will increase thermal NOx formation
- For low temperature combustion technology, always the NOx emission is decreased while CO emission is increased

Tradeoff Relation Between Soot and NOx



- Higher engine pressure ratio and bypass ratio reduce CO_2, improve fuel efficiency and facilitate noise reduction
- Higher pressure ratio requires higher flame temperature, increasing NOx formation rate

Driving Force for Soot Formation



Gas-to-Solid Transformation

- Type 1: enthalpy driven (heat release)

  metal oxides

  carbides, nitrides

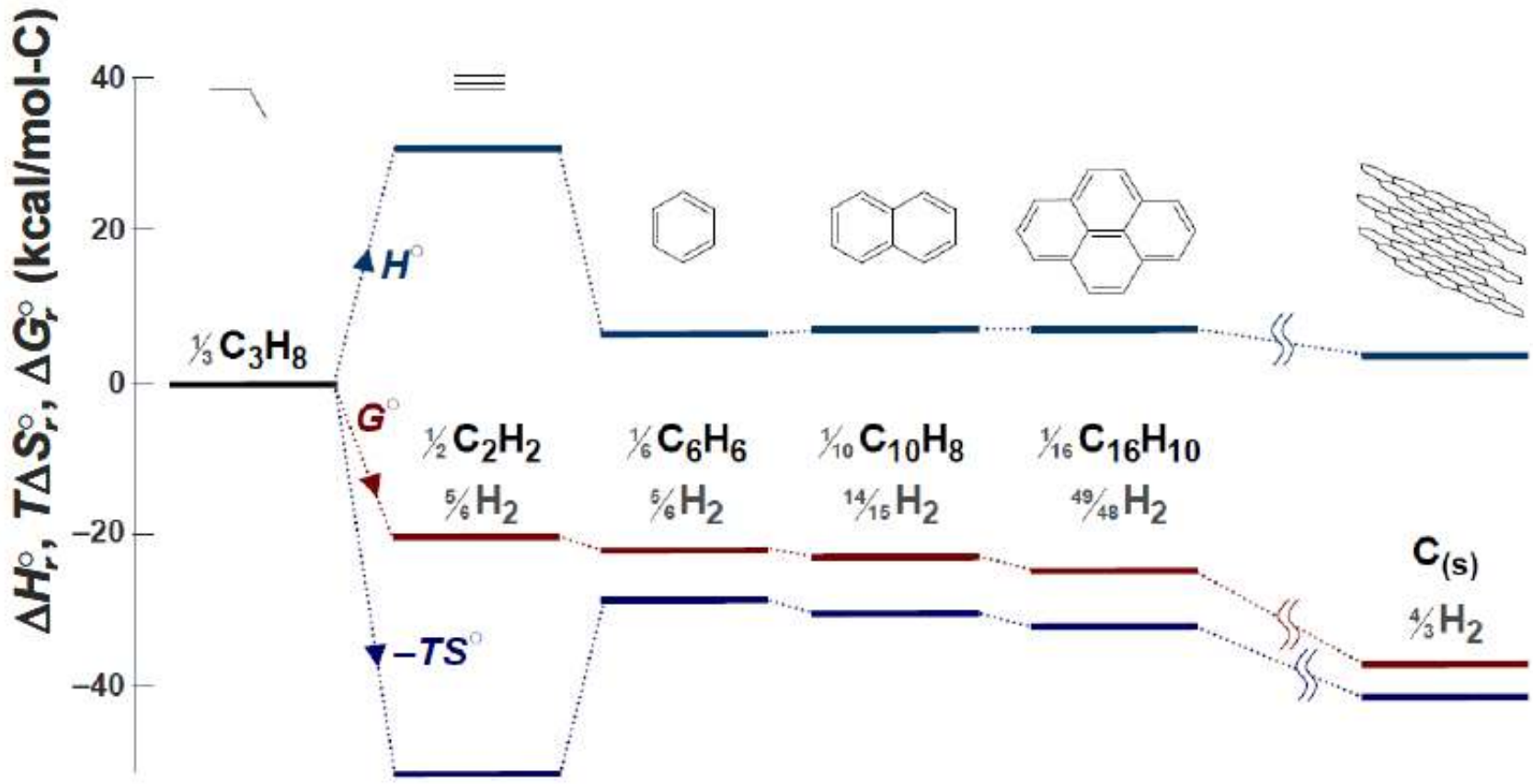
- Type 2: entropy driven

  soot

  $C_3H_8 \rightarrow$ solid carbon + $4H_2$

  ($\Delta H > 0$, but $\Delta S > 0$)

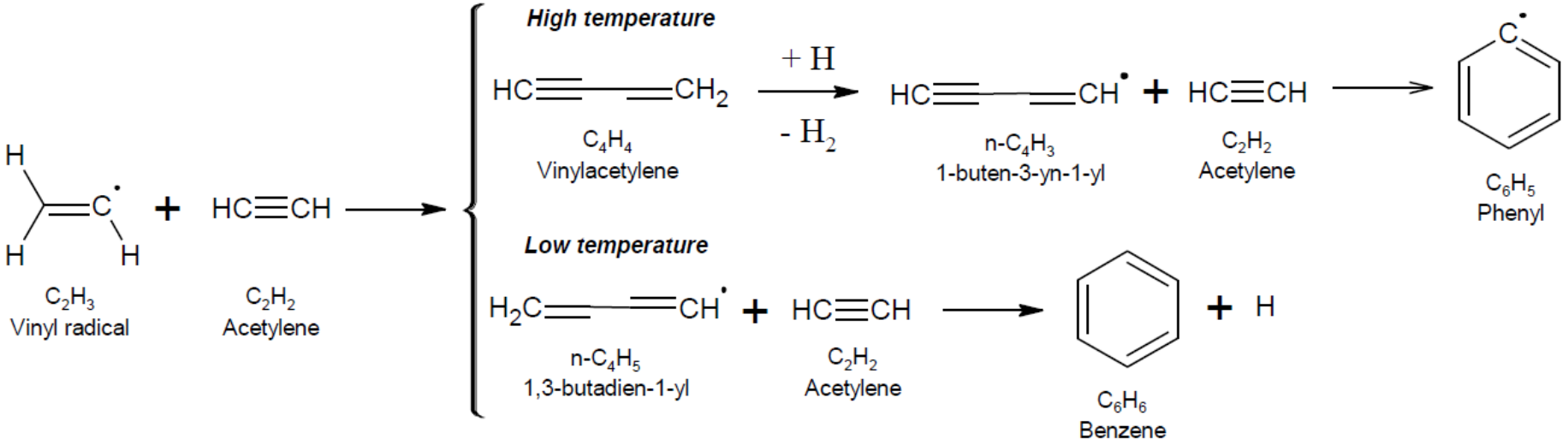
Driving Force for Soot Formation



- Soot formation is <u>entropy driven</u> (H_2 goes free)
- Condensed-phase carbon forms as an aerosol (kinetics driven)

First Aromatic Ring Formation

- Major Benzene Formation Pathways
  - Class 1: Acetylene addition
    - $C_2H_2 + n\text{-}C_4H_5 = C_6H_6 + H$
    - $C_2H_2 + n\text{-}C_4H_3 = C_6H_5$



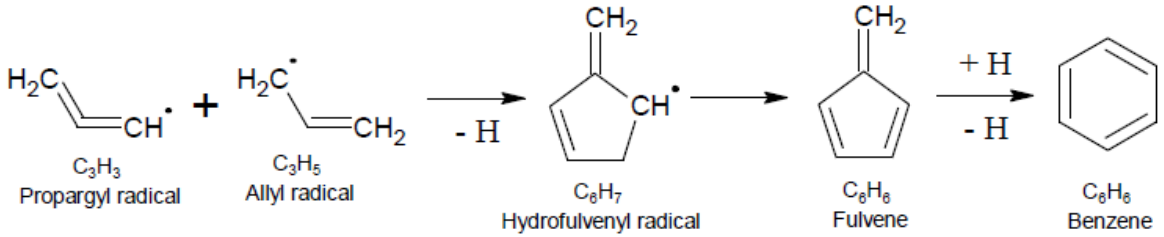
- This sub-mechanism is considered favorable in forming the first aromatic ring since it includes acetylene, which is always present in fuel-rich oxidation of hydrocarbons

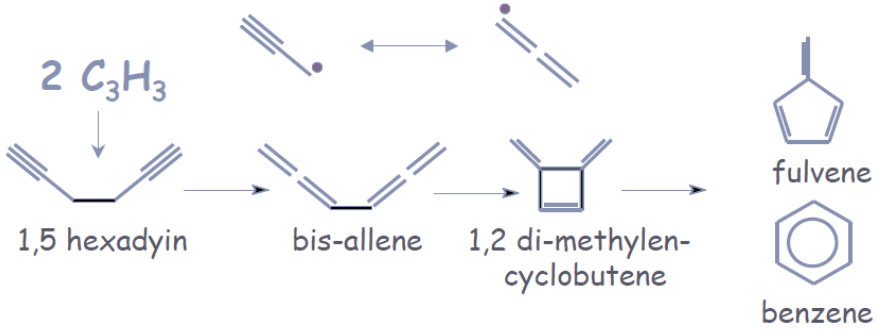
First Aromatic Ring Formation

Major Benzene Formation Pathways

Class 2: C_3 combination $C_3 + C_3$

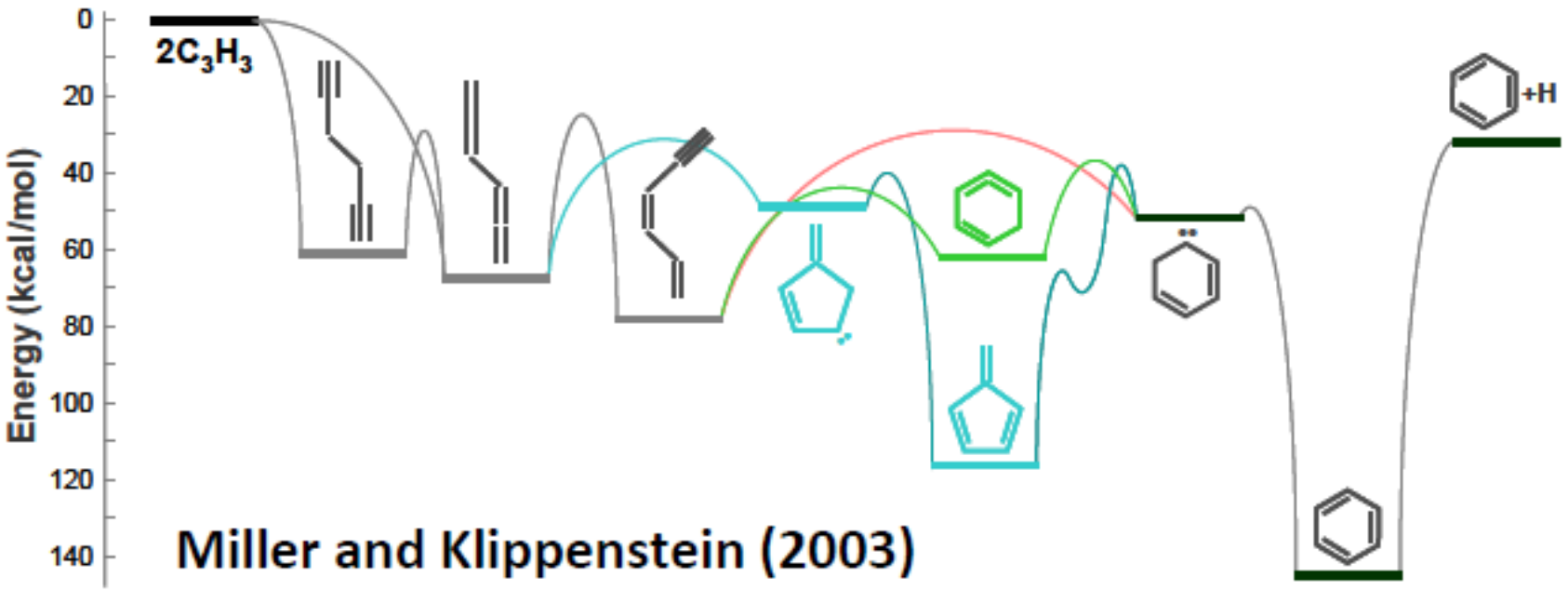
- $C_3H_3 + C_3H_3 = C_6H_6$
- $C_3H_3 + aC_3H_4 = C_6H_6 + H$
- $C_3H_3 + pC_3H_4 = C_6H_5 + H$
- $C_3H_3 + aC_3H_5 = \text{fulvene} + 2H$





First Aromatic Ring Formation

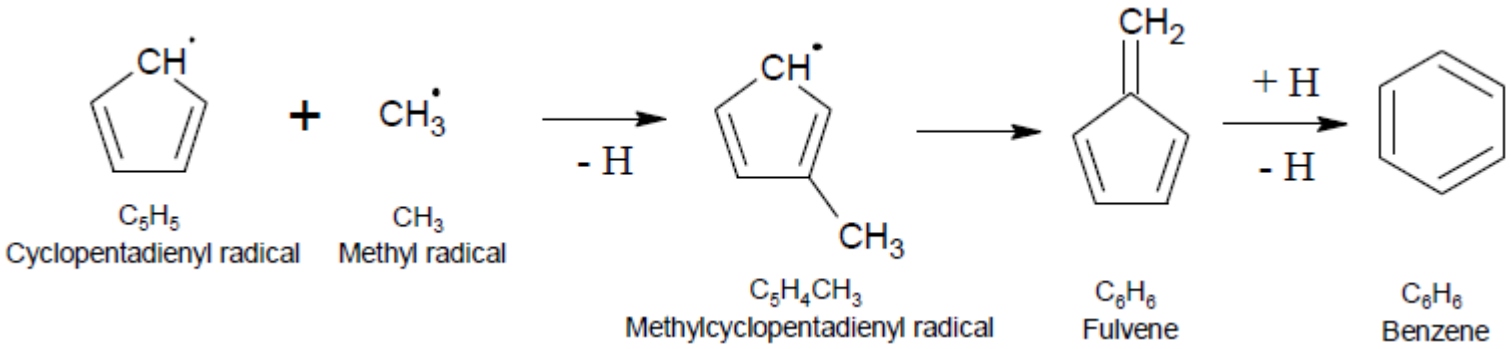
- Major Benzene Formation Pathways



- Resonantly-stabilized free radicals are present in combustion processes at high concentrations and play a crucial role in the chemical reactions of soot formation

First Aromatic Ring Formation

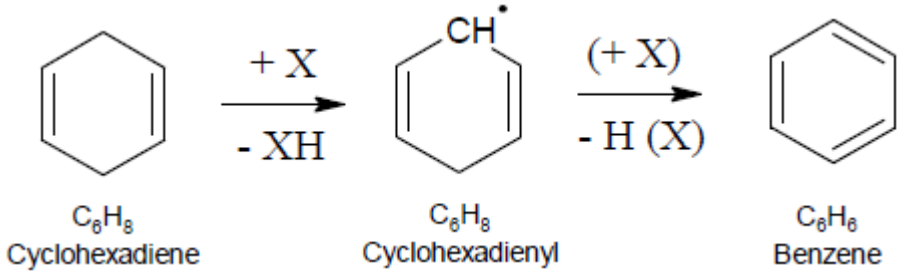
- Major Benzene Formation Pathways
  - Class 3: Combination of CH_3 and C_5H_5
    - $C_5H_5 + CH_3 = c\text{-}C_6H_8 = C_6H_6 + 2H$



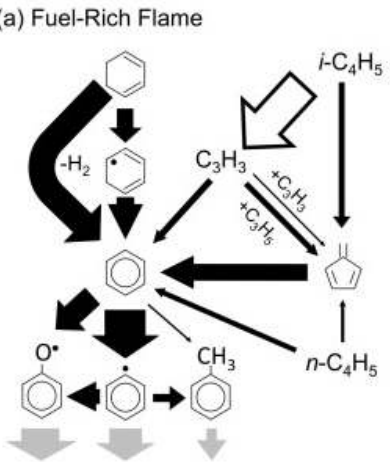
- The initial formation of methylcyclopentanienyl radical followed by its isomerization and dissociation into fulvene radical and H-atom
- Catalyzed isomerization of fulvene into benzene and atomic hydrogen

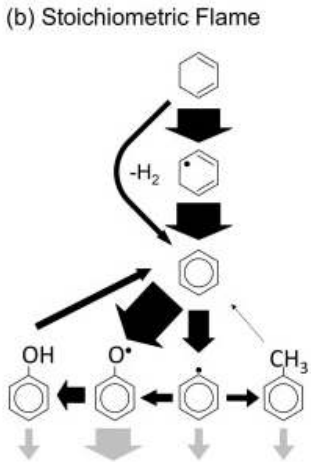
First Aromatic Ring Formation

- **Major Benzene Formation Pathways**
  - **Class 4: Cascading dehydrogenation**
    - $cycloC_6\text{-}R \rightarrow C\text{-}C_6H_{10}(-R) \rightarrow C\text{-}C_6H_8(-R) \rightarrow C_6H_6(-R)$



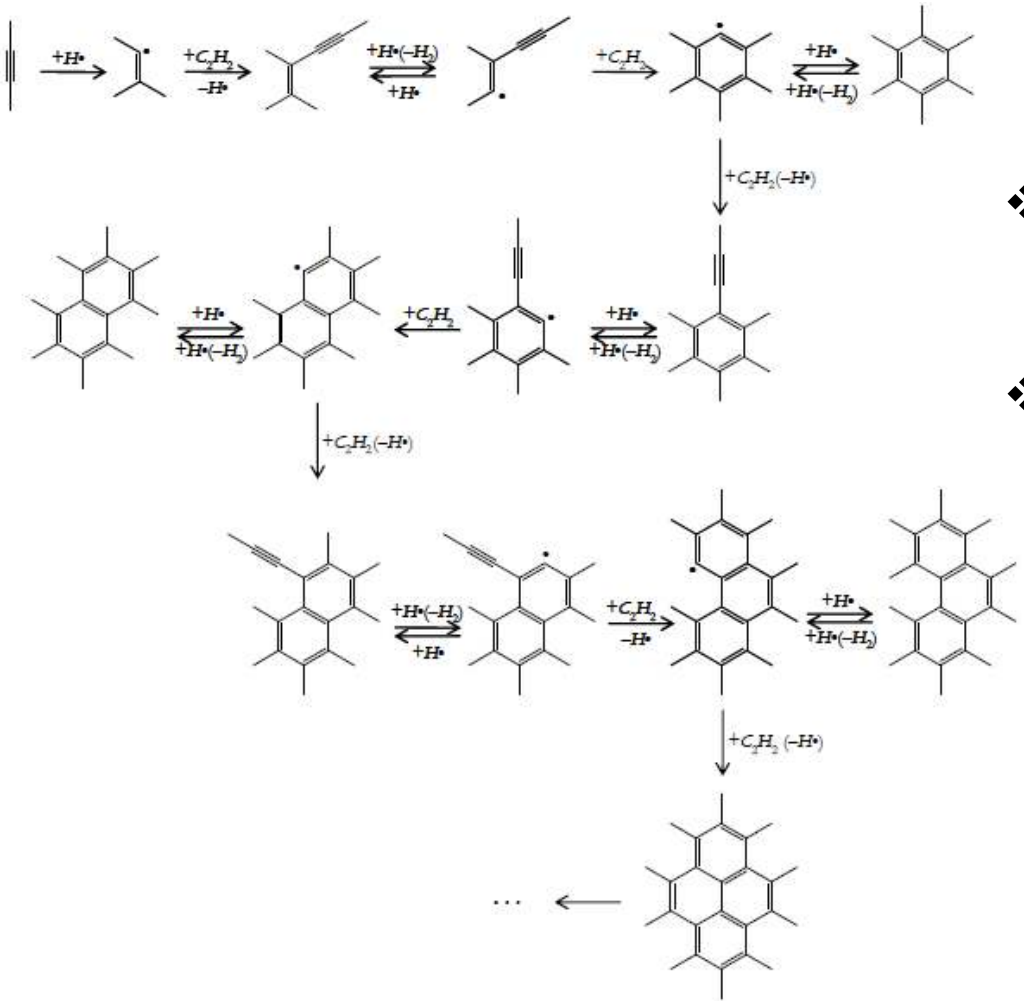
  - **Class 5: De-alkylation**
    - $C_6H_5\text{-}R + H = C_6H_6 + R$





Aromatic Growth

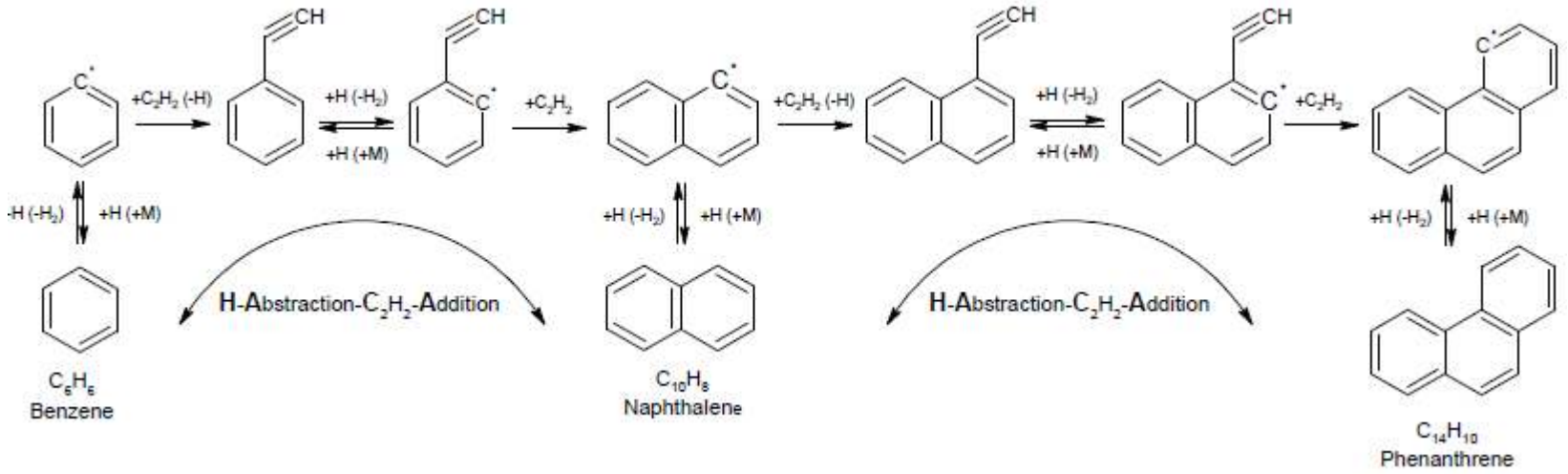
☐ The Hydrogen-Abstraction—Carbon Addition (HACA) Mechanism



- ❖ Addition of acetylene to the formed radical site
- ❖ This pathway includes two repetitive elementary steps: H-atom abstraction from aromatic species producing an aryl radical and additional aromatics.

Aromatic Growth

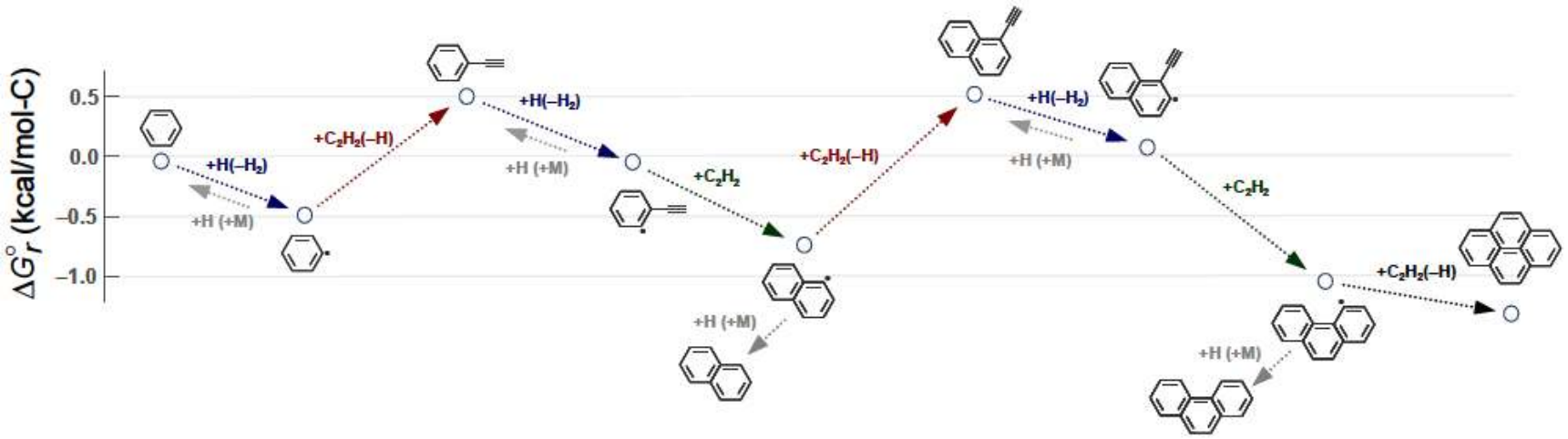
The Hydrogen-Abstraction—Carbon Addition (HACA) Mechanism



- The abundance of acetylene produced during rich-fuel combustion of hydrocarbons
- The presence of H-atoms acting as "catalytic" species during chain branching and propagation reactions
- The high temperatures that facilitate the chemistry but limits mass growth due to fragmentation make PAHs attractive soot precursors species

Aromatic Growth

The Hydrogen-Abstraction—Carbon Addition (HACA) Mechanism

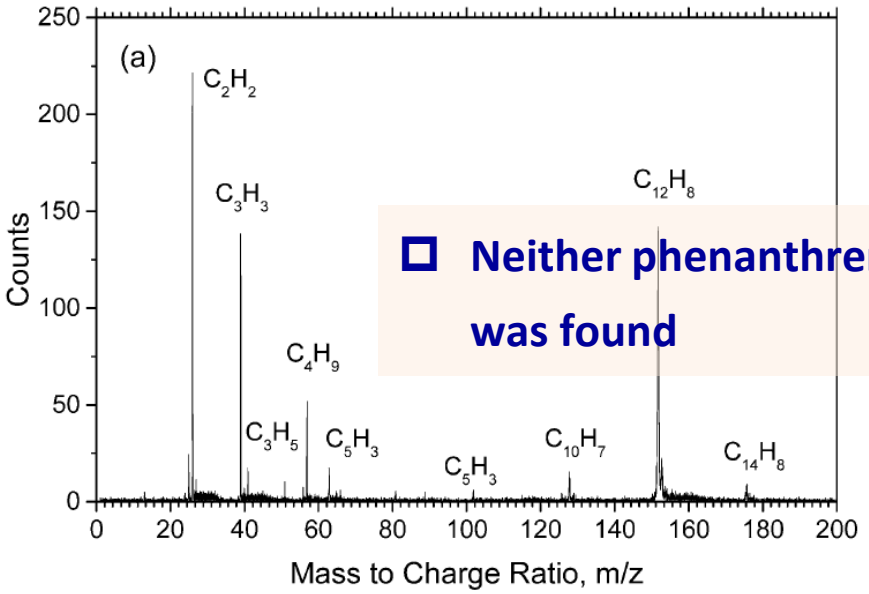


- The aromatic growth process and subsequent particle inception are characterized by high reversibility due to a modest gradual reduction in global Gibbs fee energy resulting from a mild exothermicity and relatively small entropy release
- The high degree of reversibility in the HACA mechanism does not reproduce the typical timings for soot formation process, its reversibility opens it to competitions from other pathways

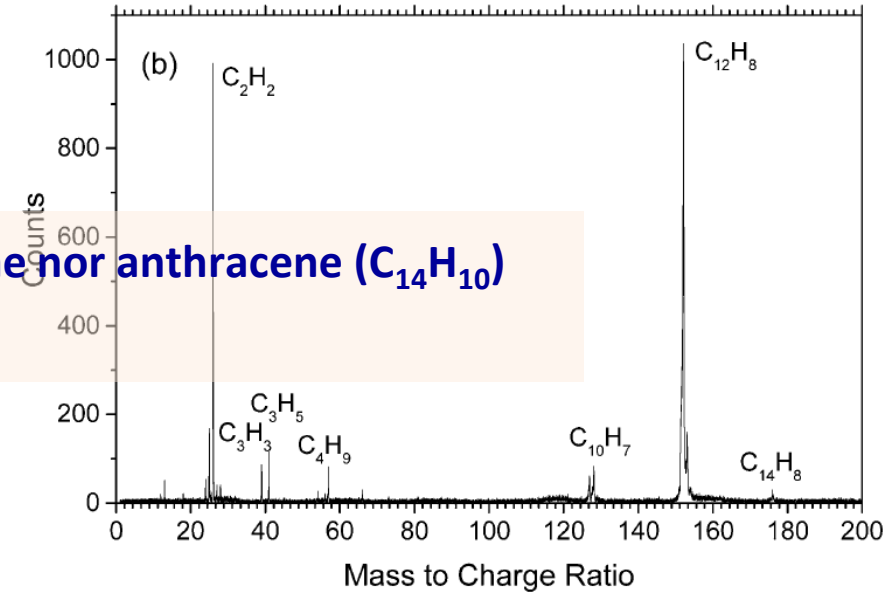
Aromatic Growth

The Hydrogen-Abstraction—Carbon Addition (HACA) Mechanism

1-naphthyl radicals with C2H2



2-naphthyl radicals with C2H2

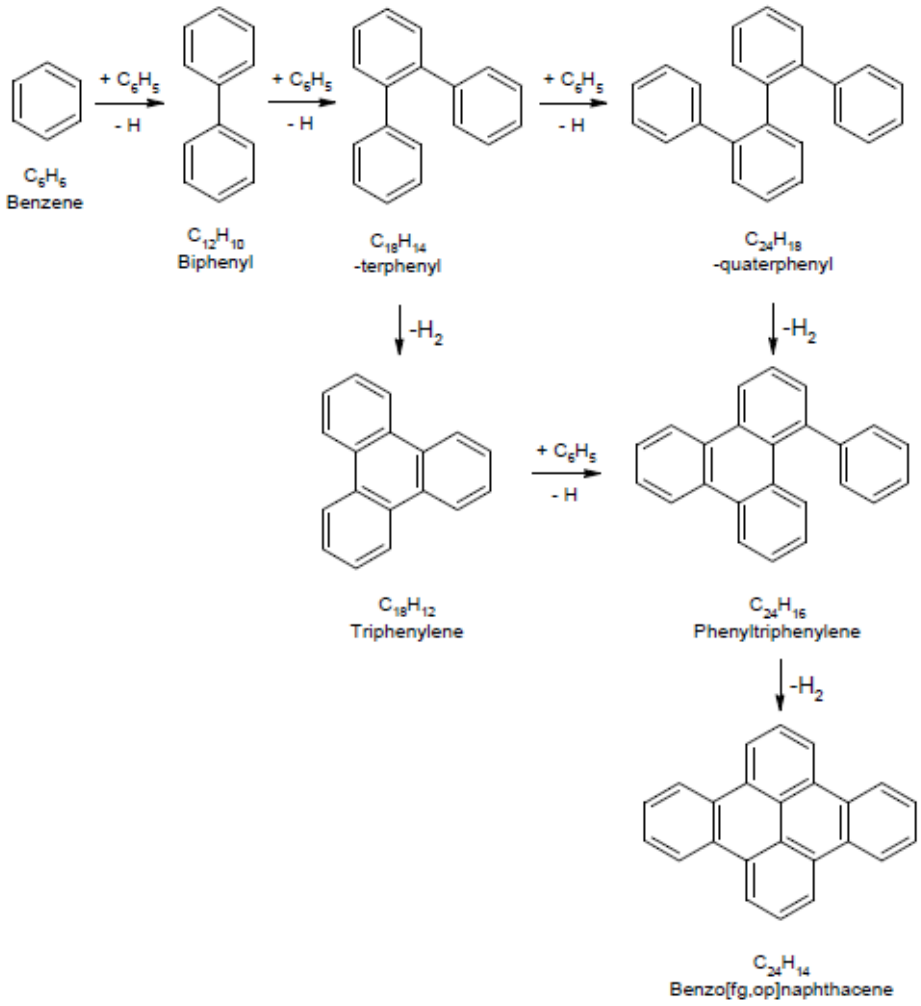


- Neither phenanthrene nor anthracene ($C_{14}H_{10}$) was found

- The HACA alone appeared to be too slow to account for the formation efficiency of heavy PAH and soot nuclei
- HACA mechanism is not the most significant mechanism in mass growth to high-mass PAHs opening the door to other competing mechanisms

Aromatic Growth

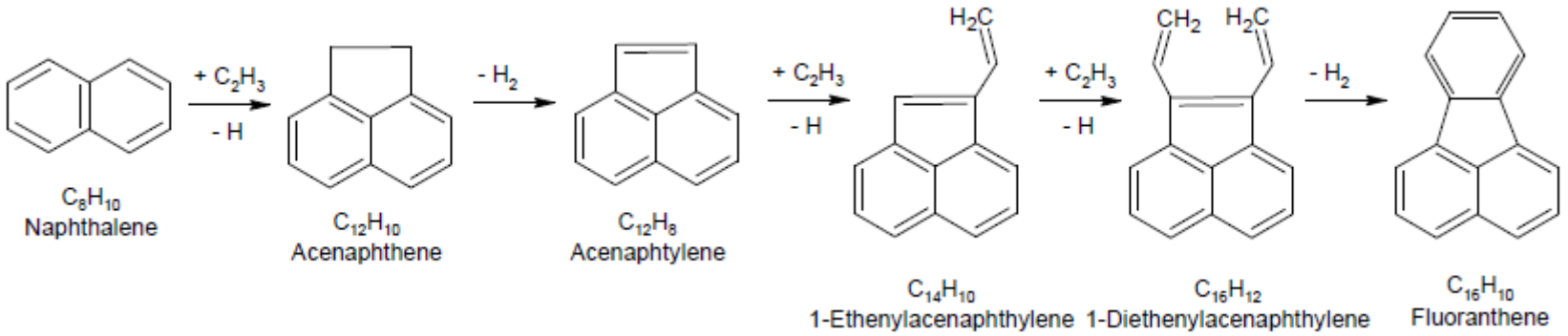
The phenyl-addition/cyclization (PAC) mechanism



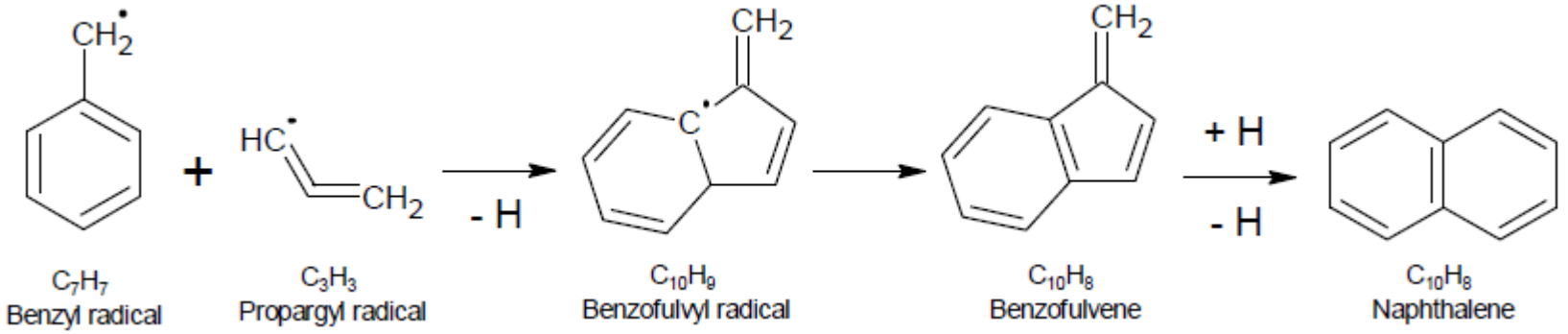
- Phenyl radical addition with an H-atom elimination to produce a phenyl-PAH, which is followed by transformation into a stable PAH by dehydrocyclization
- The kinetics of PAC pathways appear to be highly efficient for the growth of large PAHs while the HACA mechanism is better suited to form symmetrical PAHs

Aromatic Growth

The Hydrogen-Abstraction—Vinyl Addition (HAVA) mechanism

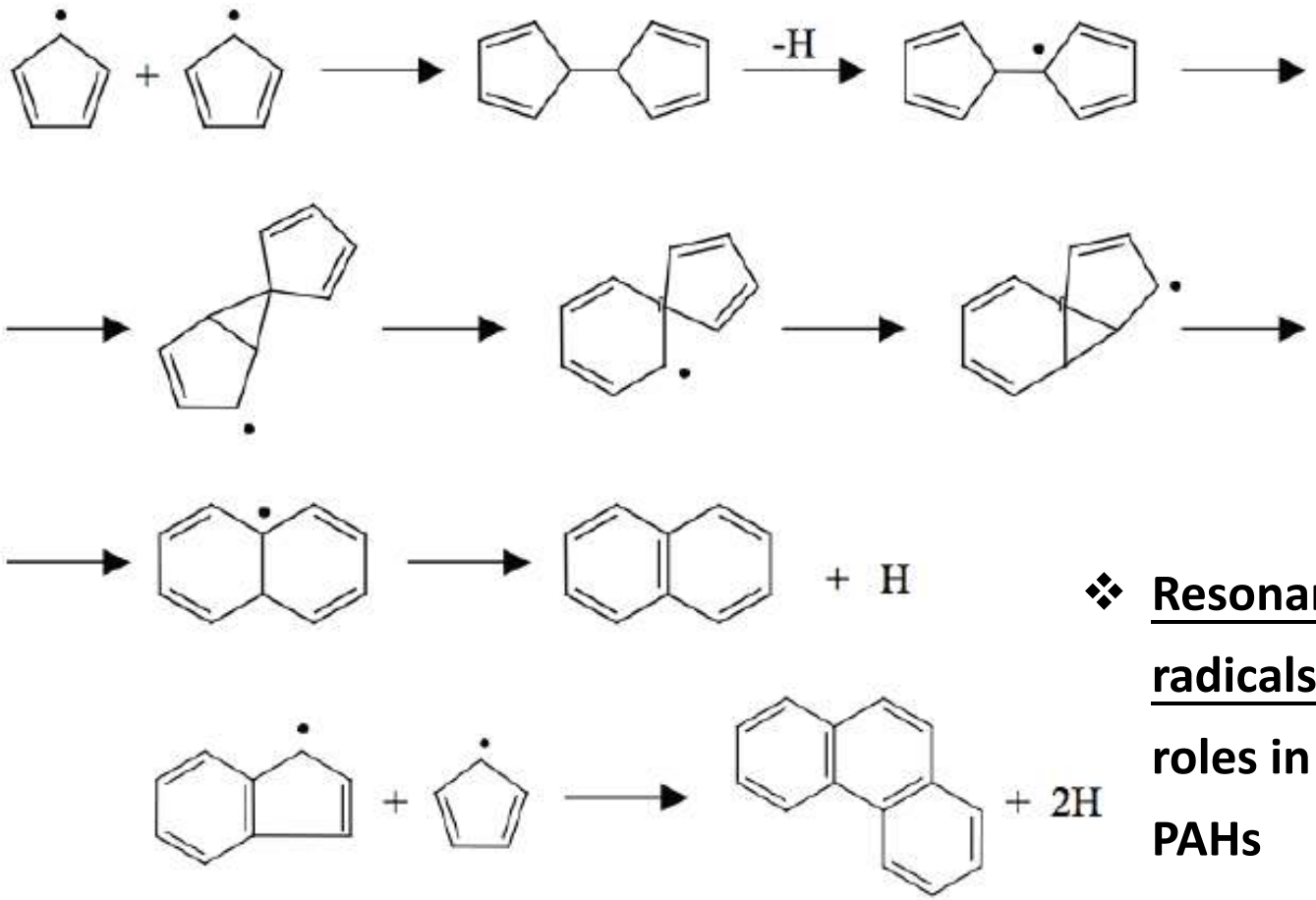


The addition of propargyl radical to benzyl radical



Aromatic Growth

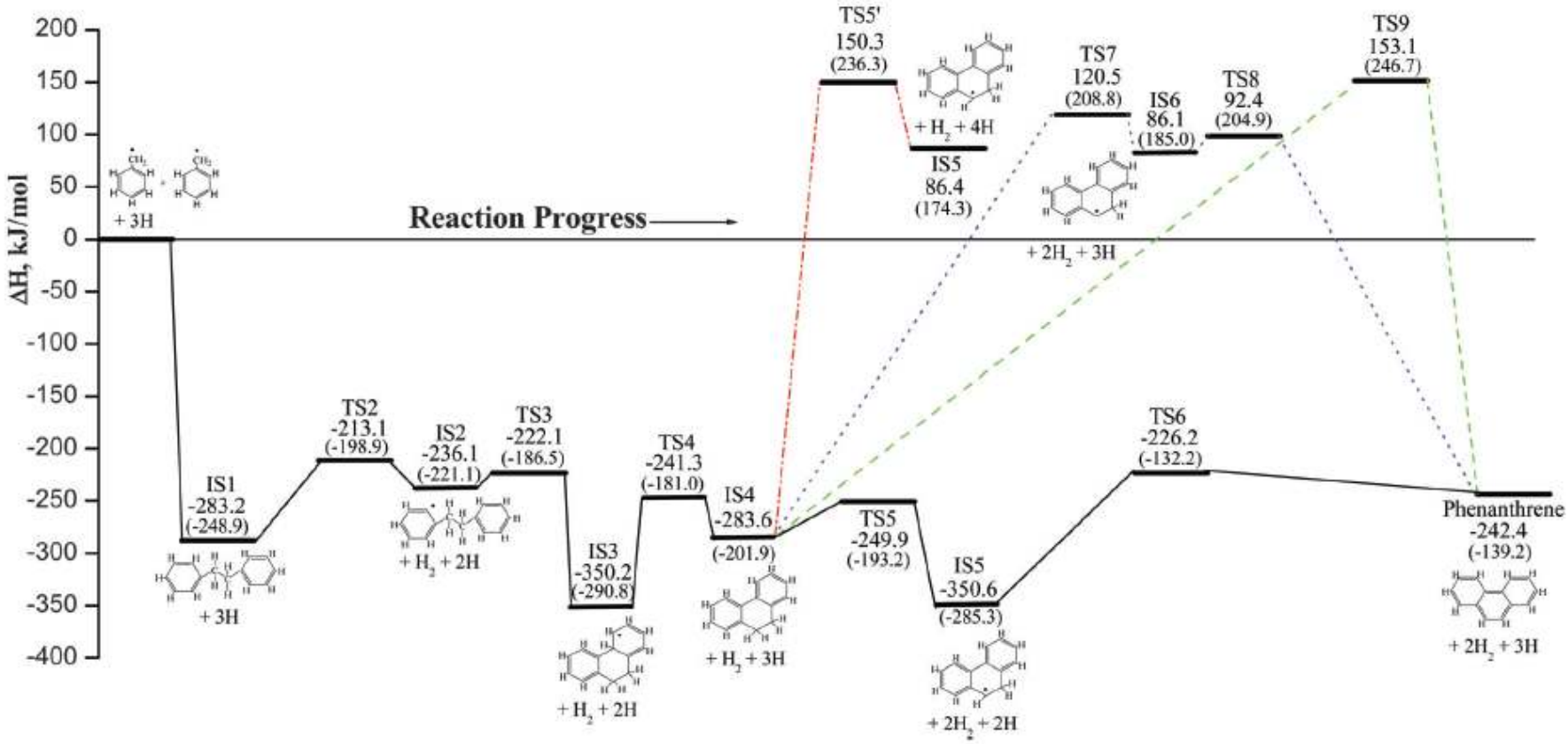
- The recombination of two cyclopentadienyl radicals



- Resonantly stabilized radicals play important roles in the growth of PAHs

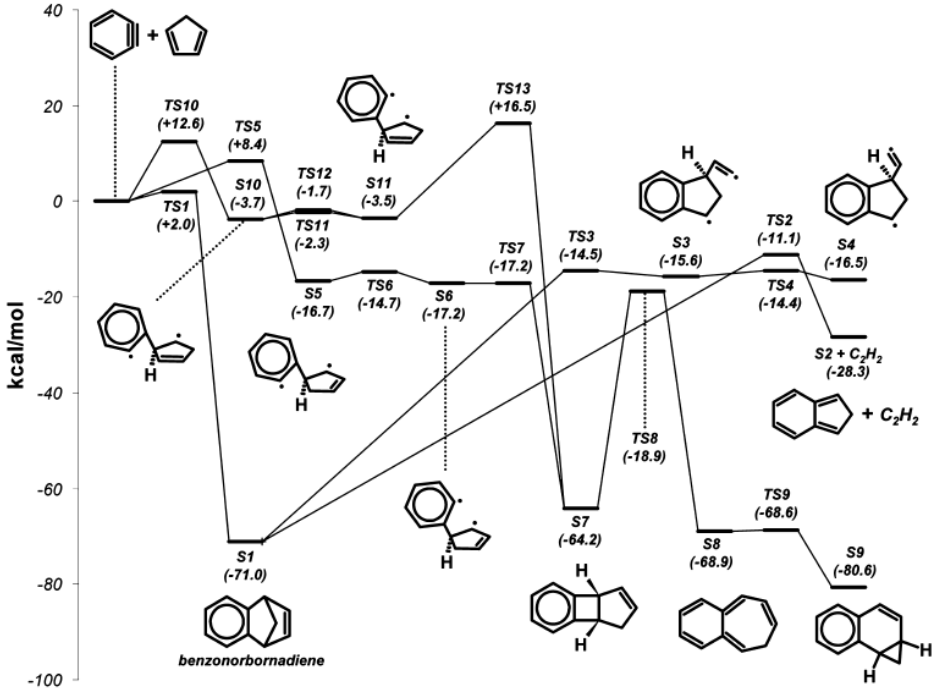
Aromatic Growth

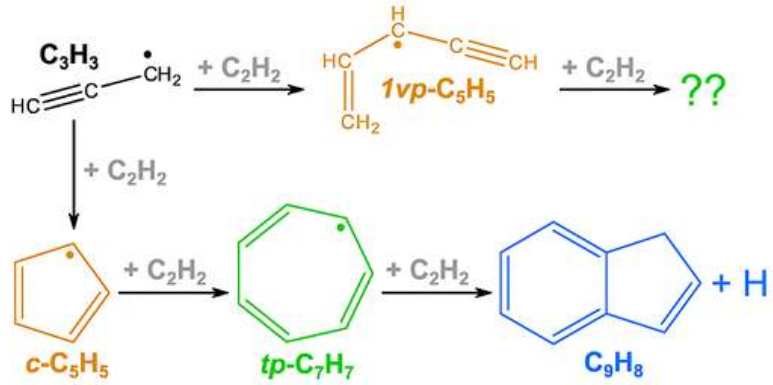
- The dehydrocyclization of diphenylalkanes (二苯烷)

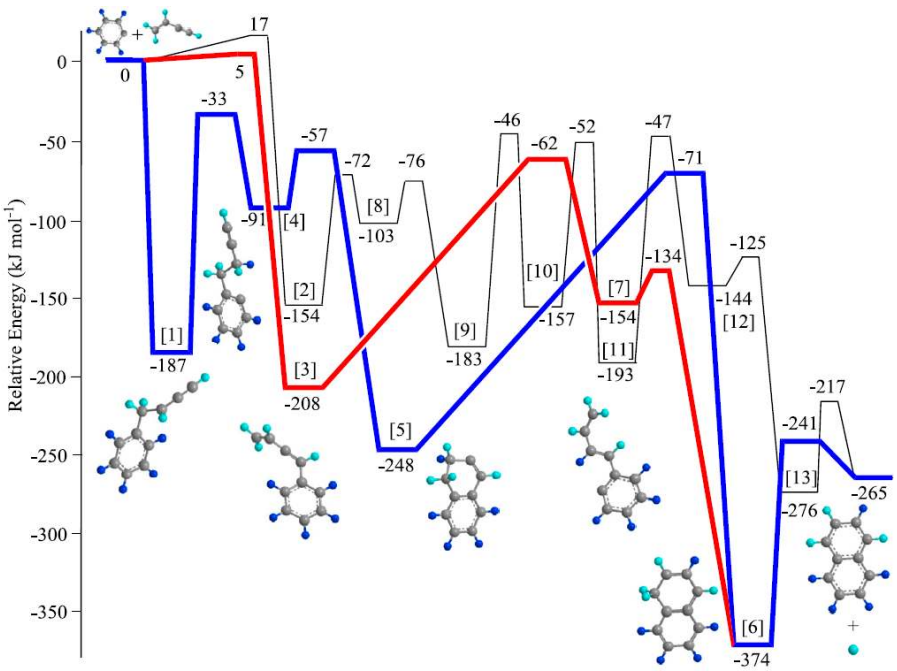


Aromatic Growth

- Other many pathways……

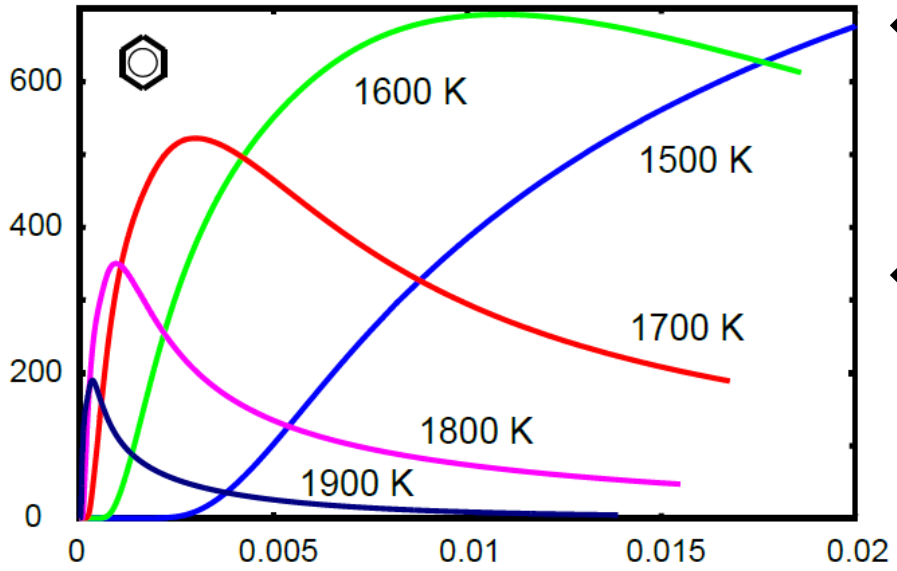


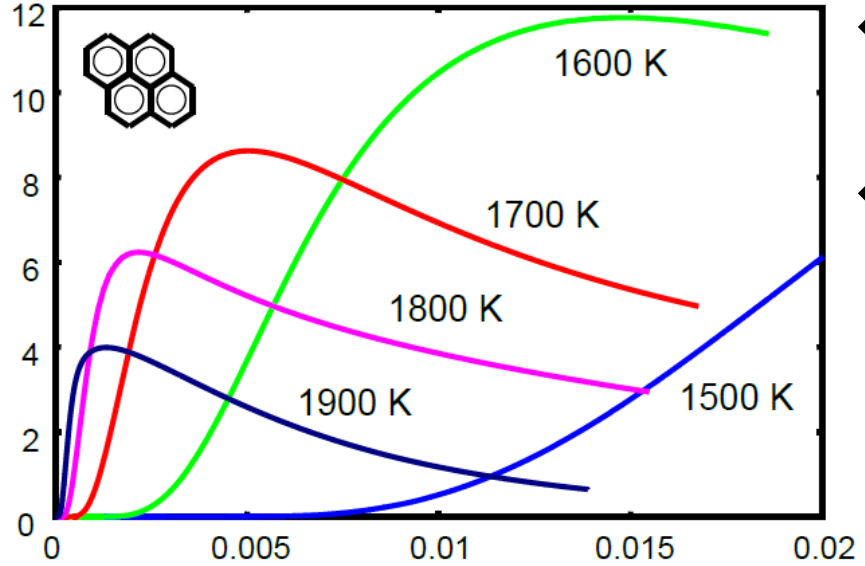




- A large number of pathways to PAHs have yet been considered, and will continue increase in the future

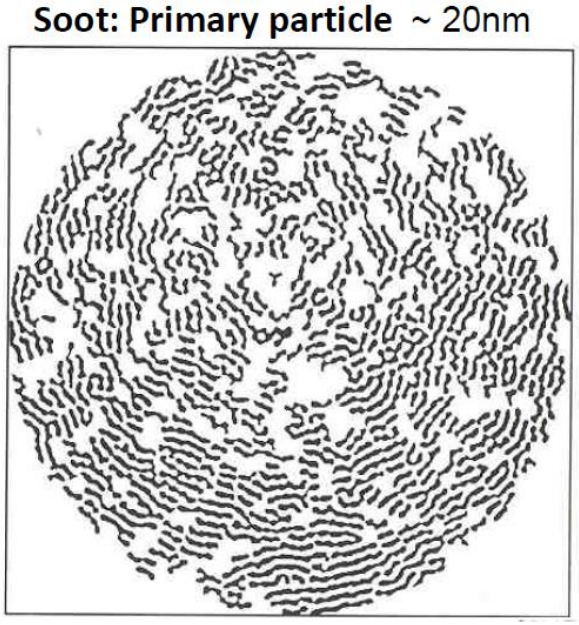
Aromatic Growth

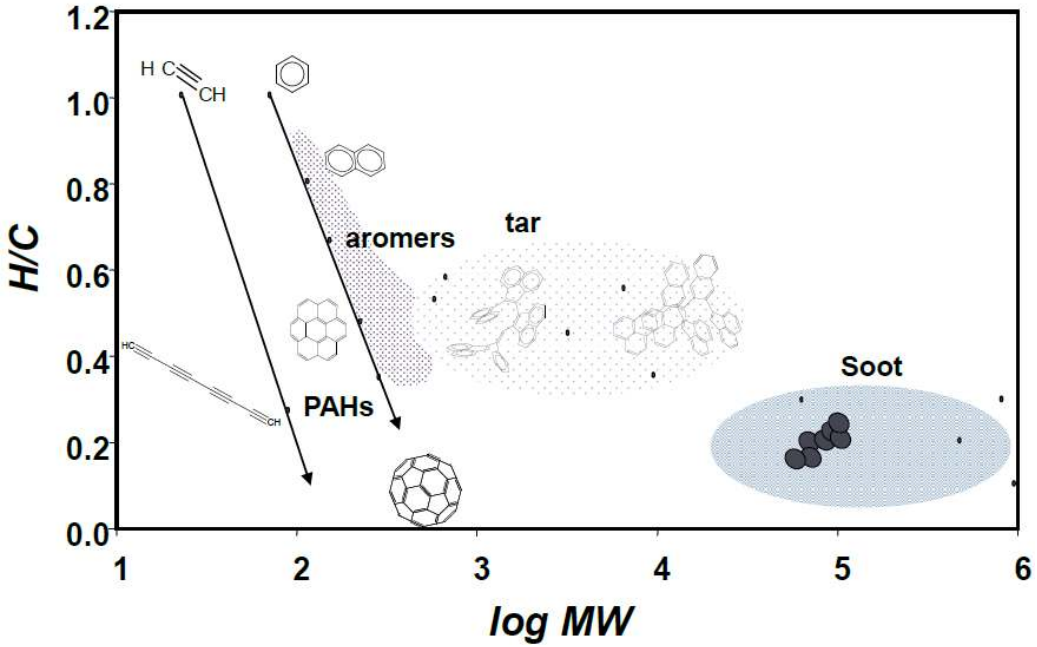




- PAH formation is sensitive to main flame chemistry, local flame conditions, fuel structure and composition
- For real fuels and their surrogate, the number of pathways to aromatics is currently undefined; and it remains to be seen whether this number is finite
- Theoretical approaches beyond one reaction-at-a-time type calculations
- Need to account for the formation of aromatic π radicals

Soot Nucleation

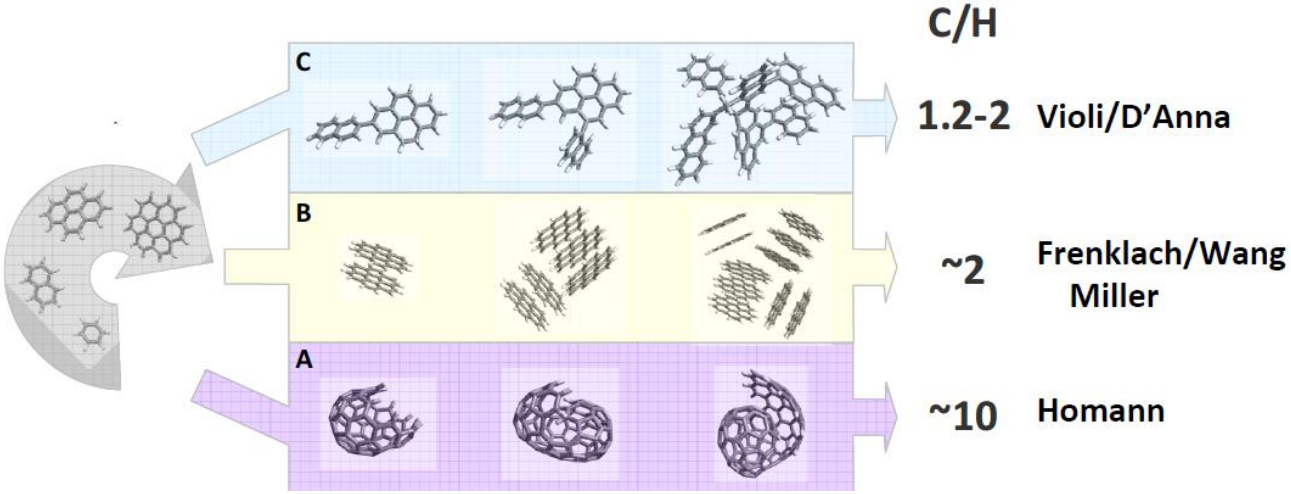




- The mechanism of soot nucleation (or particle inception) that represents the transition of gas-phase species to solid particles
- The mass of the molecules and radicals is continuously increasing above about 2000 Dalton
- The size of the solid particles is increasing by chemical reactions till about size 20 nm

Soot Nucleation

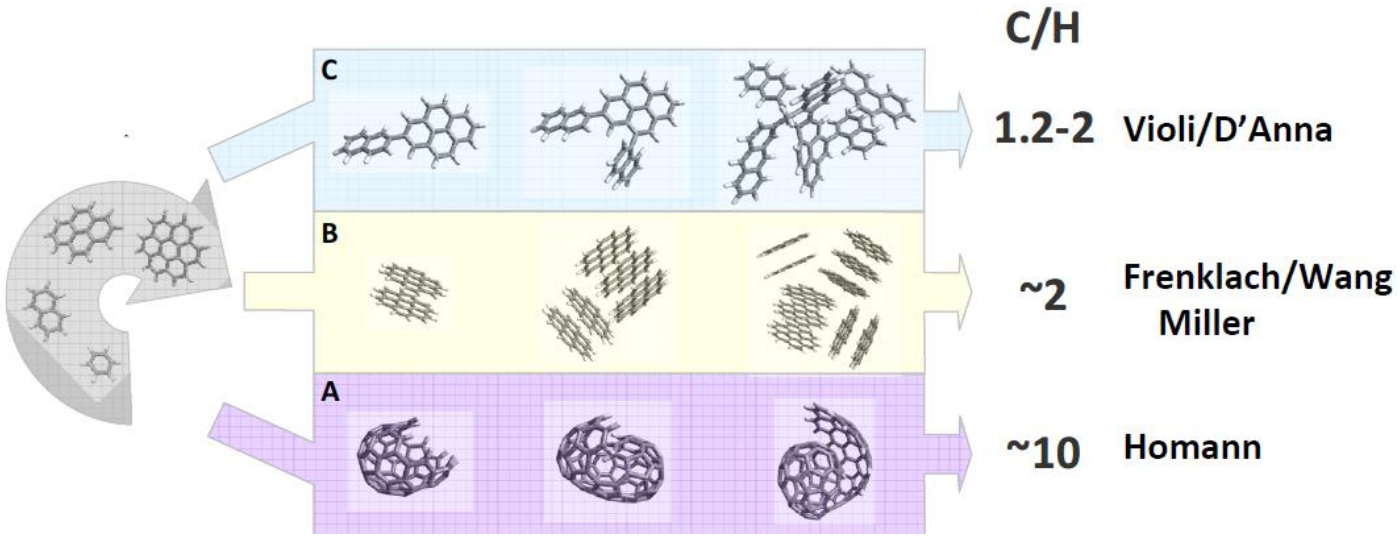
- The mechanism of soot nucleation still remains elusive



- Path A considers the growth of two-dimensional PAHs into curved, fullerene-like structures that are the nucleus of spherical particles. These particles cannot close into a layer of carbon, but develop in successive layers
- Path A is too slow to explain the time scale of soot inception phenomena, also considering the concentrations of PAHs with growing number of rings

Soot Nucleation

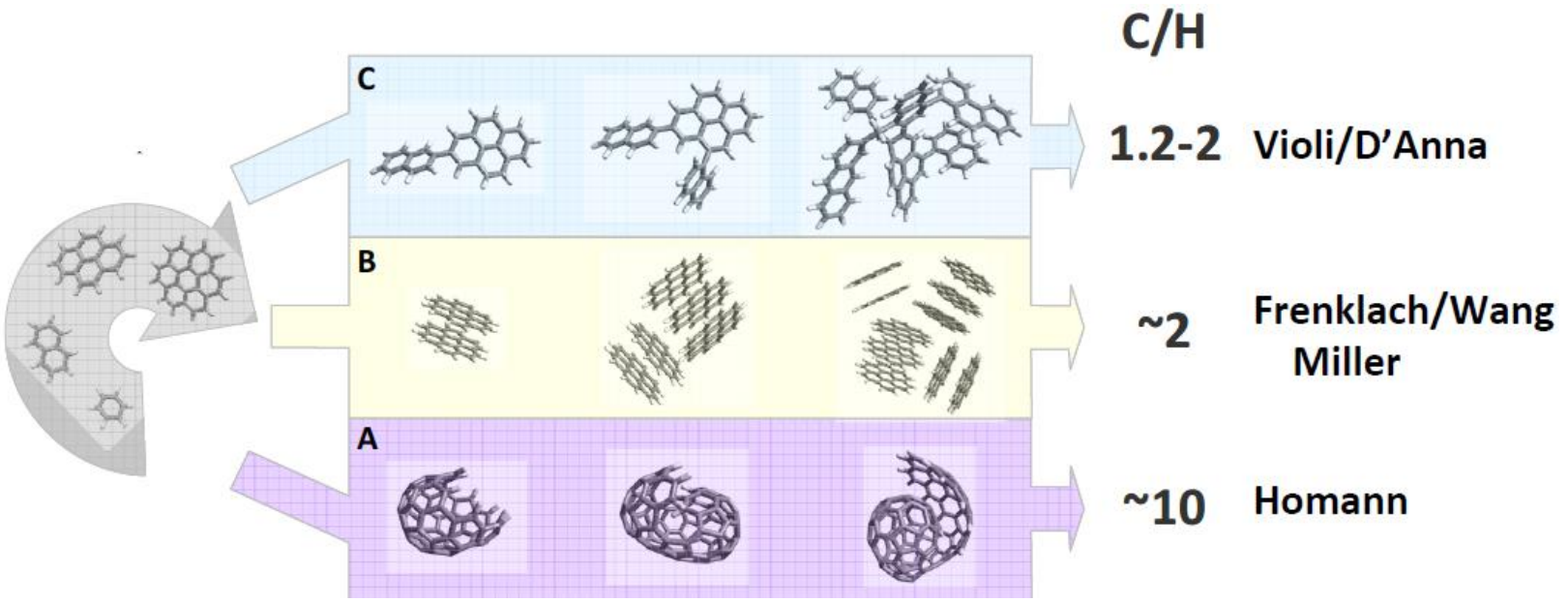
- The mechanism of soot nucleation still remains elusive



- Path B represents the nucleation as physical coalescence of moderate-sized PAHs into stacked clusters
- Path B seems to be able to describe soot formation in high-temperature combustion, when the chemical growth is limited by an high degree of reaction reversibility

Soot Nucleation

- The mechanism of soot nucleation still remains elusive



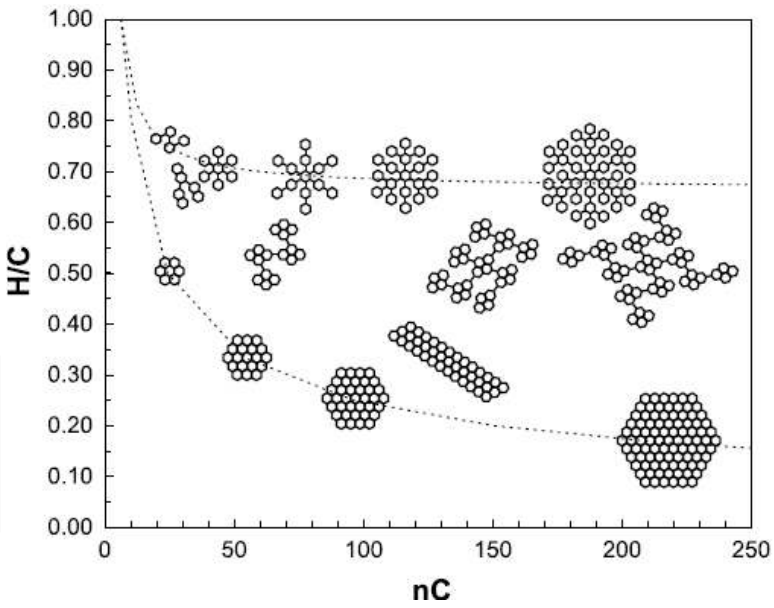
- Path C consists in the reaction or chemical coalescence of PAHs into cross-linked three-dimensional structures
- Path C becomes predominant when the formation of chemical bonds is not heavily counterbalanced by fragmentation, leading to the formation of a network of aromatic-aliphatic-linked structures

Soot Surface Growth

HACA Mechanism

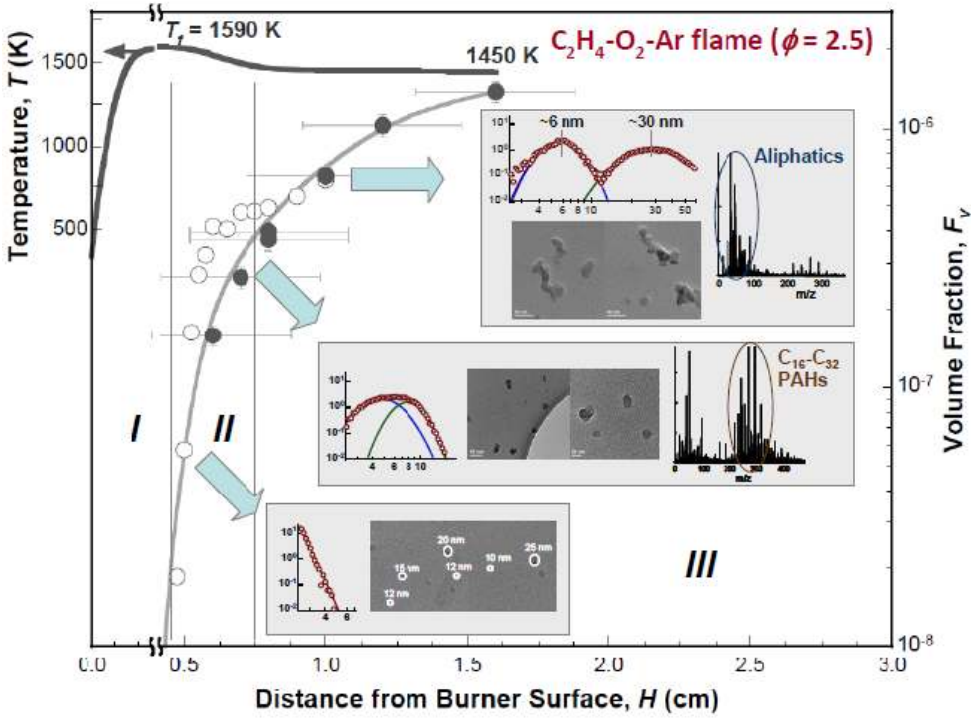
$$S_i\text{–H} + H\bullet \leftrightarrow S_i\bullet + H_2 \quad (1)$$
$$S_i\bullet + H\bullet \rightarrow S_i\text{–H} \quad (2)$$
$$S_i\bullet + C_2H_2 \rightarrow S_{i+2}\text{–H} + H\bullet \quad (3)$$

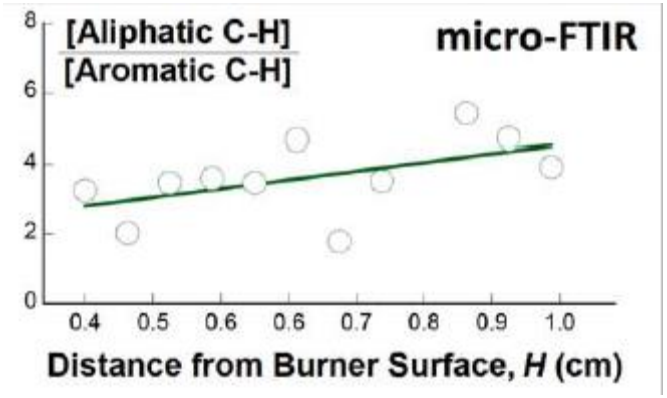
$$\omega_s\left(\frac{\text{mol C-atom}}{\text{cm}^3\text{s}}\right) \approx 2k_3\frac{k_{1f}}{k_{1b}}\frac{[H\bullet]}{[H_2]}[S_i\text{-H}][C_2H_2]$$



- The formed particles' nuclei acquire mass by means of surface growth induced by acetylene addition, by reactions with small resonant radicals or aromatic radicals and PAHs (through biaryl formation)
- Nucleation controls the number of particles, surface growth determines their mass

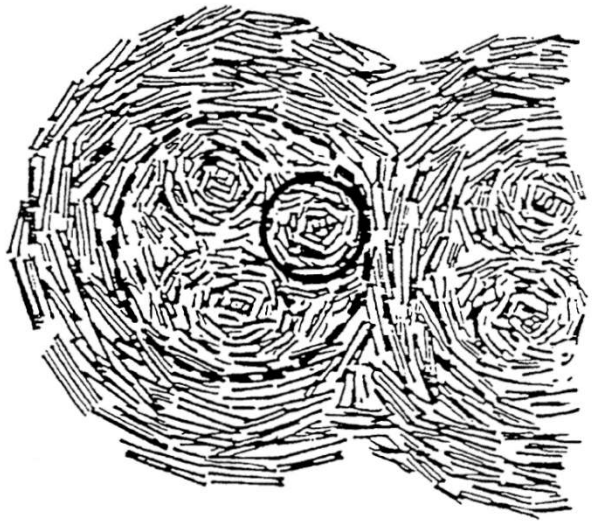
Soot Surface Growth

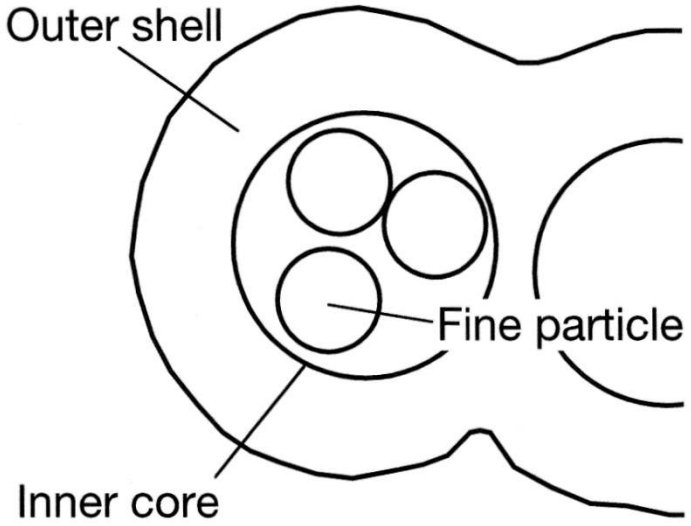




- Baby soot is entirely unlike mature soot
- Nascent soot is liquid-like rather than being carbonized and rigid
- Nascent soot has C/H ~ 1 and ρ = 1.5 g/cc
- Nascent soot is rich in aliphatics (in addition aromatics)

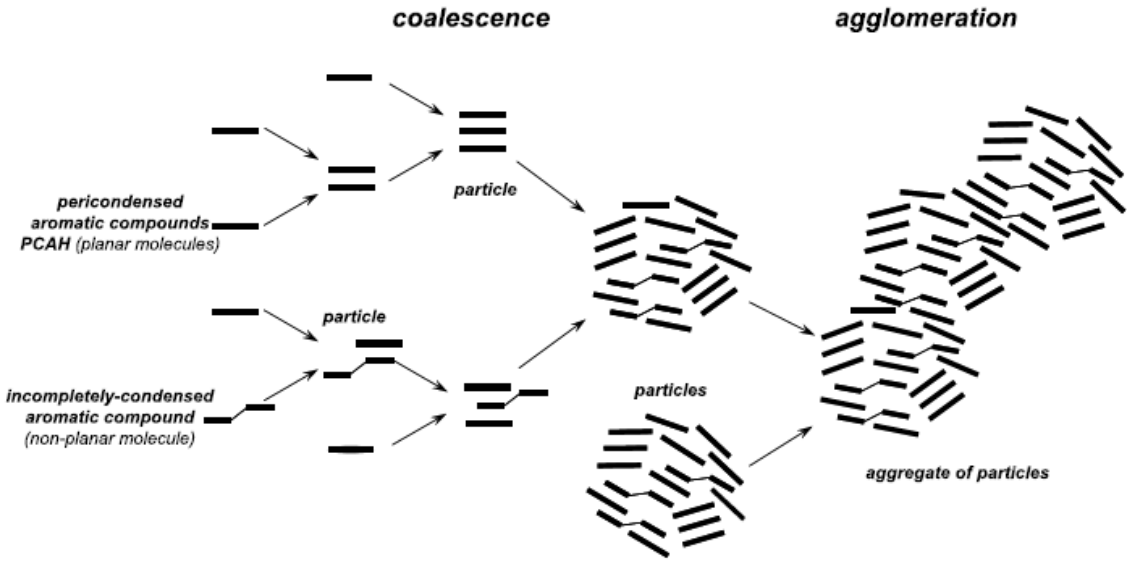
Soot Surface Growth





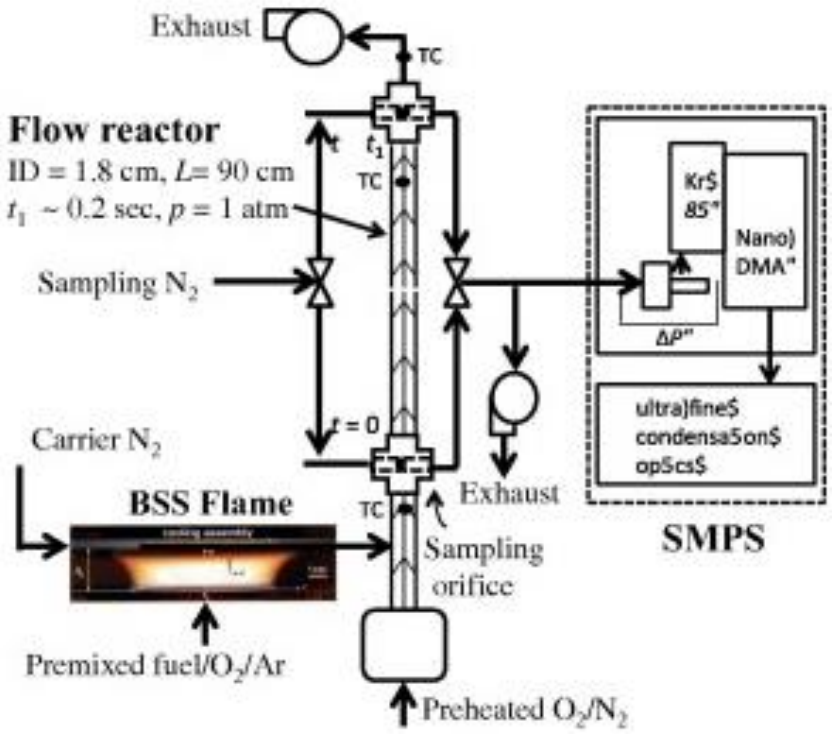
- The ultimate size of soot particles is governed by simultaneous particle-particle coagulation
- The chain-like structure of soot arises due to coagulation process
- The continuous surface growth after particle coagulation leads to a structure where the fine particles constitute the inner core and PAHs form the outer shell

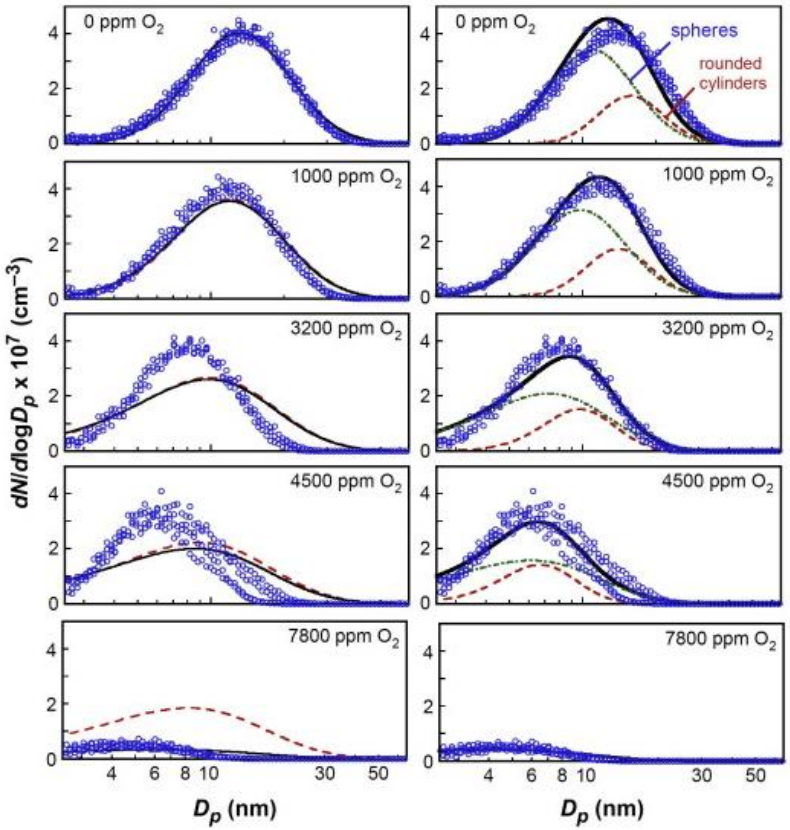
Coagulation



- There are two different types of coagulation (凝结): coalescence (聚结) and agglomeration (成团)
- *Coalescence* happens when a gas-phase molecule or a small particle collides against another particle to form a structure that, to minimize the free energy, tends to reduce its surface
- When bigger particles collide the interacting species maintain their own structure and *aggregate* into chain-like fractal structures

Soot Oxidation

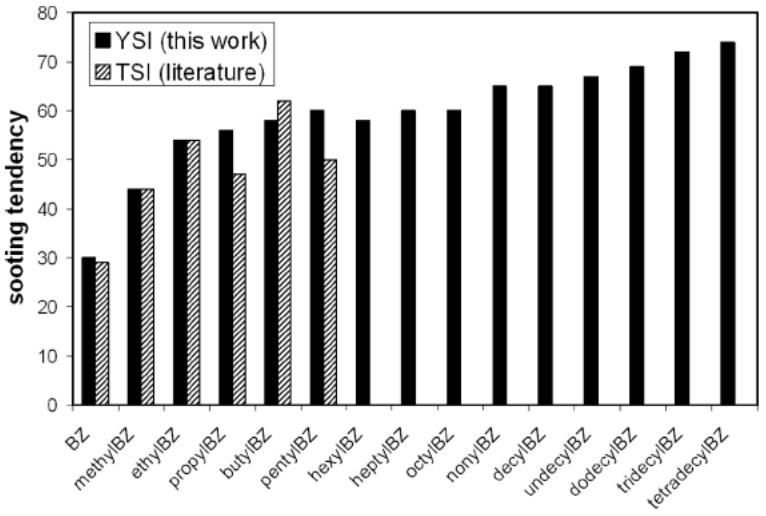




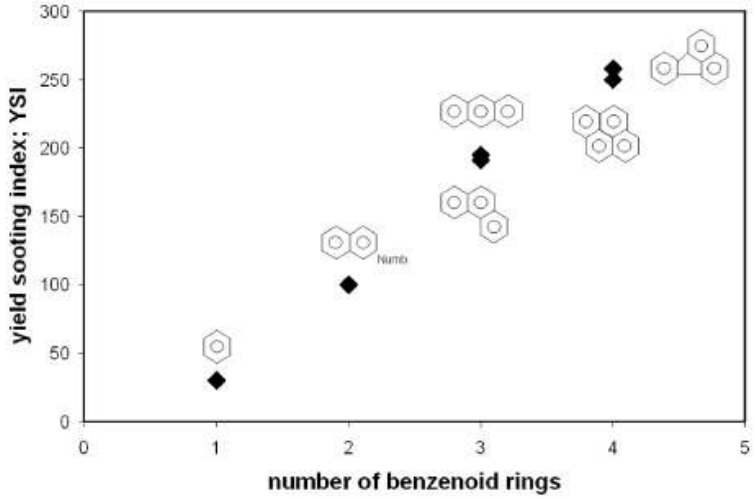
- Soot formation and growth processes are in competition with soot depletion due to oxidation reactions by OH, O and O_2
- The dominant radical in the process of soot oxidation is OH because of its higher concentration

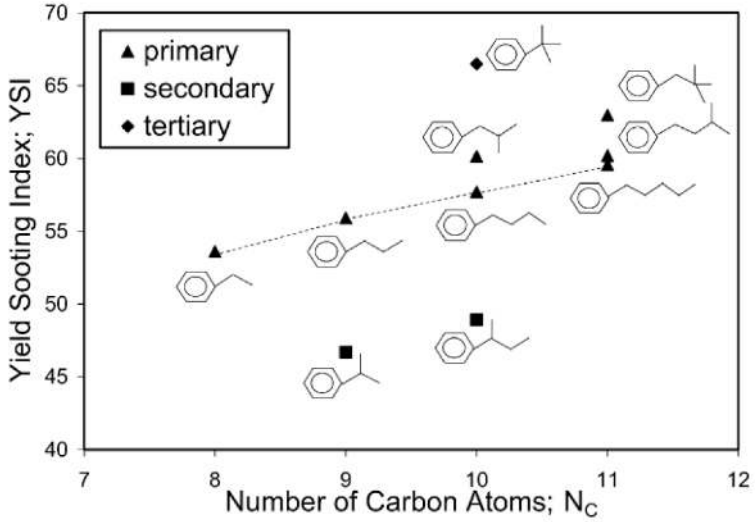
Sooting Tendency

With alkyl chain length



Aromatics



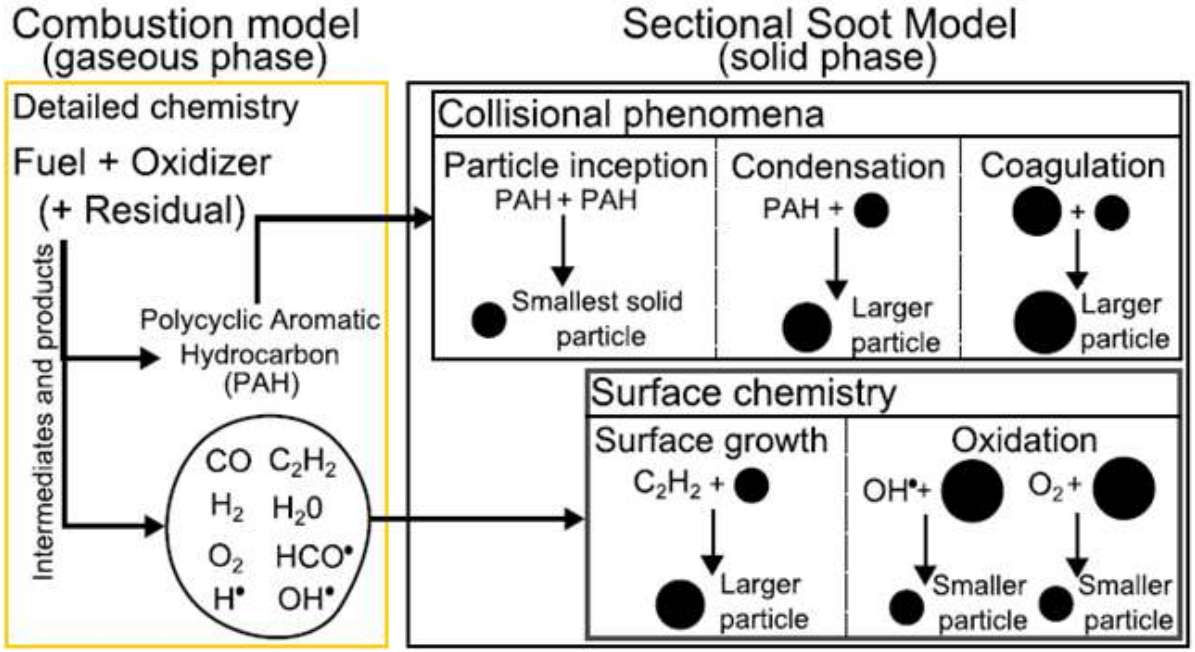


Approaches to Modeling

- The gas and solid phase kinetics of the system must be coupled with transport equations
- It is not possible to write an equation for every species, since particles' dimension, molecular weight and chemical composition span over a wide range
- An accurate description of soot is also numerically very complex and demanding in terms of computational cost
- The main parameters are considered to describe soot: soot volume fraction, number density and particle size distribution
- There have been mainly three approaches proposed in literature: the method of moments, the discrete sectional method and the stochastic approach

Approaches to Modeling

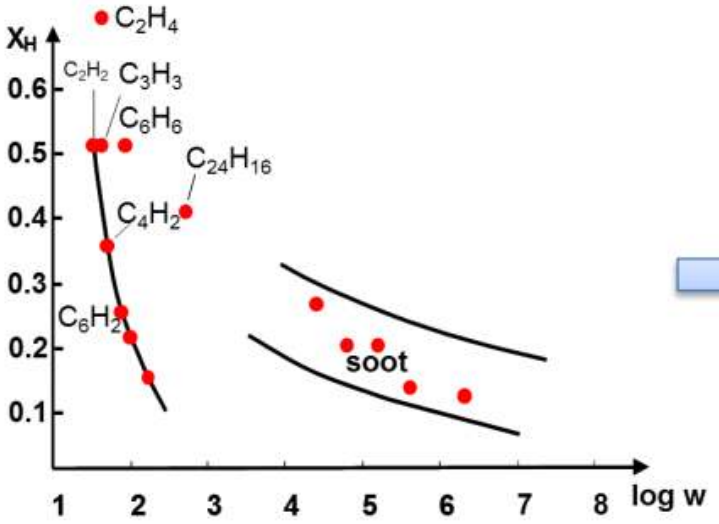
Discrete sectional method

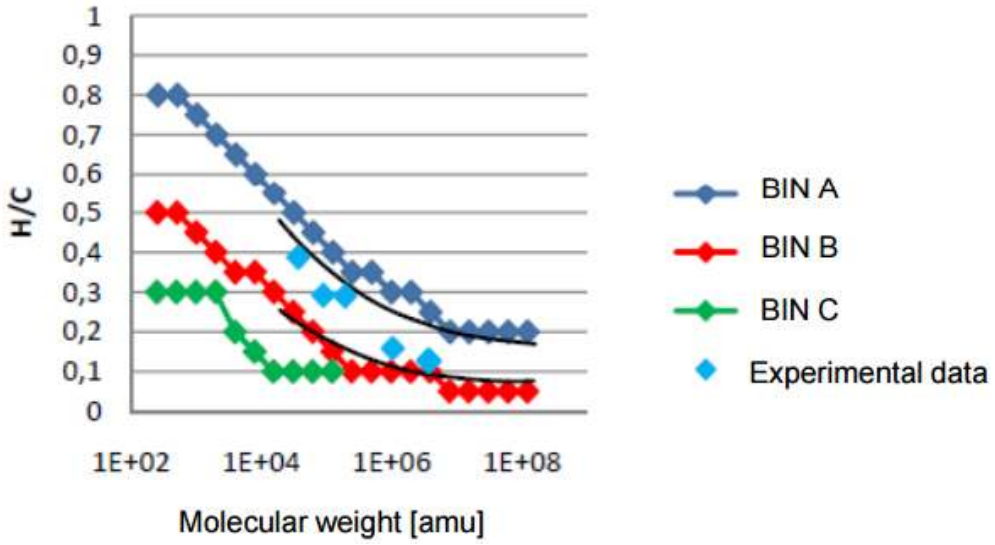


- The discrete sectional methods are based on the separation of the particle size spectrum into a set of size classes
- A transport equation is written for each of the classes
- Very high computational cost, but accurate and gives detailed information

Approaches to Modeling

Discrete sectional method





- Heavy PAHs and soot particles are divided in classes of pseudo-species(BINs)
- Molecular mass increases from a class to the next one

Approaches to Modeling

Discrete sectional method

Different H/C ratios for particles of the same class ⇒

- the **dehydrogenation** reactions
- the **aging** of the soot particles
- the different **degree of methylation** of the pericondensed aromatic species

BIN	Mass [amu]	CxHy	Mean Diameter σ [nm]	H/C A	H/C B	H/C C
1	~ 250	$C_{20}H_{16}$-$C_{20}H_{10}$-$C_{20}H_6$	0.76	0.8	0.5	0.3
2	~ 500	$C_{40}H_{32}$-$C_{40}H_{20}$-$C_{40}H_{12}$	0.96	0.8	0.5	0.3
3	~ 1000	$C_{80}H_{60}$-$C_{80}H_{36}$-$C_{80}H_{24}$	1.21	0.75	0.45	0.3
4	~ 2000	$C_{160}H_{112}$-$C_{160}H_{64}$-$C_{160}H_{48}$	1.52	0.7	0.4	0.3
5	~ 4000	$C_{320}H_{208}$-$C_{320}H_{64}$-$C_{320}H_{64}$	1.91	0.65	0.35	0.20
6	~ 8000	$C_{640}H_{384}$-$C_{640}H_{224}$-$C_{640}H_{96}$	2.41	0.6	0.35	0.15
7	~ 15500	$C_{1250}H_{688}$-$C_{1250}H_{375}$-$C_{1250}H_{125}$	3.01	0.55	0.3	0.1
8	~ 30000	$C_{2500}H_{1250}$-$C_{2500}H_{625}$-$C_{2500}H_{250}$	3.78	0.5	0.25	0.1
9	~ 61000	$C_{5000}H_{2250}$-$C_{5000}H_{1000}$-$C_{5000}H_{500}$	4.76	0.45	0.2	0.1
10	~ 121000	$C_{10000}H_{4000}$-$C_{10000}H_{1500}$-$C_{10000}H_{1000}$	5.99	0.4	0.15	0.1

First soot particle

Approaches to Modeling

☐ Discrete Sectional Method

❖ Reaction classes

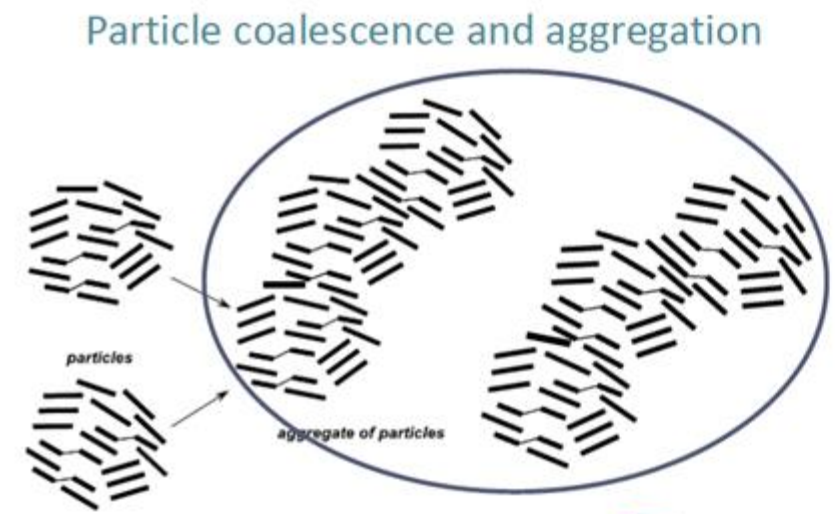
Reactions and their classes	$k = A\,T^n \exp(-E/RT)^a$ A	n	E	Reference Reaction
1. HACA mechanism				
1a. H-abstraction				
$H + BIN_i \rightarrow H_2 + BIN_i\bullet$	$5.4\cdot10^5$	2	10.5	$H + BIN_5 \rightarrow H_2 + BIN_5\bullet$
1b. Acetylene addition				
$C_2H_2 + BIN_i\bullet \rightarrow$ *products*	$1.0\cdot10^9$	0	5	$C_2H_2 + C_{10}H_7 \rightarrow C_{12}H_9\bullet$
2. Soot inception ($i,n < 5$)				
$BIN_i\bullet + BIN_n\bullet \rightarrow$ *products*	$1.0\cdot10^9$	0	6*	$C_5H_5\bullet+C_5H_5\bullet \rightarrow C_{10}H_8+2H\bullet$
$BIN_i + BIN_n\bullet \rightarrow$ *products*	$1.0\cdot10^9$	0	8*	$C_6H_5\bullet+C_8H_6 \rightarrow C_{14}H_{10}+H\bullet$
$BIN_i + BIN_n \rightarrow$ *products*	$1.0\cdot10^8$	0.5	0	$C_{16}H_{10}+C_{16}H_{10} \rightarrow$ products
3. Surface growth				
3a. Small RR• addition				
$RR\bullet + BIN_i \rightarrow$ *products*	$2.0\cdot10^9$	0	19*	$C_3H_3\bullet+C_6H_6 \rightarrow C_9H_8+H\bullet$
$RR\bullet + BIN_i\bullet \rightarrow$ *products*	$2.5\cdot10^9$	0	3*	$C_3H_3\bullet+C_6H_5\bullet \rightarrow C_9H_8$
3b. PAH condensation				
$i \geq 5$				
$PAH\bullet + BIN_i\bullet \rightarrow$ *products*	$5.0\cdot10^{11}$	0.5	0	see text
$PAH\bullet + BIN_i \rightarrow$ *products*				
$i < 5$ and $n \geq 5$				
$BIN_i\bullet + BIN_n\bullet \rightarrow$ *products*	$3.0\cdot10^{11}$	0.5	0	see text
$BIN_i + BIN_n\bullet \rightarrow$ *products*				

4. Dehydrogenation reactions				
4a. Dehydrogenation				
$BIN_i\bullet \rightarrow H\bullet + BIN_i$	$1.0\cdot10^{13}$	0	12	$C_{13}H_{11}\bullet \rightarrow C_{13}H_{10} + H\bullet$
$BIN_i \rightarrow H_2 + BIN_i$	$1.0\cdot10^8$	0	32	$C_{13}H_{12} \rightarrow C_{13}H_{10}+H_2$
4b. Demethylation (for H/C > 0.3)				
$H\bullet + BIN_i \rightarrow CH_3\bullet +$ *products*	$1.2\cdot10^{10}$	0	5	$C_{11}H_{10} + H\bullet \rightarrow C_{10}H_8 + CH_3\bullet$
4c. C-H fission/recombination				
$BIN_i \rightarrow H\bullet + BIN_i\bullet$	$1.5\cdot10^{17}$	0	114	$C_6H_6 \rightarrow C_6H_5\bullet + H\bullet$
$H\bullet + BIN_i\bullet \rightarrow BIN_i$	$1.0\cdot10^{13}$	0	0	$C_6H_5\bullet+H\bullet \rightarrow C_6H_6$
5. Particle coalescence and aggregation				
5a. Particle coalescence ($5 \leq i,n < 13$)				
$BIN_i + BIN_n \rightarrow$ *products*	$2.0\cdot10^{11}$	0.5	0	see text
5b. Particle coalescence on aggregates ($5 \leq i < 13$ and $n \geq 13$)				
$BIN_i + BIN_n \rightarrow$ *products*	$1.5\cdot10^{13}$	0.5	0	see text
5c. Particle aggregation ($i,n \geq 13$)				
$BIN_i + BIN_n \rightarrow$ *products*	$1.5\cdot10^{13}$	0.5	0	see text
6. Oxidation				
6a. Oxidation with OH•				
$OH\bullet + BIN_i \rightarrow$ *products* $+ CH_2CO$	$1.0\cdot10^{10}$	0	10	$OH\bullet+C_{16}H_{10} \rightarrow C_{14}H_9\bullet+CH_2CO$
$OH\bullet+BIN_i \rightarrow$ *products* $+CO+CH_3\bullet$	$1.0\cdot10^{10}$	0	4	$OH\bullet+C_{11}H_{10} \rightarrow C_9H_8+CO+CH_3\bullet$
$OH\bullet+BIN_i\bullet \rightarrow$ *products* $+CO+H\bullet$	$2.0\cdot10^{13}$	0	0	$OH\bullet+C_{10}H_7\bullet \rightarrow C_9H_7\bullet+CO+H\bullet$
$OH\bullet + BIN_i \rightarrow$ *products* $+ HCO\bullet$	$3.0\cdot10^9$	0.5	10.6	$OH\bullet+C_{10}H_8 \rightarrow C_9H_8+HCO\bullet$
6b. Oxidation with O•				
$O\bullet + BIN_i \rightarrow$ *products* $+ HCCO\bullet$	$2.0\cdot10^{10}$	0	4	$O\bullet+C_{16}H_{10} \rightarrow C_{14}H_9\bullet+HCCO\bullet$
$O\bullet + BIN_i\bullet \rightarrow$ *products* $+ CO$	$5.0\cdot10^{10}$	0	0	$O\bullet+C_{10}H_7\bullet \rightarrow C_9H_7\bullet+CO$
6c. Oxidation with O_2				
$O_2+BIN_i\bullet \rightarrow$ *products* $+CO+HCO\bullet$	$2.1\cdot10^9$	0	7.4	$O_2+C_{10}H_7\bullet \rightarrow C_8H_6\bullet+CO+HCO\bullet$
$O_2 + BIN_i\bullet \rightarrow$ *products* $+O\bullet+CO$	$2.6\cdot10^{10}$	0	6.12	$O_2+C_{10}H_7\bullet \rightarrow C_9H_7\bullet+O\bullet+CO$
$O_2 + BIN_i\bullet \rightarrow$ *products* $+ 2CO$	$4.2\cdot10^8$	0.5	8	$O_2+C_{16}H_9\bullet \rightarrow C_{14}H_9\bullet+2CO$

❖ HACA mechanism; soot inception; surface growth; dehydrogenation; particle coalescence and aggregation; oxidation

Approaches to Modeling

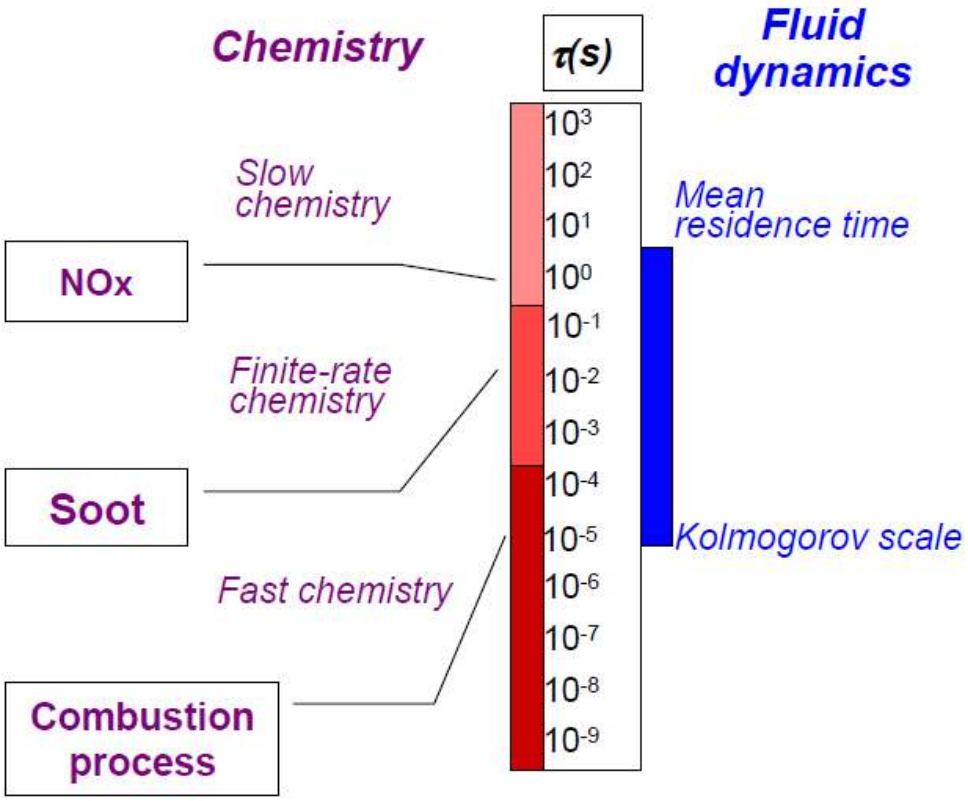
Discrete Sectional Method



Reactions and their classes	$k = A\,T^n \exp(-E/RT)$ A	n	E	Reference Reaction
1. HACA mechanism				
1a. H-abstraction				
$H + BIN_i \rightarrow H_2 + BIN_i\bullet$	$5.4\cdot10^3$	2	10.5	$H + BIN_3 \rightarrow H_2 + BIN_3\bullet$
1b. Acetylene addition				
$C_2H_2 + BIN_n\bullet \rightarrow products$	$1.0\cdot10^9$	0	5	$C_2H_2 + C_{16}H_7 \rightarrow C_{18}H_9\bullet$
2. Soot inception ($i,n < 5$)				
$BIN_i\bullet + BIN_n\bullet \rightarrow products$	$1.0\cdot10^9$	0	6*	$C_3H_3\bullet + C_3H_3\bullet \rightarrow C_6H_6 + 2H\bullet$
$BIN_i + BIN_n\bullet \rightarrow products$	$1.0\cdot10^9$	0	8*	$C_6H_5\bullet + C_4H_4 \rightarrow C_{14}H_{10} + H\bullet$
$BIN_i + BIN_n \rightarrow products$	$1.0\cdot10^9$	0.5	0	$C_{16}H_{10} + C_{16}H_{10} \rightarrow products$
3. Surface growth				
3a. Small RR• addition				
$RR\bullet + BIN_i \rightarrow products$	$2.0\cdot10^9$	0	19*	$C_6H_5\bullet + C_6H_6 \rightarrow C_6H_6 + H\bullet$
$RR\bullet + BIN_i\bullet \rightarrow products$	$2.5\cdot10^9$	0	3*	$C_3H_3\bullet + C_4H_2\bullet \rightarrow C_7H_5$
3b. PAH condensation				
$i \geq 5$				
$PAH\bullet + BIN_i\bullet \rightarrow products$	$5.0\cdot10^{11}$	0.5	0	see text
$PAH\bullet + BIN_i \rightarrow products$				
$i < 5$ and $n \geq 5$				
$BIN_i\bullet + BIN_n\bullet \rightarrow products$	$3.0\cdot10^{11}$	0.5	0	see text
$BIN_i + BIN_n\bullet \rightarrow products$				
4. Dehydrogenation reactions				
4a. Dehydrogenation				
$BIN_i\bullet \rightarrow H\bullet + BIN_i$	$1.0\cdot10^{13}$	0	12	$C_{13}H_{11}\bullet \rightarrow C_{13}H_{10} + H\bullet$
$BIN_i \rightarrow H_2 + BIN_i$	$1.0\cdot10^8$	0	32	$C_{13}H_{12} \rightarrow C_{13}H_{10} + H_2$
4b. Demethylation (for H/C > 0.3)				
$H\bullet + BIN_i \rightarrow CH_3\bullet + products$	$1.2\cdot10^{10}$	0	5	$C_{14}H_{10} + H\bullet \rightarrow C_{26}H_8 + CH_3\bullet$
4c. C-H fission/recombination				
$BIN_i \rightarrow H\bullet + BIN_i\bullet$	$1.5\cdot10^{17}$	0	114	$C_6H_6 \rightarrow C_6H_5\bullet + H\bullet$
$H\bullet + BIN_i\bullet \rightarrow BIN_i$	$1.0\cdot10^{13}$	0	0	$C_6H_5\bullet + H\bullet \rightarrow C_6H_6$
5. Particle coalescence and aggregation				
5a. Particle coalescence ($5 \leq i,n < 13$)				
$BIN_i + BIN_n \rightarrow products$	$2.0\cdot10^{11}$	0.5	0	see text
5b. Particle coalescence on aggregates ($5 \leq i < 13$ and $n \geq 13$)				
$BIN_i + BIN_n \rightarrow products$	$1.5\cdot10^{11}$	0.5	0	see text
5c. Particle aggregation ($i,n \geq 13$)				
$BIN_i + BIN_n \rightarrow products$	$1.5\cdot10^{11}$	0.5	0	see text
6b. Oxidation with O•				
$O\bullet + BIN_i \rightarrow products + HCCO\bullet$	$2.0\cdot10^{13}$	0	4	$O\bullet + C_{16}H_{10} \rightarrow C_{14}H_9\bullet + HCCO\bullet$
$O\bullet + BIN_i\bullet \rightarrow products + CO$	$5.0\cdot10^{12}$	0	0	$O\bullet + C_{10}H_7\bullet \rightarrow C_9H_7\bullet + CO$
6c. Oxidation with O_2				
$O_2 + BIN_i\bullet \rightarrow products + CO + HCO\bullet$	$2.1\cdot10^9$	0	7.4	$O_2 + C_{10}H_7\bullet \rightarrow C_8H_8\bullet + CO + HCO\bullet$
$O_2 + BIN_i\bullet \rightarrow products + O\bullet + CO$	$2.6\cdot10^{12}$	0	6.12	$O_2 + C_{20}H_7\bullet \rightarrow C_9H_7\bullet + O\bullet + CO$
$O_2 + BIN_i\bullet \rightarrow products + 2CO$	$4.2\cdot10^9$	0.5	8	$O_2 + C_{14}H_9\bullet \rightarrow C_{14}H_9\bullet + 2CO$

Approaches to Modeling

Soot modeling in turbulent flames



- The soot characteristic time is of the same order of the fluid dynamics times
- Combustion process: Thermal field and species can be successfully modeled using the flamelets approach
- Soot: strong interactions between fluid mixing and chemical reactions
- Specific approaches need to be used to model soot in turbulent flames

Thanks for your attention



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